A no-strong-triangle condition for the Méndez–Vaz Pinto–Villarreal conjecture on weighted monomial ideals

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Abstract

Let I be a nonzero proper monomial ideal satisfying the three original Méndez–Vaz Pinto–Villarreal hypotheses: I has no embedded associated primes, the radicals in its irredundant irreducible decomposition are pairwise distinct, and I is Simis. If the minimal generator supports of I and $\sqrt{I}$ coincide and their clutter has no strong 3-cycle, we prove that $I = (\sqrt{I})^{\delta_d}$ for a positive diagonal weighting δ_d and that $\sqrt{I}$ is a squarefree Simis ideal in the original ring, without adjoining variables or using a nonstandard weighting. The proof first forces one generator on each support; an isolated unequal-exponent pair then yields a monomial in $I^{(2)} \setminus I^2$, and the weighting–Simis transfer controls the radical. The boundary is sharp at this hypothesis level: $I_{r,s} = (ac, bd, abz^r, cdz^s)$ with $r \neq s$ satisfies the original hypotheses and the support equality, while admitting no same-ring squarefree positive-diagonal representation, even without requiring the source to be Simis. The theorem also covers an infinite quantified support-4 family outside the named forest and exact or cumulative support-at-most-3 classes, and examples with strong triangles still satisfying the conclusion show that the new condition is sufficient but not necessary. The main result of this paper was obtained using the Pharos system.

Key words and phrases: monomial ideals, symbolic powers, Simis ideals, weighted monomial ideals, clutters, strong cycles.

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1 Introduction

Let k be a field, let $n \geq 1$, and let $S = k[x_1, \ldots, x_n]$ be a polynomial ring in algebraically independent variables. All ideals below are monomial ideals. For $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$, write $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ and $\operatorname{supp}(x^\alpha) = \{x_i \mid \alpha_i > 0\}$. Let $\mathrm{G}(L)$ denote the minimal monomial generating set of a monomial ideal L. Put

$$\mathcal{F}(L) = \{\operatorname{supp}(u) \mid u \in \mathrm{G}(L)\}, \qquad \Delta(L) = \langle F : F \in \mathcal{F}(L)\rangle,$$

where $\langle \mathcal{F}(L)\rangle$ is the simplicial complex generated by the members of $\mathcal{F}(L)$. When $\mathcal{F}(L)$ is an antichain, its members are precisely the facets of $\Delta(L)$; in that case $\mathcal{F}(L)$ is a *clutter*. If $\mathcal{H}$ is a clutter on a finite vertex set V, its *blocker* is the clutter $b(\mathcal{H})$ of inclusion-minimal subsets of V meeting every member of $\mathcal{H}$.

For a nonempty set A of variables and positive integral exponents $a = (a_x)_{x\in A}$, put

$$Q(A, a) = (x^{a_x} : x \in A), \qquad P(A) = (x : x \in A).$$

Every nonzero proper irreducible monomial ideal is uniquely of the form $Q(A, a)$ and is $P(A)$-primary. Every nonzero proper monomial ideal L has an irredundant finite decomposition

$$L = \bigcap_{j=1}^{m} Q_j$$

into irreducible monomial ideals, unique up to reordering. The decomposition is *MPV-minimal* if the radicals $\sqrt{Q_1}, \ldots, \sqrt{Q_m}$ are pairwise distinct. Symbolic powers are always taken with respect to the minimal primes:

$$L^{(s)} = \bigcap_{P\in \operatorname{Min}(L)} (L^s S_P \cap S) \qquad (s \geq 1).$$

The ideal L is *Simis* if $L^{(s)} = L^s$ for every $s \geq 1$, and *normally torsion-free* if $\operatorname{Ass}(S/L^s) \subseteq \operatorname{Ass}(S/L)$ for every $s \geq 1$. It has no embedded primes when $\operatorname{Ass}(S/L) = \operatorname{Min}(L)$.

For $d = (d_1, \ldots, d_n) \in {\mathbb{Z}_{>0}}^n$, let $\delta_d(\alpha) = (d_1\alpha_1, \ldots, d_n\alpha_n)$. The *standard positive diagonal weighting* of a monomial ideal $L \subseteq S$ is

$$L^{\delta_d} = \Big(\Big\{x^{\delta_d(\alpha)} \,\Big|\, \alpha \in \mathbb{N}^n,\ x^\alpha \in L\Big\}\Big).$$

This operation uses the same polynomial ring; no variables may be adjoined. We call a monomial ideal squarefree when it is generated by squarefree monomials. The *three original MPV hypotheses* on a nonzero proper monomial ideal L are that L has no embedded primes, that an irredundant irreducible decomposition of L is MPV-minimal, and that L is Simis. The *MPV conclusion* holds for L if there are a squarefree Simis monomial ideal $J \subseteq S$ and $d \in {\mathbb{Z}_{>0}}^n$ such that $L = J^{\delta_d}$. Méndez, Vaz Pinto, and Villarreal conjectured that the three original hypotheses imply the MPV conclusion [MPV25, Conjecture 5.7]; the same statement is recorded as Conjecture 1.1 in [BDK26a].

For $q \geq 3$, a *strong q-cycle* in a clutter $\mathcal{H}$ is an alternating sequence

$$(v_1, E_1, v_2, E_2, \ldots, v_q, E_q, v_1)$$

of q distinct vertices and q distinct edges $E_i \in \mathcal{H}$ such that, with indices read modulo q,

$$E_i \cap \{v_1, \dots, v_q\} = \{v_i, v_{i+1}\} \qquad (1 \le i \le q).$$

The clutter is *balanced* if it has no strong cycle of odd length. We use the facet-induced convention for subcomplexes. Thus, for $\mathcal{G} \subseteq \mathcal{F}(\Gamma)$, the complex $\langle \mathcal{G} \rangle$ has facet set $\mathcal{G}$. If a complex Λ has more than one facet, a facet F is a *leaf* if some other facet G satisfies $F \cap H \subseteq F \cap G$ for every facet $H \neq F$ of Λ; a complex with one facet is itself a leaf. A nonempty finite complex is a *simplicial forest* if every nonempty facet subcomplex has a leaf; a connected forest is a *simplicial tree*. These are Faridi's notions [Far02]. For a finite complex Γ, write $\beta_1(\Gamma)$ for its first simplicial Betti number over a fixed coefficient field; the assertion $\beta_1 = 0$ below holds over every coefficient field.

Méndez, Vaz Pinto, and Villarreal also describe the canonical primary components and symbolic powers of a monomial ideal through its irredundant irreducible decomposition, prove that standard diagonal weighting preserves that primary structure, and prove that L is Simis exactly when L^{δ_d} is Simis, with no embedded-prime hypothesis in the last equivalence [MPV25, Section 1, Corollaries 4.6 and 5.5(b)]. They prove the corresponding symbolic-power classification for standardly weighted graph edge ideals, where bipartiteness is the decisive condition, and the all-symbolic-power criterion for the weighted oriented graph ideals treated there, with the full weighted-root conclusion in the sink case [MPV25, Corollary 5.8, Theorem 7.5, Corollary 7.7]. Their Example 8.5 also shows that the pairwise-distinct-radical condition cannot simply be omitted: without it, the radical of a Simis ideal need not be Simis.

The conjecture is known in several support-restricted classes. Bordoloi, Das, and Kumar prove that, for an exact support-2 monomial ideal in the original structural class, Simisness is equivalent to bipartiteness of the underlying graph together with representability by a standard linear weighting; in particular the MPV conclusion holds throughout that class [BDK26b]. Here exact support 2 means that every minimal monomial generator has support cardinality exactly 2. In subsequent work, the same authors prove the conjecture when the three original hypotheses are supplemented by $\mathcal{F}(L) = \mathcal{F}(\sqrt{L})$ and $\Delta(L)$ is a simplicial forest, and when the ideal has exact support 3 and satisfies the three original hypotheses [BDK26a, Theorem 2.3, Section 3, Theorem 3.10]. Support 3 there is exact, not cumulative support at most 3; support equality is automatic for an exact support-3 ideal.

At the squarefree level these complexes sit in a second hierarchy. Fulkerson, Hoffman, and Oppenheim prove that every balanced hypergraph is Mengerian [FHO74]; Herzog, Hibi, Trung, and Zheng record that implication and prove that a facet complex is Mengerian precisely when symbolic and ordinary powers of its squarefree facet ideal agree in every degree [HHTZ08, Section 2, Corollary 1.6]. Their results also relate simplicial forests to the absence of all strong cycles and hence give the strict hierarchy

$$\text{forest} \implies \text{balanced} \implies \text{Mengerian} \iff \text{Simis and normally torsion-free}$$

for facet ideals, with both inclusions strict in general. None of these squarefree statements itself controls the exponent rigidity of a nonsquarefree ideal, but, together with the MPV Simis transfer, they supply an essential input to the strict family stated below.

Bijender and Kumar give a component-incidence criterion of a different kind [BK25, Theorem 4.7]. Let $I \subsetneq S$ be a nonzero monomial ideal without embedded primes, and let

$$I = \bigcap_{j=1}^{r} q_j, \qquad q_j = (x_i^{w_{j,i}} : i \in A_j), \qquad p_j = \sqrt{q_j} = (x_i : i \in A_j),$$

be an irredundant irreducible decomposition, with $\{1,\dots,n\} = \bigcup_{j=1}^{r} A_j$, indexed so that

$$p_j \neq p_k \qquad (j \neq k),$$

equivalently $A_j \neq A_k$ for $j \neq k$. This pairwise-distinct-radical condition is BK's Notation 3.5 minimality convention; Set-up 4.1 and Theorem 4.7 retain it, and together with irredundance it is exactly the MPV-minimality convention used above. Put

$$X_I = \{(k,\ell) \mid 1 \leq k < \ell \leq r \text{ and some } i \in A_k \cap A_\ell \text{ satisfies } w_{k,i} \neq w_{\ell,i}\}.$$

If $A_j \not\subseteq A_k \cup A_\ell$ for every $(k,\ell) \in X_I$ and every $j \notin \{k,\ell\}$, then I is Simis if and only if it has a Simis squarefree standard positive diagonal root in S; consequently every Simis ideal satisfying the structural antecedent satisfies the MPV conclusion. The distinct-radical convention is essential, not terminology. In $k[x,y]$, the ideal $(x,y)^2 = (x^2,y) \cap (x,y^2)$ is irredundant and Simis, has no embedded primes, has equal component radicals, exhausts the variables, and has a vacuous intervening-support condition, but it has no squarefree positive-diagonal root. Section 8 records the proof. Thus deleting $p_j \neq p_k$ makes the displayed equivalence false; BK-minimality and MPV-minimality must not be separated.

The known arguments stop at a common geometric configuration. The forest and support-size arguments compare two generator supports F and G by second-symbolic monomial witnesses. A leaf ordering supplies pairs whose induced facet subcomplex on $F \cup G$ contains no third facet, while exact support 3 reduces all possible third supports to a finite catalogue. The BK component argument instead requires every exponent-mismatching component pair $(k,\ell) \in X_I$ to have no third component support inside $A_k \cup A_\ell$; its obstruction can have symbolic degree larger than two. Theorem 1.2 realizes the case in which both forms of isolation fail: the mismatching generator supports are $\{a,b,z\}$ and $\{c,d,z\}$, while the two supports $\{a,c\}$ and $\{b,d\}$ lie in their union; dually the component pair indexed by (a,b,z) and (c,d,z) has the two intervening supports (a,d) and (b,c). Theorem 1.3 further shows that this obstruction survives under an apex cone whose first Betti number vanishes. The two families just discussed lie outside the existing leaf/forest and exact-support positive classes; they therefore do not weaken those sufficient criteria. What the coned family rules out is replacing such structural criteria by support equality together with the homological condition $\beta_1(\Delta(L)) \leq 1$.

We show that excluding a strong 3-cycle from the common generator-support clutter restores the conjecture, and we delineate exactly how the resulting sufficient condition, its component-incidence companion, and the proposed boundary repairs succeed or fail.

Theorem 1.1 (no strong triangle)**.** *Let $I \subsetneq S$ be a nonzero monomial ideal, and let*

$$I = \bigcap_{j=1}^{m} Q_j$$

be its irredundant irreducible monomial decomposition. Assume that:

(a) $\mathcal{F}(I) = \mathcal{F}(\sqrt{I})$, and $\Delta(I)$ has the members of this common clutter as its facets;

(b) the radicals $\sqrt{Q_1},\dots,\sqrt{Q_m}$ are pairwise distinct;

(c) $\operatorname{Ass}(S/I) = \operatorname{Min}(I)$;

(d) I is Simis;

(e) the clutter $\mathcal{H} = \mathcal{F}(I)$ has no strong 3-cycle.

For each $F \in \mathcal{H}$, let u_F denote the minimal monomial generator of I whose support is F. Then, for every variable x_i occurring in at least one member of $\mathcal{H}$, the exponent of x_i in u_F is independent of the facet $F \in \mathcal{H}$ containing x_i; denote its common positive value by d_i. For a variable x_i occurring in no member of $\mathcal{H}$, put $d_i = 1$, and let $d = (d_1, \ldots, d_n)$. Moreover,

$$J := \sqrt{I} = \left(\left\{ \prod_{x_i \in F} x_i \;\middle|\; F \in \mathcal{H} \right\} \right)$$

is a squarefree Simis monomial ideal in S, and

$$I = J^{\delta_d}.$$

Consequently the MPV conclusion holds for I. In particular, since every balanced clutter has no strong odd cycle and hence no strong 3-cycle, the same conclusion holds when hypothesis (e) is replaced by the stronger hypothesis that $\mathcal{H}$ is balanced.

Theorem 1.1 is Theorem 6.1, proved in Section 6. Its first four hypotheses are the three original MPV hypotheses together with support preservation $\mathcal{F}(I) = \mathcal{F}(\sqrt{I})$; the fifth hypothesis excludes strong triangles from the generator-support clutter. The theorem does not prove the original conjecture, which is false by Theorem 1.2, and it proves no reverse implication. The no-strong-triangle condition is a sufficient, not a necessary, condition: Theorem 1.6 gives an ideal satisfying all the other hypotheses and the full conclusion while its generator-support clutter contains a strong triangle. Nor does the theorem contain every exact support-3 case; a support-3 clutter may itself contain a strong triangle.

Theorem 1.2 (five-variable counterexample boundary)**.** *Let k be a field, let $S = k[a, b, c, d, z]$, and fix $r, s \in \mathbb{Z}_{>0}$. In the variable order (a, b, c, d, z), put*

$$I_{r,s} = (ac, bd, abz^r, cdz^s),$$

$$Q_1 = (a, b, z^s), \quad Q_2 = (a, d), \quad Q_3 = (b, c), \quad Q_4 = (c, d, z^r),$$
$$P_1 = (a, b, z), \quad P_2 = (a, d), \quad P_3 = (b, c), \quad P_4 = (c, d, z),$$

and

$$M = (ac, bd, abz, cdz).$$

Then the following hold:

(i) The displayed decomposition

$$I_{r,s} = Q_1 \cap Q_2 \cap Q_3 \cap Q_4$$

is irredundant and irreducible. Its component radicals are $P_1, \ldots, P_4$, which are pairwise distinct and pairwise incomparable, with respective heights $(3, 2, 2, 3)$. Moreover

$$\operatorname{Ass}(S/I_{r,s}) = \operatorname{Min}(I_{r,s}) = \{P_1, P_2, P_3, P_4\},$$

and $\sqrt{I_{r,s}} = M$.

(ii) For every $\ell \geq 1$,

$${I_{r,s}}^{(\ell)} = {I_{r,s}}^{\ell} = \bigcap_{j=1}^{4} {Q_j}^{\ell}.$$

Thus every $I_{r,s}$ satisfies the three original MPV hypotheses, and

$$\mathcal{F}(I_{r,s}) = \mathcal{F}(\sqrt{I_{r,s}}) = \{\{a, c\}, \{b, d\}, \{a, b, z\}, \{c, d, z\}\}.$$

(iii) There are a squarefree monomial ideal $J \subseteq S$ and $w \in {\mathbb{Z}_{>0}}^5$ such that $I_{r,s} = J^{\delta_w}$ if and only if $r = s$. If $r = s = t$, the data are forced:

$$J = M = I_{1,1}, \qquad w = (1, 1, 1, 1, t),$$

and J is Simis. If $r \neq s$, no squarefree monomial ideal $J \subseteq S$ and positive diagonal vector w represent $I_{r,s}$, even when the requirement that J be Simis is omitted.

(iv) Suppose $r \neq s$. With

$$\mathcal{F}_{r,s} = \{\{a, c\}, \{b, d\}, \{a, b, z\}, \{c, d, z\}\}, \qquad \mathcal{A}_{r,s} = \{\{a, b, z\}, \{a, d\}, \{b, c\}, \{c, d, z\}\},$$

the sequences

$$a, \{a, c\}, c, \{c, d, z\}, z, \{a, b, z\}, a$$

and

$$a, \{a, d\}, d, \{c, d, z\}, z, \{a, b, z\}, a$$

are strong 3-cycles in $\mathcal{F}_{r,s}$ and $\mathcal{A}_{r,s}$, respectively. Consequently neither clutter is balanced, and

$$b(\mathcal{F}_{r,s}) = \mathcal{A}_{r,s}, \qquad b(\mathcal{A}_{r,s}) = \mathcal{F}_{r,s}.$$

Theorem 1.2 combines the corresponding clauses of Construction 4.1, Theorem 4.3, and Theorem 4.5, all proved in Section 4. Its off-diagonal members are counterexamples to the literal MPV implication even after $\mathcal{F}(L) = \mathcal{F}(\sqrt{L})$ is added to the original three hypotheses. The strong cycles occur in both the generator-support and component-support clutters, so the same family also explains why the terminal theorem excludes a support-theoretic local configuration rather than merely a homological one.

Theorem 1.3 (apex-cone boundary)**.** *Retain the notation of Theorem 1.2, let $R = S[t] = k[a, b, c, d, z, t]$, regard Q_j and P_j as ideals of R, and put*

$$K_{r,s} = tI_{r,s}R = (tac, tbd, tabz^r, tcdz^s), \qquad Q_0 = (t), \qquad P_0 = (t).$$

Then the following hold for all $r, s \in \mathbb{Z}_{>0}$:

(i) The decomposition

$$K_{r,s} = Q_0 \cap Q_1 \cap Q_2 \cap Q_3 \cap Q_4$$

is the unique irredundant irreducible monomial decomposition up to reordering. Its component radicals $P_0, P_1, \ldots, P_4$ are pairwise distinct and pairwise incomparable, with respective heights $(1, 3, 2, 2, 3)$, and

$$\operatorname{Ass}_R(R/K_{r,s}) = \operatorname{Min}(K_{r,s}) = \{P_0, P_1, P_2, P_3, P_4\}.$$

Thus $K_{r,s}$ has no embedded primes and is MPV-minimal.

(ii) For every $\ell \geq 1$,

$${K_{r,s}}^{(\ell)} = (t^\ell) \cap \bigcap_{j=1}^{4} {Q_j}^\ell = {K_{r,s}}^\ell.$$

Moreover

$$\sqrt{K_{r,s}} = (tac, tbd, tabz, tcdz),$$

and

$$\mathcal{F}(K_{r,s}) = \mathcal{F}(\sqrt{K_{r,s}}) = \{\{a, c, t\}, \{b, d, t\}, \{a, b, z, t\}, \{c, d, z, t\}\}.$$

(iii) $\Delta(K_{r,s})$ *is the cone with apex* t *over* $\Delta(I_{r,s})$*. It is contractible, and its first simplicial Betti number satisfies* $\beta_1(\Delta(K_{r,s})) = 0$ *over every coefficient field.*

(iv) There are a squarefree monomial ideal $J \subseteq R$ *and* $d \in {\mathbb{Z}_{>0}}^6$ *such that* $K_{r,s} = J^{\delta_d}$ *if and only if* $r = s$*. If* $r = s = \rho$*, the unique squarefree source is*

$$J = (tac, tbd, tabz, tcdz) = tM,$$

and the weights on (a, b, c, d, z, t) *are forced to be* $d = (1, 1, 1, 1, \rho, 1)$*.*

(v) Let $\mathcal{H}$ *be a clutter on a finite vertex set* W*, let* $t \notin W$*, and put* $\mathcal{H}^+ = \{E \cup \{t\} \mid E \in \mathcal{H}\}$*. Adding* t *to every edge and deleting* t *are inverse length-preserving bijections between the strong cycles of* $\mathcal{H}$ *and those of* $\mathcal{H}^+$*. In particular* $\mathcal{F}(K_{r,s}) = {\mathcal{F}_{r,s}}^+$ *preserves the generator-support strong* 3*-cycle above. The component-support clutter is*

$$\mathcal{B} = \{\{t\}, \{a, b, z\}, \{a, d\}, \{b, c\}, \{c, d, z\}\},$$

which contains the other displayed strong 3*-cycle.*

In particular, if $r \neq s$*, then* $K_{r,s}$ *satisfies the three original MPV hypotheses,* $\mathcal{F}(K_{r,s}) = \mathcal{F}(\sqrt{K_{r,s}})$*,* $\beta_1(\Delta(K_{r,s})) = 0$*, and yet has no squarefree standard positive diagonal root in* R*. Hence adding support preservation and* $\beta_1(\Delta(L)) \leq 1$ *to the original three hypotheses does not yield the MPV conclusion. Because coning preserves strong cycles, this boundary does not refute a balanced-support or no-strong-triangle strengthening imposed on the generator clutter, the component clutter, or both.*

Theorem 1.3 records the corresponding clauses of Theorem 5.3, proved in Section 5, together with the two explicit strong cycles carried over from Section 4. It is boundary evidence and is not used as a premise in the proof of Theorem 1.1.

We next show that the terminal theorem has genuine content beyond the forest and exact support-3 classes. The *balanced-support strengthening* consists of the three original MPV hypotheses, support preservation $\mathcal{F}(L) = \mathcal{F}(\sqrt{L})$, and balancedness of $\mathcal{F}(L)$. The *one-facet-deletion strengthening* consists of the three original hypotheses, the same support equality, and the requirement that $\langle \mathcal{F}(L) \setminus \{F\} \rangle$ be a simplicial forest for every $F \in \mathcal{F}(L)$ with $\mathcal{F}(L) \setminus \{F\} \neq \varnothing$. Given either strengthening, we call L a *strict quantified example* if it satisfies the complete antecedent and the MPV conclusion, while for each of the forest and exact support-3 results cited from [BDK26a] it fails at least one antecedent which that result adds to the three original MPV hypotheses.

Theorem 1.4 (strict balanced support-four family)**.** *Let* $m \geq 2$*, put* $n = 2m$*, read all subscripts below modulo* n*, and put*

$$V_m = \{x_1, \ldots, x_n, y_1, \ldots, y_n, z_1, \ldots, z_n\}, \qquad S_m = k[V_m], \qquad E_i = \{x_i, x_{i+1}, y_i, z_i\},$$

for $1 \leq i \leq n$*. Let*

$$\mathcal{H}_m = \{E_1, \ldots, E_n\}, \qquad \Delta_m = \langle E_1, \ldots, E_n \rangle, \qquad u_F = \prod_{v \in F} v$$

for $F \subseteq V_m$*. Fix the displayed variable order on* V_m *and an arbitrary* $d = (d_v)_{v \in V_m} \in {\mathbb{Z}_{>0}}^{6m}$*, and define*

$$J_m = (u_{E_1}, \ldots, u_{E_n}), \qquad I_{m,d} = ({u_{E_1}}^{\langle d \rangle}, \ldots, {u_{E_n}}^{\langle d \rangle}), \qquad {u_F}^{\langle d \rangle} = \prod_{v \in F} v^{d_v}.$$

Then:

(i) *The displayed generators of $I_{m,d}$ are its $2m$ minimal monomial generators and have support cardinality 4. Moreover*

$$\sqrt{I_{m,d}} = J_m = I_{m,\mathbf{1}}, \qquad \mathcal{F}(I_{m,d}) = \mathcal{F}(\sqrt{I_{m,d}}) = \mathcal{H}_m, \qquad \Delta(I_{m,d}) = \Delta_m.$$

(ii) *Both J_m and $I_{m,d}$ have no embedded primes, are MPV-minimal, Simis, and normally torsion-free, and for every $q \geq 1$*

$$I_{m,d}{}^{(q)} = I_{m,d}{}^{q}, \qquad \operatorname{Ass}_{S_m}(S_m/I_{m,d}{}^{q}) = \operatorname{Ass}_{S_m}(S_m/I_{m,d}).$$

Furthermore $I_{m,d} = J_m{}^{\delta_d}$, so every $I_{m,d}$ satisfies the full MPV conclusion in its own ring, with the same-ring squarefree Simis source J_m.

(iii) *The clutter $\mathcal{H}_m$ is balanced and 4-uniform. The complex Δ_m is connected and leaf-free, hence not a simplicial forest, while every nonempty proper facet subcomplex is a forest; equivalently, Δ_m is minimally nonforest.*

Consequently, for every $m \geq 2$ and every $d \neq \mathbf{1}$, $I_{m,d}$ is a nonsquarefree support-4 strict quantified example for both the balanced-support and one-facet-deletion strengthenings: it satisfies both complete antecedents and the MPV conclusion, but fails the whole-complex forest antecedent, fails the exact support-3 antecedent of the second BDK26 positive case, and also lies outside the broader cumulative support-at-most-3 class. For each $m \geq 2$, let $d_{y_1}^{(m)} = 2$ and $d_v^{(m)} = 1$ for every $v \neq y_1$. As m varies, the ambient pairs $(S_m, I_{m,d^{(m)}})$ are pairwise nonisomorphic because S_m has $6m$ variables. No pairwise-nonisomorphism assertion is made here for fixed m and varying weight vectors.

Theorem 1.4 is the corresponding clause set of Definition 7.6, Construction 7.7, and Theorem 7.8, proved in Section 7. Its squarefree Simis and normal-torsion-free assertions are inherited from Fulkerson–Hoffman–Oppenheim through Herzog–Hibi–Trung–Zheng, and their weighted counterparts use the MPV Simis transfer [FHO74, HHTZ08, MPV25]; they are not a new independent squarefree Simis construction. The family is leafless, so the forest propagation has no starting leaf, and its support cardinality is 4, so the exact support-3 intersection catalogue does not apply. Balancedness gives the stronger absence of all strong odd cycles, hence in particular the absence of strong triangles required by Theorem 1.1; applying that theorem is an a priori verification of exponent rigidity, although the displayed construction already exhibits the resulting weighted representation.

Theorem 1.5 (one-intervenor root criterion and sharp bound)**.** *Let $I \subsetneq S$ be a nonzero monomial ideal without embedded primes, and let*

$$I = \bigcap_{j=1}^{m} q_j, \qquad q_j = (x_r^{a_{j,r}} : r \in A_j),$$

be an irredundant irreducible decomposition such that every A_j is a nonempty subset of $\{1, \ldots, n\}$, every $a_{j,r}$ is positive, and $A_1, \ldots, A_m$ are pairwise distinct. For $k < \ell$, put

$$N(k, \ell) = \{j \notin \{k, \ell\} \mid A_j \subseteq A_k \cup A_\ell\}.$$

Then the following hold:

(i) Assume that $|N(k,\ell)| \le 1$ *whenever* $A_k \cap A_\ell \neq \varnothing$. *If* $I^{(2)} = I^2$, *then every occurrence set*

$$C_i = \{a_{j,i} \,|\, 1 \le j \le m,\ i \in A_j\}$$

has at most one element. For i *in* $V = \bigcup_{j=1}^m A_j$, *let* c_i *be its unique element; for* $i \notin V$, *let* c_i *be an arbitrary positive integer. With*

$$P_j = (x_r : r \in A_j), \qquad J = \bigcap_{j=1}^m P_j, \qquad c = (c_1, \dots, c_n),$$

the displayed decomposition of J *is irredundant and* $I = J^{\delta_c}$. *The ideal* J *is the unique squarefree monomial ideal admitting a positive diagonal representation of* I, *and another* $d \in {\mathbb{Z}_{>0}}^n$ *satisfies* $I = J^{\delta_d}$ *exactly when* $d_i = c_i$ *for every* $i \in V$; *the coordinates outside* V *are arbitrary. If in addition* I *is Simis, then* J *is Simis and the full MPV conclusion holds for* I. *The intervening-support inequality holds automatically when every* $|A_j| = 2$.

(ii) The bound 1 *is sharp. If the inequality in (i) is relaxed to* $|N(k,\ell)| \le 2$ *for all intersecting pairs, then for all distinct* $r, s \in \mathbb{Z}_{>0}$ *the ideal*

$$I_{r,s} = (ac, bd, abz^r, cdz^s) \subseteq k[a,b,c,d,z]$$

satisfies all the resulting hypotheses and ${I_{r,s}}^{(2)} = {I_{r,s}}^2$, *while its exponent-mismatching component pair* (Q_1, Q_4) *has exactly the two intervening supports* $\{a,d\}$ *and* $\{b,c\}$, *and* $I_{r,s}$ *has no squarefree positive diagonal source in its own ring.*

Theorem 1.5 is Theorem 8.3 together with Corollary 8.4, proved in Section 8. Unlike the terminal support theorem, this criterion does not assume $\mathcal{F}(I) = \mathcal{F}(\sqrt{I})$; it assumes the second symbolic identity rather than all-power Simisness in order to construct the root, and adds all-power Simisness only to transfer Simisness back to the forced source.

Theorem 1.6 (nonnecessity of the no-strong-triangle condition)**.** *Let* k *be a field and put*

$$S = k[u_1, v_1, w_1, u_2, v_2, u_3, v_3, u_4, v_4].$$

Let

$$B_1 = \{u_1, v_1, w_1\}, \quad B_2 = \{u_2, v_2\},$$
$$B_3 = \{u_3, v_3\}, \quad B_4 = \{u_4, v_4\},$$
$$V = B_1 \sqcup B_2 \sqcup B_3 \sqcup B_4,$$

take $a_z = 2$ *for every* $z \in V$, *and define*

$$I_a = \bigcap_{h=1}^4 (z^2 : z \in B_h), \qquad J = \bigcap_{h=1}^4 (z : z \in B_h), \qquad a = (a_z)_{z \in V}.$$

Then I_a *has* 24 *nonsquarefree minimal monomial generators, all of support cardinality* 4, *and its component heights are* $(3,2,2,2)$. *Both* I_a *and* J *have no embedded primes, are MPV-minimal, Simis, and normally torsion-free, and*

$$\sqrt{I_a} = J, \qquad \mathcal{F}(I_a) = \mathcal{F}(J), \qquad I_a = J^{\delta_a}.$$

The ideal J is the unique squarefree ideal representing I_a by a positive diagonal weighting, and the weights are forced to equal a, so the full MPV conclusion holds. The common facet complex $\Delta(I_a) = \Delta(J)$ is leaf-free and contains the strong triangle

$$u_1, F_1, u_2, F_2, u_3, F_3, u_1,$$
$$F_1 = \{u_1, u_2, v_3, u_4\}, \quad F_2 = \{v_1, u_2, u_3, u_4\},$$
$$F_3 = \{u_1, v_2, u_3, u_4\}.$$

Thus I_a satisfies every other hypothesis and the full conclusion of Theorem 1.1, while $\mathcal{F}(I_a)$ contains a strong triangle. It lies outside the support-at-most-3, forest, balanced/no-strong-triangle, and pure component-height-two criteria. Hence absence of a strong triangle is sufficient but not necessary for the conclusion, and the boundary package does not exclude every possible weakening of the hypotheses.

Theorem 1.6 is Corollary 8.7, with the supporting Construction 8.5 and Theorem 8.6, all proved in Section 8. Its four component supports are pairwise disjoint, so there are no exponent-mismatching component pairs: $X_{I_a} = \varnothing$ and every $N(k, \ell)$ is already empty. Consequently this example is a hypothesis-level separator but not a separator of represented Simis MPV successes from the BK25 criterion restated in Theorem 8.8; the additional hypothesis of that theorem is vacuous on the example.

Several mechanisms and inputs carry the preceding results. (*i*) The central obstacle is isolated at symbolic degree two. Under support preservation, MPV-minimality, and Simisness, if a support F carries at least two minimal generators, choose two such generators x^β, x^γ whose total degrees are the two least total degrees on F, with ties broken arbitrarily. Then $\prod_{x_i \in F} x_i^{\max\{2\min(\beta_i,\gamma_i),\max(\beta_i,\gamma_i)\}} \in I^{(2)} \setminus I^2$, so every support carries exactly one minimal generator. If F, G are the only supports contained in $F \cup G$ and their unique generators x^a, x^b have unequal exponents at some $x_r \in F \cap G$, then, with $m_i = \min\{a_i, b_i\}$ and $M_i = \max\{a_i, b_i\}$ on $F \cap G$,

$$f = \Big(\prod_{x_i \in F \cap G} x_i^{\max\{2m_i, M_i\}}\Big)\Big(\prod_{x_i \in F \setminus G} x_i^{a_i}\Big)\Big(\prod_{x_i \in G \setminus F} x_i^{b_i}\Big)$$

lies in $\bigcap_j Q_j^2 = I^{(2)}$, because each irreducible component meeting F and G has two coordinate witnesses, possibly the same coordinate repeated, yet support isolation leaves no product x^{2a}, x^{a+b}, or x^{2b} dividing f. This contradicts $I^{(2)} = I^2$ and forces equality of the incident exponents. (*ii*) The combinatorial engine is the no-strong-triangle isolation lemma: if a labeling of the edges through a vertex is nonconstant, some unequal-label pair F, G through that vertex has no third support contained in $F \cup G$; otherwise a minimal counterexample pair produces a strong triangle. This replaces the forest leaf ordering and the exact support-3 intersection catalogue by a local clutter condition, which is precisely what lets the balanced minimally nonforest family pass the barrier despite being leaf-free and support four. (*iii*) The generator and component dictionaries make root existence a priori and exact. For a monomial ideal L, let $D_i(L)$ record the positive exponents of x_i among its minimal generators; then a squarefree positive diagonal root exists exactly when $|D_i(L)| \le 1$ for every i. For an irredundant decomposition $L = \bigcap_j (x_r^{a_{j,r}} : r \in A_j)$, let $C_i = \{a_{j,i} \mid i \in A_j\}$; then a root exists exactly when $|C_i| = 1$ for every used variable, equivalently when there is no pair of intersecting component supports carrying unequal exponents for a shared variable. In both criteria the source is unique and the weights on used variables are forced, while unused weights are arbitrary. These criteria do not require the three MPV hypotheses; without

Simisness they do not by themselves give the full MPV conclusion. When I is Simis, the diagonal Simis transfer of [MPV25, Corollary 5.5(b)] makes $\sqrt{I}$ Simis, and the full MPV conclusion follows. (*iv*) The strict family combines an even clutter whose only cycles are the full even cycle and its reverse with the balanced-to-Mengerian theorem of [FHO74], the Mengerian–Simis equivalence of [HHTZ08, Corollary 1.6], and the diagonal Simis transfer of [MPV25, Corollary 5.5(b)]; its minimally nonforest leaf-free geometry is exactly outside the forest and support-three methods but inside the no-strong-triangle theorem. (*v*) The boundary package proves that each apparent shortcut is sharp at its own quantifier: off-diagonal five-variable members refute the original implication even before source Simisness is requested; their apex cones add support equality and vanishing first Betti number without producing a root; the component one-intervenor theorem is sharp when its bound is relaxed from one to two; and the transversal $(3, 2, 2, 2)$ example shows that no-strong-triangle is not necessary without separating BK25's represented-success class. These boundaries constrain the positive theorem but none is a premise of its proof.

Section 2 records the standing notation and conventions. Section 3 proves the irreducible, generator, and component dictionaries and records the imported MPV primary and transfer statements. Section 4 constructs the five-variable family, proves its power identities and exact root boundary, and displays the two strong triangles. Section 5 records the generalized Krull height bound, enters the new-variable cone, and closes the support-equality plus first-Betti-number repair while preserving the strong-cycle obstruction. Section 6 proves Theorem 6.1. Section 7 records the forest, exact support-three, and balanced/Mengerian interfaces and proves the strict support-four package. Finally, Section 8 proves the one-intervenor criterion and its sharpness, the disjoint-block nonnecessity boundary, and the BK25 comparison with its posterior-vacuity consequence.

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2 Notation and terminologies

Throughout, k is a field, $n \geq 1$, and $S = k[x_1, \dots, x_n]$ is a polynomial ring in algebraically independent variables. All ideals under discussion are monomial ideals, and all nonzero ideals are proper unless the contrary is explicit. Put $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{Z}_{>0} = \{1, 2, \dots\}$. For $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$, write $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ and

$$\operatorname{supp}(x^\alpha) = \{x_i \,|\, \alpha_i > 0\}\,.$$

For a monomial ideal L, $\mathrm{G}(L)$ denotes its minimal monomial generating set,

$$\mathcal{F}(L) = \{\operatorname{supp}(u) \,|\, u \in \mathrm{G}(L)\}\,, \qquad \Delta(L) = \langle F \,|\, F \in \mathcal{F}(L)\rangle\,.$$

Thus $\Delta(L)$ is the simplicial complex generated by the members of $\mathcal{F}(L)$. When the members of $\mathcal{F}(L)$ form an antichain, they are precisely the facets of $\Delta(L)$; in that case $\mathcal{F}(L)$ is a clutter. For a clutter $\mathcal{H}$ on a finite vertex set V, its *blocker* is

$$b(\mathcal{H}) = \{C \subseteq V \,|\, C \cap E \neq \varnothing \text{ for every } E \in \mathcal{H}, \text{ and } C \text{ is inclusion-minimal with this property}\}.$$

A nonzero proper ideal $Q \subseteq S$ is *irreducible* if it is not the intersection of two strictly larger ideals. For a nonempty $A \subseteq \{1, \ldots, n\}$ and positive integers a_i indexed by $i \in A$, put

$$Q(A, a) = (x_i^{a_i} : i \in A), \qquad P(A) = (x_i : i \in A).$$

Every irreducible monomial ideal is uniquely of the form $Q(A, a)$; it is $P(A)$-primary and has radical $P(A)$. Every nonzero proper monomial ideal L admits an irredundant finite decomposition $L = \bigcap_{j=1}^{m} Q_j$ into irreducible monomial ideals. Its *irredundant irreducible monomial decomposition* is unique up to reordering, and its displayed decomposition is *MPV-minimal* if $\sqrt{Q_j} \neq \sqrt{Q_\ell}$ whenever $j \neq \ell$.

For a monomial ideal L, $\operatorname{Ass}(S/L)$ and $\operatorname{Min}(L)$ denote its associated and minimal primes. Symbolic powers are always taken with respect to the minimal primes:

$$L^{(s)} = \bigcap_{P \in \operatorname{Min}(L)} (L^s S_P \cap S) \qquad (s \geq 1).$$

The ideal L is *Simis* if $L^{(s)} = L^s$ for every $s \geq 1$, and is *normally torsion-free* if

$$\operatorname{Ass}(S/L^s) \subseteq \operatorname{Ass}(S/L) \qquad \text{for every } s \geq 1.$$

When $\operatorname{Min}(L) = \operatorname{Ass}(S/L)$, we say that L has no embedded primes. The phrase *three original MPV hypotheses* means that L has no embedded primes, that a displayed irredundant irreducible decomposition of L is MPV-minimal, and that L is Simis. The *MPV conclusion* holds for L if there are a squarefree monomial ideal $J \subseteq S$, necessarily in the same polynomial ring, and $e \in {\mathbb{Z}_{>0}}^n$ such that

$$L = J^{\delta_e}$$

and J is Simis.

For $d = (d_1, \ldots, d_n) \in {\mathbb{Z}_{>0}}^n$, the *standard positive diagonal weighting* determined by d is the exponent map

$$\delta_d(\alpha) = (d_1\alpha_1, \ldots, d_n\alpha_n),$$

and, for a monomial ideal $L \subseteq S$,

$$L^{\delta_d} = \left(\left\{ x^{\delta_d(\alpha)} \,\middle|\, \alpha \in \mathbb{N}^n,\ x^\alpha \in L \right\}\right).$$

A monomial ideal is *squarefree* if it can be generated by squarefree monomials. A squarefree monomial ideal $J \subseteq S$ is a *squarefree diagonal root* of L if $L = J^{\delta_d}$ for some $d \in {\mathbb{Z}_{>0}}^n$; neither variable extensions nor nonstandard linear weightings are allowed. For a fixed i, the exponents of x_i occurring among the components of a displayed irreducible decomposition, or among the minimal generators, are called its *occurrence exponents*; the set they form is its *occurrence set*.

Let $\mathcal{H}$ be a clutter on a finite vertex set. For $q \geq 3$, a *strong q-cycle* is an alternating sequence

$$(v_1, E_1, v_2, E_2, \ldots, v_q, E_q, v_1)$$

of q distinct vertices and q distinct edges $E_i \in \mathcal{H}$ such that, with indices read modulo q,

$$v_i, v_{i+1} \in E_i, \qquad E_i \cap \{v_1, \dots, v_q\} = \{v_i, v_{i+1}\} \qquad (1 \le i \le q).$$

The clutter $\mathcal{H}$ is *balanced* if it has no strong cycle of odd length.

Following the facet-subcomplex convention, a subcomplex Λ of a finite simplicial complex Γ is determined by an arbitrary subset $\mathcal{G} \subseteq \mathcal{F}(\Gamma)$ and is the complex generated by $\mathcal{G}$; notation is $\mathcal{F}(\Lambda) = \mathcal{G}$. If Λ has more than one facet, a facet F is a *leaf* if there is another facet G such that

$$F \cap H \subseteq F \cap G \qquad \text{for every facet } H \neq F \text{ of } \Lambda.$$

A complex with one facet is itself a leaf. A nonempty finite complex is a *simplicial forest* if every nonempty facet subcomplex has a leaf; a connected forest is a *simplicial tree*. These are Faridi's notions [Far02]. A complex Γ is *minimally nonforest* if Γ is not a forest but every nonempty proper facet subcomplex of Γ is a forest.

For a finite simplicial complex Δ with vertex set V and facets $\mathcal{F}(\Delta)$, let M be its facet-by-vertex incidence matrix, with rows indexed by $\mathcal{F}(\Delta)$ and columns by V. The complex Δ is *Mengerian* if, for every $c \in \mathbb{N}^V$,

$$\min\left\{c \cdot a \,\middle|\, a \in \mathbb{N}^V,\ Ma \ge \mathbf{1}\right\} = \max\left\{b \cdot \mathbf{1} \,\middle|\, b \in \mathbb{N}^{\mathcal{F}(\Delta)},\ M^T b \le c\right\}.$$

Its *facet ideal* in $k[V]$ is

$$I(\Delta) = \left(\textstyle\prod_{v \in F} v \,\middle|\, F \in \mathcal{F}(\Delta)\right).$$

3 Weighting and component dictionaries

We retain the notation of Section 2. This section has three parts. We first record the irreducible-component bookkeeping for monomial ideals. We then quote, without proof, the three results imported from [MPV25]. Finally, we prove exact diagonal-root criteria, one read from the minimal generators and one from an irredundant irreducible decomposition. The two criteria are purely monomial criteria: neither their hypotheses nor their proofs use the three original MPV hypotheses.

We begin with the elementary monomial-generator facts used throughout the section.

Lemma 3.1 (minimal monomial generators after a diagonal weighting)**.** *Let $L \subseteq S$ be a monomial ideal and let $r \in {\mathbb{Z}_{>0}}^n$. Then:*

(i) every monomial in L is divisible by a member of $\mathrm{G}(L)$;

(ii) no two distinct members of $\mathrm{G}(L)$ divide one another;

(iii)

$$\mathrm{G}(L^{\delta_r}) = \left\{x^{\delta_r(\alpha)} \,\middle|\, x^\alpha \in \mathrm{G}(L)\right\}, \tag{3.1}$$

and $\alpha \mapsto \delta_r(\alpha)$ is a bijection from the exponent vectors of the members of $\mathrm{G}(L)$ to those of the members of $\mathrm{G}(L^{\delta_r})$;

(iv) if L is generated by squarefree monomials, then every member of $\mathrm{G}(L)$ is squarefree.

Proof. If $x^\beta \in L$, write x^β as a finite S-linear combination of the members of $\mathrm{G}(L)$. One product in that sum must contribute a nonzero coefficient to the monomial x^β; if that product involves x^α, then $\alpha \le \beta$ coordinatewise, so $x^\alpha \mid x^\beta$. This proves (i). If distinct $x^\alpha, x^\beta \in \mathrm{G}(L)$ satisfy $x^\alpha \mid x^\beta$, then x^β can be omitted from the minimal monomial generating set. This proves (ii).

Let H denote the set on the right-hand side of (3.1). The inclusion $(H) \subseteq L^{\delta_r}$ follows from the definition of the weighted ideal. Conversely, take a monomial $x^\beta \in L$. By (i), some $x^\alpha \in \mathrm{G}(L)$ divides x^β. Since every r_i is positive,

$$x^{\delta_r(\alpha)} \mid x^{\delta_r(\beta)}.$$

Thus $L^{\delta_r} = (H)$.

For any exponent vectors α and β, again because every r_i is positive,

$$x^{\delta_r(\alpha)} \mid x^{\delta_r(\beta)} \quad \Longleftrightarrow \quad x^\alpha \mid x^\beta.$$

Together with (ii), this shows that H is an antichain and that distinct members of $\mathrm{G}(L)$ have distinct weighted exponent vectors. Hence H is the minimal monomial generating set displayed in (3.1), and the map in (iii) is a bijection.

Finally, suppose that L is generated by a set $\mathcal{B}$ of squarefree monomials and let $x^\alpha \in \mathrm{G}(L)$. Since $x^\alpha \in L = (\mathcal{B})$, some $x^\sigma \in \mathcal{B}$ divides x^α. Since $x^\sigma \in L$, part (i) gives some $x^\tau \in \mathrm{G}(L)$ dividing x^σ. Then x^τ is squarefree and divides x^α. By (ii), $\tau = \alpha$, so x^α is squarefree. This proves (iv). □

Proposition 3.2 (irreducible-component dictionary)**.** *Let $R = k[x_1, \dots, x_n]$. For every nonempty set of variables E and every family $e = (e_x)_{x \in E}$ of positive integers, write*

$$Q(E, e) = (x^{e_x} : x \in E), \qquad P(E) = (x : x \in E).$$

Let $L \subseteq R$ be a nonzero proper ideal. Then every irreducible component in an irredundant irreducible decomposition of L is uniquely of the form $Q(E, e)$; conversely, every such $Q(E, e)$ is irreducible, $P(E)$-primary, has radical $P(E)$, and satisfies

$$\mathrm{Ass}_R(R/Q(E, e)) = \{P(E)\}.$$

If $L = \bigcap_{\ell=1}^{t} Q(E_\ell, e_\ell)$ is an irredundant irreducible decomposition and $P_\ell = P(E_\ell)$, then

$$\mathrm{Ass}_R(R/L) = \{P_\ell \mid 1 \le \ell \le t\}.$$

In particular, the set $\{P_1, \dots, P_t\}$ of component radicals is independent of the chosen irredundant irreducible decomposition of L.

Proof. The ring R in the statement is the ring S fixed in Section 2. We first analyze the pure-power ideals $Q(E, e)$; we then classify irreducible monomial ideals; finally, we compute the associated primes of an irredundant decomposition.

Pure-power components are primary and irreducible. Fix E and $e = (e_x)_{x \in E}$, and write $Q = Q(E, e)$ and $P = P(E)$. Put

$$Y = \{x_1, \dots, x_n\} \setminus E, \qquad B = k[Y], \qquad \tau = \prod_{x \in E} x^{e_x - 1}.$$

As a B-module, R/Q is free with basis the monomials $\prod_{x\in E} x^{\rho_x}$, where $0 \le \rho_x < e_x$ for every $x \in E$. Write a nonzero class $\overline{F} \in R/Q$ uniquely as

$$\overline{F} = \sum_{\rho} f_\rho(Y) \prod_{x\in E} x^{\rho_x}.$$

Every polynomial has one such normal form because division by the pure powers x^{e_x} reduces exactly the E-exponents, and uniqueness follows by expanding in the polynomial ring $B[x : x \in E]$. Moreover $\sqrt{Q} = P$: the inclusion $\sqrt{Q} \subseteq P$ follows from $Q \subseteq P$, while $x^{e_x} \in Q$ for every $x \in E$ gives the reverse inclusion.

Choose a coordinatewise minimal index ρ_0 with $f_{\rho_0} \neq 0$, and put

$$T = \prod_{x\in E} x^{e_x-1-\rho_{0,x}}.$$

If $\rho \neq \rho_0$ and $\rho_x > \rho_{0,x}$ for some $x \in E$, then the corresponding term of $T\overline{F}$ is zero in R/Q. If instead $\rho_x \le \rho_{0,x}$ for every x, then $\rho \neq \rho_0$ would contradict the coordinatewise minimality of ρ_0 among the indices occurring in $\overline{F}$. Consequently

$$T\overline{F} = f_{\rho_0}(Y)\tau \neq 0. \tag{3.2}$$

Suppose now that $\overline{F}\,\overline{G} = 0$ in R/Q with $\overline{F} \neq 0$. Write

$$\overline{G} = \sum_{\eta} g_\eta(Y) \prod_{x\in E} x^{\eta_x}$$

in the same B-basis, taking $g_0 = 0$ if no constant term occurs. Multiplying $\overline{F}\,\overline{G} = 0$ by the monomial T from (3.2) and observing that τ annihilates every nonconstant monomial in the variables from E gives

$$0 = f_{\rho_0}(Y)g_0(Y)\tau.$$

Freeness over B gives $f_{\rho_0}(Y)g_0(Y) = 0$. The ring B is a domain and $f_{\rho_0} \neq 0$, so $g_0 = 0$. Thus every term of $\overline{G}$ is divisible by some variable in E, which means that $G \in P$.

The preceding zero-divisor calculation shows that, if $FG \in Q$ and $F \notin Q$, then $G \in \sqrt{Q}$; some power of G belongs to Q, so Q is P-primary. Moreover,

$$\operatorname{ann}_R(\overline{\tau}) = P.$$

Indeed, multiplication by τ kills every monomial divisible by a variable from E. If $H \notin P$, its part $h_0(Y)$ involving none of the variables from E is nonzero, and $H\tau = h_0(Y)\tau$ is nonzero by freeness over B. Hence $P \in \operatorname{Ass}_R(R/Q)$. Conversely, let $\mathfrak{p} \in \operatorname{Ass}_R(R/Q)$. Choose $0 \neq \overline{F} \in R/Q$ with $\mathfrak{p} = \operatorname{ann}_R(\overline{F})$. Then $Q \subseteq \mathfrak{p}$; since $\mathfrak{p}$ is prime, $P = \sqrt{Q} \subseteq \mathfrak{p}$. Every element of $\mathfrak{p}$ annihilates the nonzero class $\overline{F}$, and hence belongs to P by the zero-divisor calculation above. Thus

$$\operatorname{Ass}_R(R/Q) = \{P\}. \tag{3.3}$$

It remains in this part to prove that Q is irreducible. Let H_1 and H_2 be ideals strictly containing Q. Choose nonzero classes $\overline{F}_s \in H_s/Q$ for $s = 1, 2$. By (3.2), after multiplying $\overline{F}_s$ by a monomial in the variables from E, we obtain a nonzero class of the form $f_s(Y)\tau \in H_s/Q$, with $f_s(Y) \in B \setminus \{0\}$. The elements $f_2(Y)\big(f_1(Y)\tau\big)$ and $f_1(Y)\big(f_2(Y)\tau\big)$ are equal, so

$$0 \neq f_1(Y)f_2(Y)\tau \in H_1 \cap H_2.$$

This element is not in Q, again by freeness over the domain B. Thus no two ideals strictly containing Q intersect in Q, and Q is irreducible.

Pure powers are the only irreducible monomial ideals. Let $W \subseteq R$ be a nonzero proper irreducible monomial ideal. Suppose that one of its minimal monomial generators m is divisible by two distinct variables x and y, and put $u = m/x$ and $v = m/y$. If $u \in W$, then by part (i) of Lemma 3.1 some minimal monomial generator g of W divides u, hence divides m. Part (ii) of that lemma forces $g = m$, but m does not divide u, a contradiction. Thus $u \notin W$, and similarly $v \notin W$.

Consequently $W + (u)$ and $W + (v)$ strictly contain W. If a monomial w belongs to their intersection but not to W, then $u \mid w$ and $v \mid w$. Hence

$$m = \operatorname{lcm}(u, v) \mid w,$$

so $w \in W$, a contradiction. Because all three ideals involved are monomial ideals, this proves

$$W = (W + (u)) \cap (W + (v)),$$

contrary to irreducibility. Every minimal monomial generator of W is therefore a positive power of a single variable. There can be at most one such power for each variable, by the antichain property in part (ii) of Lemma 3.1. Since W is nonzero and proper, it is generated by at least one nontrivial pure power, so

$$W = Q(E, e)$$

for a nonempty set of variables E and positive exponents e_x. The set E is the union of the supports of the minimal monomial generators of W, and e_x is the unique exponent with which x occurs; hence (E, e) is unique. Together with the preceding part, this proves the classification asserted in the proposition.

The component radicals are the associated primes. Let

$$L = \bigcap_{\ell=1}^{t} Q_\ell$$

be an irredundant irreducible decomposition of the nonzero proper monomial ideal L. Each Q_ℓ contains the nonzero ideal L, so $Q_\ell \neq 0$. It is also proper: if $t = 1$, this follows from $L \neq S$, while if $t > 1$ and $Q_\ell = S$, that term could be omitted. The classification just proved supplies unique data (E_ℓ, e_ℓ) with $Q_\ell = Q(E_\ell, e_\ell)$; put $P_\ell = P(E_\ell)$.

Let $\mathfrak{p} \in \operatorname{Ass}_R(R/L)$, say $\mathfrak{p} = \operatorname{ann}_R(\overline{f})$ with $f \notin L$. The natural map

$$R/L \longrightarrow \bigoplus_{\ell=1}^{t} R/Q_\ell$$

is injective. The annihilator in R of the ℓ-th component of $\overline{f}$ is $Q_\ell : f$, with the convention that $Q_\ell : f = R$ when $f \in Q_\ell$. Therefore

$$\mathfrak{p} = \bigcap_{\ell=1}^{t} (Q_\ell : f).$$

If no $Q_\ell : f$ were contained in $\mathfrak{p}$, choose $h_\ell \in (Q_\ell : f) \setminus \mathfrak{p}$ for every ℓ. Then $\prod_\ell h_\ell$ belongs to every $Q_\ell : f$, hence to $\mathfrak{p}$, contradicting primality. Thus $Q_j : f \subseteq \mathfrak{p}$ for some j. The reverse inclusion

always holds, so $\mathfrak{p} = Q_j : f$. In particular $f \notin Q_j$, and $\mathfrak{p}$ is the annihilator in R of the nonzero class $f + Q_j$. By (3.3), $\mathfrak{p} = P_j$. This gives

$$\operatorname{Ass}_R(R/L) \subseteq \{P_1, \dots, P_t\}.$$

For the reverse inclusion, fix j and put

$$K_j = \bigcap_{\ell \neq j} Q_\ell,$$

with $K_j = R$ if $t = 1$. Irredundance gives a monomial $w = x^\omega \in K_j \setminus L$. Since $w \notin Q_j$,

$$\omega_x < e_{j,x} \qquad (x \in E_j).$$

Define

$$v = w \prod_{x \in E_j} x^{e_{j,x} - \omega_x - 1}.$$

Then $v \in K_j$, because $w \in K_j$ and K_j is an ideal. On the other hand, the exponent of every $x \in E_j$ in v is $e_{j,x} - 1$, so $v \notin Q_j$ and therefore $v \notin L$. For $\ell \neq j$, we have $v \in Q_\ell$, hence $Q_\ell : v = R$. It follows that

$$L : v = \bigcap_{\ell=1}^{t} (Q_\ell : v) = Q_j : v.$$

A monomial x^β satisfies $x^\beta v \in Q_j$ exactly when $\beta_x \geq 1$ for some $x \in E_j$. Since multiplication by the monomial v does not identify distinct monomials, the same criterion holds for an arbitrary polynomial. Thus $Q_j : v = P_j$. Because $v \notin L$,

$$P_j = L : v = \operatorname{ann}_R(v + L),$$

so $P_j \in \operatorname{Ass}_R(R/L)$. This proves the displayed equality of associated primes. Its right-hand side is the set of radicals of the components, while its left-hand side depends only on L; the set of component radicals is therefore independent of the chosen irredundant irreducible decomposition. □

The last display is the bookkeeping used throughout the sequel: for any irredundant irreducible decomposition, the associated primes of S/L are exactly the component radicals, with repetitions deleted. A component radical is an embedded prime exactly when it is a nonminimal member of that set of distinct radicals. Repetition by itself has no associated-prime effect: a repeated minimal component radical is still a minimal prime. Repetition does matter to MPV-minimality, which asks the radicals of the displayed components to be pairwise distinct; the set-valued identity discards multiplicity and therefore does not decide that property. No decomposition choice is involved when later arguments identify an associated prime as a component radical, but arguments about component multiplicity or MPV-minimality must retain the component-level data.

The next theorem is [MPV25, Section 1, equations (1.1)–(1.2)], together with the displayed symbolic-power formula immediately following those equations.

Theorem 3.3 (MPV canonical primary and symbolic components)**.** *Let $I \subseteq S$ be a nonzero proper monomial ideal, and let*

$$I = \bigcap_{Q \in \mathcal{Q}(I)} Q$$

be its irredundant irreducible monomial decomposition. For each $P \in \operatorname{Ass}(S/I)$*, put*

$$\mathfrak{q}_P = \bigcap_{\substack{Q \in \mathcal{Q}(I) \\ \sqrt{Q}=P}} Q.$$

Then $\mathfrak{q}_P$ *is* P*-primary,*

$$I = \bigcap_{P \in \operatorname{Ass}(S/I)} \mathfrak{q}_P$$

is the canonical minimal primary decomposition of I*, and*

$$I^{(s)} = \bigcap_{P \in \operatorname{Min}(I)} \mathfrak{q}_P^s \qquad \text{for every } s \geq 1.$$

The following compatibility is [MPV25, Corollary 4.6].

Theorem 3.4 (MPV weight-primary compatibility)**.** *Let* $L \subsetneq S$ *be a nonzero monomial ideal, let* $w \in {\mathbb{Z}_{>0}}^n$*, and let*

$$L = \bigcap_{P \in \operatorname{Ass}(S/L)} \mathfrak{q}_P$$

be the canonical minimal primary decomposition of Theorem 3.3. Then

$$L^{\delta_w} = \bigcap_{P \in \operatorname{Ass}(S/L)} {\mathfrak{q}_P}^{\delta_w}$$

is the canonical minimal primary decomposition of L^{δ_w}*, and* ${\mathfrak{q}_P}^{\delta_w}$ *is* P*-primary for every* $P \in \operatorname{Ass}(S/L)$*. In particular, the weighting preserves every component radical, so*

$$\operatorname{Ass}(S/L^{\delta_w}) = \operatorname{Ass}(S/L).$$

The following equivalence is [MPV25, Corollary 5.5(b)]. It is stated in the source for every monomial ideal and every standard positive linear weighting; no embedded-prime, irredundant-component, or Simis preliminary hypothesis is part of the import.

Theorem 3.5 (MPV Simis transfer)**.** *Let* $L \subseteq S$ *be a monomial ideal and let* $w \in {\mathbb{Z}_{>0}}^n$*. Then* L *is Simis if and only if* L^{δ_w} *is Simis.*

We now turn to the two project-owned root criteria. They prove only when a squarefree standard diagonal root exists and identify the root and the exact freedom in the weights. Their arguments do not use Theorem 3.4 or Theorem 3.5; those cited results become relevant later, when component and Simis information is transported between the two sides of a representation.

Theorem 3.6 (generator root criterion)**.** *Let* $I \subsetneq S$ *be a nonzero monomial ideal, and for* $1 \leq i \leq n$ *put*

$$D_i(I) = \{\nu_i \,|\, x^\nu \in \mathrm{G}(I),\ \nu_i > 0\}.$$

Let

$$U = \{i \,|\, D_i(I) \neq \varnothing\}, \qquad J_* = \Big(\{x^{\operatorname{supp}(\nu)} \mid x^\nu \in \mathrm{G}(I)\}\Big),$$

where $\operatorname{supp}(\nu)_i = 1$ *if* $\nu_i > 0$ *and* $\operatorname{supp}(\nu)_i = 0$ *if* $\nu_i = 0$*. The following are equivalent:*

(i) there are a squarefree monomial ideal $J \subseteq S$ *and* $d \in {\mathbb{Z}_{>0}}^n$ *such that* $I = J^{\delta_d}$*;*

(ii) $|D_i(I)| \leq 1$ *for every* $1 \leq i \leq n$.

Suppose condition (ii) holds, and for $i \in U$ *let* e_i *be the unique element of* $D_i(I)$. *Then, for every* $d \in {\mathbb{Z}_{>0}}^n$,

$$I = {J_*}^{\delta_d} \quad \Longleftrightarrow \quad d_i = e_i \text{ for every } i \in U.$$

Thus the coordinates d_i *with* $i \notin U$ *are arbitrary, and* J_* *is the unique squarefree monomial ideal that can occur in a representation* $I = J^{\delta_d}$.

Proof. The proof has three parts. We first show that a squarefree representation forces one occurrence exponent and the corresponding weight coordinate. We then normalize supports to construct the unique candidate root. Finally, we identify every possible root and determine exactly which weight coordinates are free.

A squarefree root forces one positive exponent and its weight. Assume first that

$$I = J^{\delta_d}$$

with J squarefree and $d \in {\mathbb{Z}_{>0}}^n$. By parts (iii) and (iv) of Lemma 3.1, every $x^\nu \in \mathrm{G}(I)$ has the form

$$x^\nu = x^{\delta_d(\alpha)}$$

for a unique $\{0,1\}$-vector α indexing a member of $\mathrm{G}(J)$. For every coordinate i,

$$\nu_i = d_i\alpha_i \in \{0, d_i\}.$$

Hence $D_i(I) \subseteq \{d_i\}$, so condition (ii) holds. If $D_i(I) = \{e_i\}$, some $x^\nu \in \mathrm{G}(I)$ has $\nu_i > 0$; for the corresponding α we have $\alpha_i = 1$ and $\nu_i = d_i$, so

$$d_i = e_i. \tag{3.4}$$

Support normalization constructs the root. Conversely, assume condition (ii). For $i \in U$, retain the notation e_i for the unique element of $D_i(I)$. Every $x^\nu \in \mathrm{G}(I)$ then satisfies

$$\nu_i = \begin{cases} e_i, & i \in U \text{ and } \nu_i > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{3.5}$$

The ideal J_* is generated by squarefree monomials. We claim that its displayed generating set is $\mathrm{G}(J_*)$. Suppose that $x^\nu, x^\mu \in \mathrm{G}(I)$ and

$$x^{\mathrm{supp}(\nu)} \mid x^{\mathrm{supp}(\mu)}.$$

If $\nu_i > 0$, then $i \in U$ and the displayed divisibility gives $\mu_i > 0$; equation (3.5) gives $\nu_i = \mu_i = e_i$. If $\nu_i = 0$, then $\nu_i \leq \mu_i$. Thus $\nu \leq \mu$ coordinatewise, so $x^\nu \mid x^\mu$. The antichain property in Lemma 3.1(ii) gives $\nu = \mu$. Therefore the displayed squarefree generators of J_* form $\mathrm{G}(J_*)$.

Let $r \in {\mathbb{Z}_{>0}}^n$ satisfy $r_i = e_i$ for every $i \in U$, with arbitrary positive values off U. Equation (3.5) gives

$$\delta_r(\mathrm{supp}(\nu)) = \nu \qquad (x^\nu \in \mathrm{G}(I)).$$

Applying (3.1) to J_* and r therefore yields

$$\mathrm{G}({J_*}^{\delta_r}) = \{x^\nu \,|\, x^\nu \in \mathrm{G}(I)\} = \mathrm{G}(I).$$

Hence $I = {J_*}^{\delta_r}$. In particular, condition (ii) implies condition (i), and every positive extension of the prescribed weights from U to all of $\{1, \dots, n\}$ represents I.

The root and the admissible weights are determined. It remains to identify an arbitrary squarefree representation

$$I = K^{\delta_b}, \qquad b \in {\mathbb{Z}_{>0}}^n.$$

The necessity argument already gives $|D_i(I)| \leq 1$ and, whenever $i \in U$, forces $b_i = e_i$ as in (3.4). By parts (iii) and (iv) of Lemma 3.1, the map $x^\alpha \mapsto x^{\delta_b(\alpha)}$ is a bijection from $\mathrm{G}(K)$ to $\mathrm{G}(I)$, and every such α is a $\{0,1\}$-vector. For $\nu = \delta_b(\alpha)$,

$$\mathrm{supp}(\nu) = \alpha.$$

As x^ν ranges through $\mathrm{G}(I)$, we obtain

$$\mathrm{G}(K) = \left\{ x^{\mathrm{supp}(\nu)} \,\middle|\, x^\nu \in \mathrm{G}(I) \right\} = \mathrm{G}(J_*).$$

Thus $K = J_*$.

Now let $d \in {\mathbb{Z}_{>0}}^n$. If $I = {J_*}^{\delta_d}$, the necessity argument applied to this representation forces $d_i = e_i$ for every $i \in U$. The converse, with the coordinates off U arbitrary, was proved in the construction. This is exactly the displayed equivalence and weight freedom in the theorem. □

The criterion imposes no condition beyond I being a nonzero proper monomial ideal. In particular, no Simis property, associated-prime hypothesis, or component hypothesis is used. When a root exists, support normalization recovers it uniquely, and only variables absent from every minimal generator have unforced weights.

Theorem 3.7 (component root criterion)**.** *Let $I \subsetneq S$ be a nonzero monomial ideal with an irredundant irreducible decomposition*

$$I = \bigcap_{j=1}^{m} q_j, \qquad q_j = (x_i^{a_{j,i}} : i \in A_j),$$

where $m \geq 1$, every A_j is a nonempty subset of $\{1, \dots, n\}$, and every $a_{j,i}$ is a positive integer. Put

$$C_i = \{a_{j,i} \,|\, 1 \leq j \leq m,\ i \in A_j\}, \qquad V = \bigcup_{j=1}^{m} A_j = \{i \,|\, C_i \neq \varnothing\},$$

and

$$X_I = \{(k,\ell) \,|\, 1 \leq k < \ell \leq m \text{ and some } i \in A_k \cap A_\ell \text{ satisfies } a_{k,i} \neq a_{\ell,i}\}.$$

The following are equivalent:

(i) there are a squarefree monomial ideal $J_0 \subseteq S$ and $d \in {\mathbb{Z}_{>0}}^n$ such that $I = {J_0}^{\delta_d}$;

(ii) $|C_i| = 1$ for every $i \in V$;

(iii) $X_I = \varnothing$.

Suppose these conditions hold. For $i \in V$, let c_i be the common element of C_i; for $i \notin V$, let c_i be an arbitrary positive integer. Put

$$P_j = (x_i : i \in A_j), \qquad J = \bigcap_{j=1}^{m} P_j.$$

Then $I = J^{\delta_c}$, the displayed decomposition of J is an irredundant decomposition into monomial prime ideals, and J is the unique squarefree monomial ideal that can represent I diagonally. Moreover, if $K \subseteq S$ is squarefree and $e \in {\mathbb{Z}_{>0}}^n$, then

$$I = K^{\delta_e} \quad \Longleftrightarrow \quad K = J, \quad e_i = c_i \text{ for every } i \in V;$$

the coordinates e_i with $i \notin V$ are arbitrary. These conclusions require only irredundance of the displayed decomposition: they do not require I to have no embedded primes, the A_j to be pairwise distinct initially, equidimensionality, $V = \{1, \dots, n\}$, or $m > 1$. Under the equivalent conditions, the A_j are pairwise distinct.

Proof. Throughout the proof put

$$P_j = (x_i : i \in A_j) = \sqrt{q_j}\,.$$

Proposition 3.2 gives the component-radical bookkeeping

$$\operatorname{Ass}(S/I) = \{P_j \,|\, 1 \le j \le m\}\,, \tag{3.6}$$

independently of the displayed ordering and of repetitions among the P_j. We stress that repeated and comparable component radicals are allowed at this stage; neither a no-embedded-prime nor an MPV-minimality hypothesis is being imposed.

The proof first identifies the component occurrence sets C_i with the generator occurrence sets $D_i(I)$. It then reduces the three criteria to Theorem 3.6. A floor formula for diagonal weightings transports the whole irreducible decomposition to its radical decomposition, after which the irredundance, forced weights, uniqueness, and final pairwise distinctness are read off.

Component and generator occurrence sets agree. Put

$$D_i = \{\nu_i \,|\, x^\nu \in \mathrm{G}(I),\ \nu_i > 0\}\,.$$

We prove that

$$D_i = C_i \qquad (1 \le i \le n). \tag{3.7}$$

For an exponent vector $\beta \in \mathbb{N}^n$ and an index j, membership in q_j is characterized by

$$x^\beta \in q_j \quad \Longleftrightarrow \quad \beta_i \ge a_{j,i} \text{ for some } i \in A_j. \tag{3.8}$$

We first prove $C_t \subseteq D_t$. Fix j and $t \in A_j$, and put

$$K_j = \bigcap_{\ell \ne j} q_\ell,$$

with $K_j = S$ if $m = 1$. Irredundance gives $K_j \not\subseteq q_j$: otherwise q_j could be omitted from the decomposition of I. Choose a polynomial in $K_j \setminus q_j$. All its monomial terms lie in K_j, and at least one term lies outside the monomial ideal q_j, so there is a monomial $x^\beta \in K_j \setminus q_j$. By (3.8),

$$\beta_i < a_{j,i} \qquad (i \in A_j).$$

Let ϵ_t be the t-th coordinate vector and define

$$\gamma_h = \begin{cases} a_{j,h} - 1, & h \in A_j, \\ \beta_h, & h \notin A_j, \end{cases} \qquad y = x^\gamma, \qquad z = x_t y = x^{\gamma + \epsilon_t}.$$

Then $y \notin q_j$, because its exponent at every $i \in A_j$ is $a_{j,i} - 1$. Since $\gamma \geq \beta$ coordinatewise and $x^\beta \in K_j$, we also have $y \in K_j$; hence $y \notin I$. On the other hand, the t-th exponent of z is $a_{j,t}$, so $z \in q_j$, while $z \in K_j$ because $y \in K_j$. Thus $z \in I$.

By Lemma 3.1(i), some $x^\nu \in \mathrm{G}(I)$ divides z. It cannot divide y, since $y \notin I$. If $\nu_t = 0$, then $\nu \leq \gamma$, because $z = x^{\gamma+\epsilon_t}$ differs from y only at t; then x^ν would divide y. Hence $\nu_t > 0$. We also have $\nu \not\leq \gamma$, while $\nu \leq \gamma + \epsilon_t$. These two vectors differ only at t, so

$$\gamma_t < \nu_t \leq \gamma_t + 1.$$

Since ν_t is an integer,

$$\nu_t = \gamma_t + 1 = a_{j,t}.$$

Thus $a_{j,t} \in D_t$, proving $C_t \subseteq D_t$ for every t.

For the reverse inclusion, fix $x^\nu \in \mathrm{G}(I)$ and a coordinate i with $\nu_i > 0$. Put $\eta = \nu - \epsilon_i$. If $x^\eta \in I$, then Lemma 3.1(i) gives some $x^\mu \in \mathrm{G}(I)$ dividing x^η. Then $\mu \leq \eta \leq \nu$ and $\mu_i < \nu_i$, so x^μ is a proper divisor of x^ν, contrary to the antichain property in part (ii) of the same lemma. Hence $x^\eta \notin I$, and there is an index j with $x^\eta \notin q_j$.

Because $x^\nu \in I \subseteq q_j$, formula (3.8) gives an $h \in A_j$ with $\nu_h \geq a_{j,h}$. If $h \neq i$, then $\eta_h = \nu_h$, so the same formula gives $x^\eta \in q_j$, a contradiction. Therefore $h = i$ and $i \in A_j$. We have $\nu_i \geq a_{j,i}$, while $x^\eta \notin q_j$ gives $\nu_i - 1 = \eta_i < a_{j,i}$. Hence

$$\nu_i = a_{j,i} \in C_i.$$

This proves $D_i \subseteq C_i$ and therefore (3.7).

The three criteria are equivalent. By Theorem 3.6, I has a squarefree diagonal root exactly when $|D_i| \leq 1$ for every i. By (3.7), this is equivalent to $|C_i| \leq 1$ for every i, hence to $|C_i| = 1$ for every $i \in V$. Thus conditions (i) and (ii) are equivalent.

For a fixed i, the set C_i has at least two elements exactly when there are indices $k \neq \ell$ with $i \in A_k \cap A_\ell$ and $a_{k,i} \neq a_{\ell,i}$. After interchanging k and ℓ if necessary, this is exactly the condition that $(k, \ell) \in X_I$. Therefore some C_i has cardinality at least two exactly when $X_I \neq \varnothing$. Conditions (ii) and (iii) are equivalent, and all three conditions are equivalent. The case $m = 1$ is included: then $X_I = \varnothing$ and every C_i has at most one element.

Diagonal weighting commutes with intersections and reflects inclusions. Fix $r \in {\mathbb{Z}_{>0}}^n$. For $\beta = (\beta_1, \ldots, \beta_n) \in \mathbb{N}^n$, put

$$\left\lfloor \frac{\beta}{r} \right\rfloor = \left(\left\lfloor \frac{\beta_1}{r_1} \right\rfloor, \ldots, \left\lfloor \frac{\beta_n}{r_n} \right\rfloor \right).$$

We claim that, for every monomial ideal $L \subseteq S$,

$$x^\beta \in L^{\delta_r} \quad \Longleftrightarrow \quad x^{\lfloor \beta/r \rfloor} \in L. \tag{3.9}$$

If the left-hand membership holds, then

$$\beta = \delta_r(\alpha) + \varepsilon$$

for some $x^\alpha \in L$ and $\varepsilon \in \mathbb{N}^n$. Taking coordinatewise floors gives

$$\left\lfloor \frac{\beta}{r} \right\rfloor = \alpha + \left\lfloor \frac{\varepsilon}{r} \right\rfloor \geq \alpha,$$

so $x^{\lfloor \beta/r \rfloor} \in L$. Conversely, write $\beta_i = r_i q_i + s_i$ with $q_i = \lfloor \beta_i / r_i \rfloor$ and $0 \le s_i < r_i$. If $x^q \in L$, then $x^{\delta_r(q)} \in L^{\delta_r}$ and

$$x^\beta = x^s x^{\delta_r(q)} \in L^{\delta_r}.$$

This proves (3.9).

We use the convention that an empty intersection of ideals of S is S. Since $S^{\delta_r} = S$, formula (3.9) gives

$$\left(\bigcap_\ell L_\ell \right)^{\delta_r} = \bigcap_\ell {L_\ell}^{\delta_r} \tag{3.10}$$

for every finite family of monomial ideals. It also shows that diagonal weighting preserves and reflects inclusions of monomial ideals:

$$L^{\delta_r} \subseteq M^{\delta_r} \quad \Longleftrightarrow \quad L \subseteq M. \tag{3.11}$$

Indeed, the right-to-left direction follows immediately from the definition of the weighted ideals. Conversely, if the weighted inclusion holds and $x^\alpha \in L$, then $x^{\delta_r(\alpha)} \in M^{\delta_r}$; applying (3.9) to $\beta = \delta_r(\alpha)$ gives $x^\alpha \in M$.

The prime intersection constructs a squarefree root. Assume now that the equivalent conditions hold. Let c_i be the common element of C_i for $i \in V$, and choose $c_i \in \mathbb{Z}_{>0}$ arbitrarily for $i \notin V$. Put

$$J = \bigcap_{j=1}^m P_j.$$

Each P_j is prime because

$$S/P_j \simeq k[x_i : i \notin A_j]$$

is a domain. Hence J is radical. It is proper because $J \subseteq P_1 \subsetneq S$, and it is nonzero because, after choosing $i_j \in A_j$ for every j, the monomial $\prod_{j=1}^m x_{i_j}$ belongs to every P_j.

Every member of $\mathrm{G}(J)$ is squarefree. If $x^\alpha \in \mathrm{G}(J)$ had $\alpha_t \ge 2$, set $\sigma = \alpha - \epsilon_t$. Then

$$2\sigma_h = 2\alpha_h \ge \alpha_h \quad (h \ne t), \qquad 2\sigma_t = 2\alpha_t - 2 \ge \alpha_t.$$

Thus $2\sigma \ge \alpha$, so $(x^\sigma)^2 = x^{2\sigma} \in J$. Since J is radical, $x^\sigma \in J$, contradicting the minimality of the properly smaller divisor $x^\sigma \mid x^\alpha$. Thus J is squarefree.

For every j,

$${P_j}^{\delta_c} = q_j. \tag{3.12}$$

If $x^\alpha \in P_j$, some $i \in A_j$ has $\alpha_i > 0$; since $c_i = a_{j,i}$, the i-th exponent of $x^{\delta_c(\alpha)}$ is at least $a_{j,i}$, so $x^{\delta_c(\alpha)} \in q_j$. This proves the inclusion from left to right. Conversely, for each $i \in A_j$,

$$x_i^{a_{j,i}} = x^{\delta_c(\epsilon_i)}, \qquad x_i \in P_j,$$

so every displayed generator of q_j belongs to ${P_j}^{\delta_c}$.

Combining (3.10) and (3.12) gives

$$J^{\delta_c} = \left(\bigcap_{j=1}^m P_j \right)^{\delta_c} = \bigcap_{j=1}^m {P_j}^{\delta_c} = \bigcap_{j=1}^m q_j = I.$$

If $r \in {\mathbb{Z}_{>0}}^n$ is any vector with $r_i = c_i$ for every $i \in V$, the same proof of (3.12) applies with r in place of c: no A_j contains a variable outside V. Hence

$$J^{\delta_r} = I \tag{3.13}$$

for every positive extension of the prescribed weights from V to $\{1, \dots, n\}$.

The prime decomposition is irredundant and the representation data are unique. We prove that the displayed decomposition of J is irredundant. Suppose that for some j,

$$\bigcap_{\ell \neq j} P_\ell \subseteq P_j,$$

where the left-hand side is S if $m = 1$. Weighting by c and using the preservation of inclusions, (3.10), and (3.12) gives

$$\bigcap_{\ell \neq j} q_\ell = \left(\bigcap_{\ell \neq j} P_\ell \right)^{\delta_c} \subseteq {P_j}^{\delta_c} = q_j.$$

Then $\bigcap_{\ell \neq j} q_\ell \subseteq q_j$, so q_j could be omitted from the displayed decomposition of I. This contradicts irredundance, including the case $m = 1$, where the displayed inclusion would say $S \subseteq q_1$ and q_1 is proper. Thus

$$J = \bigcap_{j=1}^{m} P_j$$

is irredundant.

By (3.7), the set U from Theorem 3.6 is exactly V, and the common generator occurrence exponent at each $i \in V$ is c_i. We have just proved that J is a squarefree root of I. The uniqueness clause of Theorem 3.6 therefore identifies J with the support-normalized root J_* of that theorem. Applying its exact weight statement to an arbitrary squarefree $K \subseteq S$ and an arbitrary $e \in {\mathbb{Z}_{>0}}^n$ gives

$$I = K^{\delta_e} \quad \Longleftrightarrow \quad K = J, \quad e_i = c_i \text{ for every } i \in V.$$

Together with (3.13), this also shows that the coordinates e_i with $i \notin V$ are arbitrary.

Finally, if $A_k = A_\ell$ for distinct k and ℓ, then $P_k = P_\ell$, so one of these two identical terms is redundant in the irredundant decomposition $J = \bigcap_j P_j$. Hence the A_j are pairwise distinct. Conversely, $A_j = \{i \mid x_i \in P_j\}$, so the component radicals P_j are also pairwise distinct. Thus the displayed irreducible decomposition of I is MPV-minimal, and the associated primes in (3.6) have no repetitions among the displayed components.

Under the equivalent conditions, the argument also rules out strict inclusions among the component supports. If $A_j \subsetneq A_\ell$, then $P_j \subsetneq P_\ell$ and

$$\bigcap_{h \neq \ell} P_h \subseteq P_j \subsetneq P_\ell,$$

so the component P_ℓ would be redundant in the irredundant decomposition $J = \bigcap_h P_h$ already proved above. Hence the A_j are pairwise incomparable as well as pairwise distinct. Proposition 3.2 therefore gives

$$\operatorname{Ass}(S/I) = \{P_1, \dots, P_m\} = \operatorname{Min}(I), \qquad \operatorname{Ass}(S/J) = \{P_1, \dots, P_m\} = \operatorname{Min}(J).$$

Thus MPV-minimality and absence of embedded primes are consequences of irredundance together with exponent coherence in the equivalent root case; neither property was assumed as a hypothesis of the theorem.

All degenerate cases have been included: for $m = 1$, the mismatch set and all comparisons are vacuous; a variable occurring in only one component has its weight forced by that component's threshold; and a variable occurring in no component is unused by J and has a completely free weight. □

Thus the component criterion is an exact reformulation of the generator criterion: the component thresholds of a variable are precisely the positive exponents with which that variable occurs among the minimal monomial generators. The only structural input on the component side is irredundance of the displayed irreducible decomposition.

4 A five-variable obstruction

We retain the notation of Section 2. This section isolates a five-variable family for which the three original MPV hypotheses hold at every parameter value, while a same-ring squarefree diagonal root exists exactly on the parameter diagonal. We also show that both the generator-support clutter and the component-support clutter contain a strong triangle and that the two clutters are mutual blockers.

Construction 4.1 (base family)**.** Let $S = k[a, b, c, d, z]$, fix $r, s \in \mathbb{Z}_{>0}$, and use the variable order (a, b, c, d, z). Put

$$I_{r,s} = (ac, bd, abz^r, cdz^s),$$

$$Q_1 = (a, b, z^s), \quad Q_2 = (a, d), \quad Q_3 = (b, c), \quad Q_4 = (c, d, z^r),$$

$$P_1 = (a, b, z), \quad P_2 = (a, d), \quad P_3 = (b, c), \quad P_4 = (c, d, z),$$

$$M = (ac, bd, abz, cdz), \qquad H = (ac, bd, ab, cd),$$

and

$$\mathcal{F}_{r,s} = \{\{a, c\}, \{b, d\}, \{a, b, z\}, \{c, d, z\}\}, \qquad \mathcal{A}_{r,s} = \{\{a, b, z\}, \{a, d\}, \{b, c\}, \{c, d, z\}\}.$$

The following capacity lemma controls every power of the base ideal.

Lemma 4.2 (component capacities and descent)**.** *Retain Construction 4.1. Put $D_0 = S$ and, for $m \geq 1$,*

$$D_m = \bigcap_{j=1}^{4} {Q_j}^m.$$

For a monomial

$$u = a^\alpha b^\beta c^\gamma d^\delta z^\zeta \qquad (\alpha, \beta, \gamma, \delta, \zeta \in \mathbb{N}),$$

define

$$L_1(u) = \alpha + \beta + \left\lfloor \frac{\zeta}{s} \right\rfloor, \quad L_2(u) = \alpha + \delta,$$

$$L_3(u) = \beta + \gamma, \qquad L_4(u) = \gamma + \delta + \left\lfloor \frac{\zeta}{r} \right\rfloor.$$

Then, for every $m \geq 1$:

(a) $u \in D_m$ *if and only if* $L_j(u) \geq m$ *for* $1 \leq j \leq 4$*;*

(b) if $u \in D_m$ *and* $ac \mid u$*, then* $u/(ac) \in D_{m-1}$*, and if* $u \in D_m$ *and* $bd \mid u$*, then* $u/(bd) \in D_{m-1}$*;*

(c) if $u \in D_m$ *and neither* ac *nor* bd *divides* u*, then*

$$(abz^r)^m \mid u \qquad \text{or} \qquad (cdz^s)^m \mid u.$$

In the first alternative $\gamma = \delta = 0$*,* $\alpha, \beta \geq m$*, and* $\zeta \geq rm$*; in the second alternative* $\alpha = \beta = 0$*,* $\gamma, \delta \geq m$*, and* $\zeta \geq sm$*.*

Proof. We first derive the membership criterion. The ideal ${Q_1}^m$ is generated by the monomials

$$a^i b^j z^{sk}, \qquad i, j, k \in \mathbb{N}, \quad i + j + k = m.$$

Such a monomial divides u precisely when $i \leq \alpha$, $j \leq \beta$, and $k \leq \lfloor \zeta/s \rfloor$ for some triple with sum m. This is possible exactly when $L_1(u) \geq m$: necessity follows by adding the three bounds, while sufficiency follows by choosing

$$k = \min\left\{ \left\lfloor \frac{\zeta}{s} \right\rfloor, m \right\}, \quad i = \min\{\alpha, m-k\}, \quad j = m - k - i.$$

If $j > 0$, then $k < m$ and $i = \alpha$, so

$$j = m - \left\lfloor \frac{\zeta}{s} \right\rfloor - \alpha \leq \beta$$

because $L_1(u) \geq m$. Thus $u \in {Q_1}^m$ exactly when $L_1(u) \geq m$.

The ideal ${Q_2}^m$ is generated by $a^i d^{m-i}$ for $0 \leq i \leq m$. Such a monomial divides u exactly when the interval $[m - \delta, \alpha]$ meets $[0, m]$, which is equivalent to $\alpha + \delta \geq m$. Interchanging the variable pairs (a, d) and (b, c) proves the criterion for Q_3. Applying the argument for Q_1 to the variables c, d, z and the exponent r on z proves the criterion for Q_4. Intersecting the four criteria proves (a).

Division by ac decreases each of α and γ by one and leaves the other exponents unchanged. The quantities L_1 and L_2 contain α but not γ, while L_3 and L_4 contain γ but not α. Hence every L_j decreases by one. For $m = 1$, the quotient is a monomial and therefore lies in $D_0 = S$; for $m \geq 2$, (a) gives $u/(ac) \in D_{m-1}$. The same calculation for the pair (β, δ) proves descent by bd.

Finally suppose that neither ac nor bd divides u. Then

$$(\alpha = 0 \text{ or } \gamma = 0), \qquad (\beta = 0 \text{ or } \delta = 0).$$

If $\alpha = 0$, then $L_2(u) = \delta \geq m > 0$, so $\delta > 0$ and therefore $\beta = 0$. The inequalities $L_1(u) \geq m$ and $L_3(u) \geq m$ now give $\zeta \geq sm$ and $\gamma \geq m$, and $L_2(u) \geq m$ gives $\delta \geq m$. Thus $(cdz^s)^m \mid u$, with the asserted exponent bounds.

If $\alpha > 0$, the first displayed alternative forces $\gamma = 0$. Then $L_3(u) = \beta \geq m > 0$, so $\delta = 0$. The inequalities $L_2(u) \geq m$ and $L_4(u) \geq m$ respectively give $\alpha \geq m$ and $\zeta \geq rm$, while $L_3(u) \geq m$ gives $\beta \geq m$. Thus $(abz^r)^m \mid u$, again with the asserted bounds. □

Theorem 4.3 (base-family package)**.** *Retain Construction 4.1. Then the following hold for all* $r, s \in \mathbb{Z}_{>0}$*:*

(a) *The monomials* ac, bd, abz^r, cdz^s *are precisely the minimal monomial generators of* $I_{r,s}$*, and*

$$I_{r,s} = Q_1 \cap Q_2 \cap Q_3 \cap Q_4$$

is its irredundant irreducible decomposition.

(b) *For every* $n \geq 1$*,*

$$I_{r,s}{}^{(n)} = I_{r,s}{}^n = \bigcap_{j=1}^{4} Q_j{}^n.$$

Thus $I_{r,s}$ *has no embedded primes, its four component radicals* $P_1, \ldots, P_4$ *are pairwise distinct, and it is Simis; equivalently, it satisfies the three original MPV hypotheses.*

(c) *There are a squarefree monomial ideal* $J \subseteq S$ *and* $w \in \mathbb{Z}_{>0}{}^5$ *such that* $I_{r,s} = J^{\delta_w}$ *if and only if* $r = s$*. If* $r = s = t$*, the data are forced:*

$$J = (ac, bd, abz, cdz) = I_{1,1} = M, \qquad w = (1, 1, 1, 1, t),$$

and J *is Simis. If* $r \neq s$*, no squarefree monomial ideal* $J \subseteq S$ *and positive diagonal weight vector* w *represent* $I_{r,s}$*, even when the requirement that* J *be Simis is omitted.*

(d) *For every* $t \in \mathbb{Z}_{>0}$ *and* $d = (1, 1, 1, 1, t)$*,*

$$H^{\delta_d} = H \neq I_{t,t}.$$

Consequently z *cannot be omitted from the two cubic squarefree generators: the forced root of* $I_{t,t}$ *is* (ac, bd, abz, cdz)*, not* (ac, bd, ab, cd)*.*

Proof. We first identify the components and generators. We then use Lemma 4.2 to identify simultaneously the ordinary and symbolic powers. The two root criteria locate the exact representable diagonal and force the source there. Finally, we compare that source with H.

(a) *Components and generators.* Put $I = I_{r,s}$. Each displayed generator of I lies in every Q_j; the entries in the following table are generators of the corresponding component dividing the monomial at the left:

	Q_1	Q_2	Q_3	Q_4
ac	a	a	c	c
bd	b	d	b	d
abz^r	a	a	b	z^r
cdz^s	z^s	d	c	c

Hence $I \subseteq \bigcap_{j=1}^{4} Q_j$.

For the reverse inclusion, let

$$u = a^\alpha b^\beta c^\gamma d^\delta z^\zeta \in \bigcap_{j=1}^{4} Q_j.$$

Membership in Q_2 and Q_3 gives

$$(\alpha > 0 \text{ or } \delta > 0) \quad \text{and} \quad (\beta > 0 \text{ or } \gamma > 0).$$

If $\alpha, \gamma > 0$, then $ac \mid u$; if $\beta, \delta > 0$, then $bd \mid u$. If $\alpha, \beta > 0$, membership in Q_4 gives $\gamma > 0$, $\delta > 0$, or $\zeta \geq r$, and respectively ac, bd, or abz^r divides u. If $\gamma, \delta > 0$, membership in Q_1

similarly gives $\alpha > 0$, $\beta > 0$, or $\zeta \geq s$, and respectively ac, bd, or cdz^s divides u. Thus every monomial in the intersection lies in I, and

$$I = \bigcap_{j=1}^{4} Q_j. \tag{4.1}$$

Every Q_j is generated by pure positive powers of the variables in its support. Proposition 3.2 therefore shows that Q_j is irreducible and P_j-primary and that P_j is its radical. To see that the intersection in (4.1) is irredundant, put

$$v_1 = cdz^{s-1}, \qquad v_2 = bc, \qquad v_3 = ad, \qquad v_4 = abz^{r-1},$$

with $z^0 = 1$. Then

$$v_1 \notin Q_1, \quad v_2 \notin Q_2, \quad v_3 \notin Q_3, \quad v_4 \notin Q_4,$$

while every other v_ℓ belongs to Q_j for $j \neq \ell$. For instance, v_1 has no a or b and has z-exponent $s-1 < s$, but belongs to Q_2 and Q_4 through d and c and to Q_3 through c; the other three checks are the same. Hence omitting any one component strictly enlarges the intersection.

The supports of ac, bd, abz^r, cdz^s, in this order, form an antichain. For the six pairs with increasing indices, the respective pairs of variables

$$(a,b),\ (c,b),\ (a,d),\ (d,a),\ (b,c),\ (a,c)$$

exhibit one variable in each difference of supports. Consequently none of the four displayed monomials divides another, so they are precisely the minimal monomial generators of I. Their supports are the members of $\mathcal{F}_{r,s}$, while the supports of $Q_1, \ldots, Q_4$, equivalently of $P_1, \ldots, P_4$, are the members of $\mathcal{A}_{r,s}$.

(b) *All powers and the MPV hypotheses.* The component primes are pairwise incomparable: the six pairs of variables

$$(b,d),\ (a,c),\ (a,c),\ (a,b),\ (a,c),\ (b,d)$$

exhibit both differences of supports for the pairs (P_i, P_j) with $i < j$. Proposition 3.2, applied to the decomposition (4.1), gives

$$\operatorname{Ass}(S/I) = \{P_1, P_2, P_3, P_4\}.$$

Every minimal prime of I is associated to S/I, and pairwise incomparability prevents one of these four primes from containing another. Therefore

$$\operatorname{Min}(I) = \operatorname{Ass}(S/I) = \{P_1, P_2, P_3, P_4\}. \tag{4.2}$$

In particular, I has no embedded primes and the radicals of its irreducible components are pairwise distinct.

We next prove that

$$I^n = \bigcap_{j=1}^{4} {Q_j}^n = D_n \tag{4.3}$$

for every $n \geq 0$, where D_n is the ideal of Lemma 4.2 and $D_0 = S = I^0$. The containment $I^n \subseteq D_n$ follows from $I \subseteq \bigcap_j Q_j$. For the reverse containment, it suffices to consider a monomial $u \in D_n$. We argue by induction on n, the case $n = 0$ being the definition of D_0.

Suppose $n \geq 1$. If $ac \mid u$, part (b) of Lemma 4.2 gives $u/(ac) \in D_{n-1} = I^{n-1}$, so $u \in ac\, I^{n-1} \subseteq I^n$. If $ac \nmid u$ but $bd \mid u$, descent by bd gives the same conclusion. If neither ac nor bd divides u, part (c) of the lemma gives $(abz^r)^n \mid u$ or $(cdz^s)^n \mid u$. Both displayed divisors are products of n minimal generators of I, so $u \in I^n$. This proves (4.3).

Because the four component radicals are pairwise distinct and (4.2) identifies them with $\mathrm{Min}(I)$, the canonical primary components of Theorem 3.3 are $Q_1, \ldots, Q_4$. Combining the symbolic-power formula in that theorem with the ordinary-power intersection (4.3) gives

$$I^{(n)} = \bigcap_{j=1}^{4} {Q_j}^n = I^n \qquad (n \geq 1).$$

Thus I is Simis. Together with (4.2) and the pairwise distinct component radicals, this proves all three original MPV hypotheses.

(c) *The exact diagonal and the forced root.* The two root criteria detect the same mismatch between r and s on the component side and the generator side. In the irreducible decomposition (4.1), the component occurrence sets are

$$C_a = C_b = C_c = C_d = \{1\}, \qquad C_z = \{s, r\}.$$

Theorem 3.7 therefore gives a squarefree monomial ideal $J \subseteq S$ and a weight vector $w \in {\mathbb{Z}_{>0}}^5$ with $I = J^{\delta_w}$ if and only if $r = s$. It assumes no Simis condition on J. Thus, when $r \neq s$, no such representation exists even if source Simisness is omitted.

The generator criterion shows the same boundary in the support data. The variables a, b, c, d occur in the four minimal generators only with exponent 1, while the occurrences of z have exponents r and s. Hence the generator criterion independently rules out every squarefree diagonal source when $r \neq s$.

Suppose now that $r = s = t$. Every variable occurs in the decomposition, and the common component exponents are

$$c = (1, 1, 1, 1, t).$$

Theorem 3.7 gives the unique squarefree source

$$P_1 \cap P_2 \cap P_3 \cap P_4.$$

Formula (4.1), specialized to $(r, s) = (1, 1)$, has $Q_j = P_j$ for every j and identifies this intersection with

$$I_{1,1} = (ac, bd, abz, cdz) = M.$$

Thus $I_{t,t} = M^{\delta_c}$, and both the source and every weight coordinate are forced. On the generator side, Theorem 3.6 gives the same conclusion directly: taking squarefree supports of the four minimal generators forces $J = M$, and the unique positive occurrence exponents force $w = c$. Part (b), applied to $(r, s) = (1, 1)$, shows that $M = I_{1,1}$ is Simis.

(d) *The abbreviation that loses z.* Fix $t \in \mathbb{Z}_{>0}$ and put $d = (1, 1, 1, 1, t)$. The four displayed generators of $H = (ac, bd, ab, cd)$ have z-exponent zero and all other exponents equal to 1, so each is fixed by δ_d. If $u = x^\nu \in H$ is a monomial, one of these generators g divides u, and the fixed monomial g divides $x^{\delta_d(\nu)}$. Hence $H^{\delta_d} \subseteq H$. Conversely, applying δ_d to each displayed generator of H shows that every displayed generator of H lies in H^{δ_d}, so $H \subseteq H^{\delta_d}$. Therefore $H^{\delta_d} = H$.

We have

$$I_{t,t} = (ac, bd, abz^t, cdz^t) \subseteq H,$$

because $ab \mid abz^t$ and $cd \mid cdz^t$. The containment is strict: $ab \in H$, but none of the four minimal generators ac, bd, abz^t, cdz^t of $I_{t,t}$ divides ab. Therefore

$$H^{\delta_d} = H \neq I_{t,t}.$$

The variable z is consequently indispensable in the two cubic minimal generators of the forced root; H is not a squarefree diagonal source for $I_{t,t}$.

□

The representability boundary is therefore exactly the diagonal $r = s$. Throughout this family, each $I_{r,s}$ has no embedded primes, its component radicals are pairwise distinct, and it is Simis. For $r \neq s$, the obstruction is instead the simultaneous occurrence of z with exponents s and r in the components and with exponents r and s in the generators.

For the second theorem we use the following elementary duality.

Lemma 4.4 (blocker involution)**.** *Let $\mathcal{H}$ be a nonempty clutter of nonempty subsets of a finite vertex set V. Then $b(\mathcal{H})$ is a clutter on V, and*

$$b(b(\mathcal{H})) = \mathcal{H}.$$

Proof. The ambient set V is a transversal of $\mathcal{H}$, and the inclusion-minimal transversals form an antichain, so $b(\mathcal{H})$ is a clutter.

We first show that every $E \in \mathcal{H}$ is a minimal transversal of $b(\mathcal{H})$. It meets every member of $b(\mathcal{H})$ by the definition of a transversal. Suppose that $C \subsetneq E$ also meets every member of $b(\mathcal{H})$. Because $\mathcal{H}$ is a clutter, no member of $\mathcal{H}$ is contained in C. Hence $V \setminus C$ meets every edge of $\mathcal{H}$ and therefore contains some $B \in b(\mathcal{H})$. Then $C \cap B = \varnothing$, contradicting the choice of C. Thus $\mathcal{H} \subseteq b(b(\mathcal{H}))$.

Conversely, let T be a minimal transversal of $b(\mathcal{H})$. If T contained no member of $\mathcal{H}$, then $V \setminus T$ would meet every edge of $\mathcal{H}$ and would contain a member B of $b(\mathcal{H})$ disjoint from T. This is impossible. Therefore some $E \in \mathcal{H}$ satisfies $E \subseteq T$. We have already shown that every edge of $\mathcal{H}$ is a transversal of $b(\mathcal{H})$, so minimality of T forces $T = E$. □

Theorem 4.5 (strong-triangle obstruction)**.** *Let $r, s \in \mathbb{Z}_{>0}$ be distinct and retain the clutters $\mathcal{F}_{r,s}$ and $\mathcal{A}_{r,s}$ of Construction 4.1 on the vertex set $\{a, b, c, d, z\}$. Both clutters have four edges, with edge-cardinality vectors $(2, 2, 3, 3)$ and $(3, 2, 2, 3)$ in their displayed orders. The sequences*

$$a, \{a, c\}, c, \{c, d, z\}, z, \{a, b, z\}, a$$

and

$$a, \{a, d\}, d, \{c, d, z\}, z, \{a, b, z\}, a$$

are strong 3-cycles in $\mathcal{F}_{r,s}$ and $\mathcal{A}_{r,s}$, respectively. In the first cycle the noncycle edge $\{b, d\}$ is disjoint from $\{a, c, z\}$, and in the second cycle the noncycle edge $\{b, c\}$ is disjoint from $\{a, d, z\}$. Consequently neither clutter is balanced, and

$$b(\mathcal{F}_{r,s}) = \mathcal{A}_{r,s}, \qquad b(\mathcal{A}_{r,s}) = \mathcal{F}_{r,s}.$$

Proof. By part (a) of Theorem 4.3, $\mathcal{F}_{r,s}$ is the generator-support clutter and $\mathcal{A}_{r,s}$ is the component-support clutter of $I_{r,s}$. We verify the two strong triangles, then determine both blockers.

Complementary strong triangles. Write the members of the two clutters in their displayed orders as

$$F_1 = \{a, c\}, \quad F_2 = \{b, d\}, \quad F_3 = \{a, b, z\}, \quad F_4 = \{c, d, z\},$$
$$A_1 = \{a, b, z\}, \quad A_2 = \{a, d\}, \quad A_3 = \{b, c\}, \quad A_4 = \{c, d, z\}.$$

Both displayed families are antichains. For $\mathcal{F}_{r,s}$, the two two-element edges are incomparable; neither two-element edge is contained in either three-element edge, as the respective missing variables are

	F_3	F_4
F_1	c	a
F_2	d	b

and $a \in F_3 \setminus F_4$ while $c \in F_4 \setminus F_3$. For $\mathcal{A}_{r,s}$, the two-element edges A_2 and A_3 are disjoint; neither is contained in A_1 or A_4, as the missing-variable table is

	A_1	A_4
A_2	d	a
A_3	c	b

and $a \in A_1 \setminus A_4$ while $c \in A_4 \setminus A_1$. Thus both families have four edges and their cardinality vectors are respectively $(2, 2, 3, 3)$ and $(3, 2, 2, 3)$.

The two cycles asserted in the theorem are

$$C_{\text{gen}} = (a, F_1, c, F_4, z, F_3, a), \qquad C_{\text{comp}} = (a, A_2, d, A_4, z, A_1, a).$$

The vertices and edges in each cycle are pairwise distinct, and the consecutive vertices lie in the intermediate edge. For $V_{\text{gen}} = \{a, c, z\}$ and $V_{\text{comp}} = \{a, d, z\}$, their exact intersections with the cycle vertices are

$$F_1 \cap V_{\text{gen}} = \{a, c\}, \quad F_4 \cap V_{\text{gen}} = \{c, z\}, \quad F_3 \cap V_{\text{gen}} = \{z, a\},$$
$$A_2 \cap V_{\text{comp}} = \{a, d\}, \quad A_4 \cap V_{\text{comp}} = \{d, z\}, \quad A_1 \cap V_{\text{comp}} = \{z, a\}.$$

The one edge omitted from each cycle is disjoint from its cycle vertices:

$$F_2 \cap V_{\text{gen}} = \varnothing, \qquad A_3 \cap V_{\text{comp}} = \varnothing.$$

Hence C_{gen} and C_{comp} are strong 3-cycles. Since a balanced clutter has no strong cycle of odd length, neither $\mathcal{F}_{r,s}$ nor $\mathcal{A}_{r,s}$ is balanced.

Mutual blockers. We compute $b(\mathcal{F}_{r,s})$. In the following table, the entry in row A_i and column F_j is a vertex of $A_i \cap F_j$:

	F_1	F_2	F_3	F_4
A_1	a	b	a	z
A_2	a	d	a	d
A_3	c	b	b	c
A_4	c	d	z	c

Thus every A_i is a transversal of $\mathcal{F}_{r,s}$.

Conversely, let T be an $\mathcal{F}_{r,s}$-transversal. Choose

$$x \in T \cap F_1, \qquad y \in T \cap F_2.$$

If $(x,y)=(c,b)$, then $A_3=\{b,c\}\subseteq T$; if $(x,y)=(a,d)$, then $A_2=\{a,d\}\subseteq T$. If $(x,y)=(a,b)$, transversality for $F_4=\{c,d,z\}$ gives a vertex $q\in T\cap F_4$, and

$$\begin{cases} A_3\subseteq T, & q=c,\\ A_2\subseteq T, & q=d,\\ A_1\subseteq T, & q=z. \end{cases}$$

If $(x,y)=(c,d)$, transversality for $F_3=\{a,b,z\}$ similarly gives

$$\begin{cases} A_2\subseteq T, & q=a,\\ A_3\subseteq T, & q=b,\\ A_4\subseteq T, & q=z. \end{cases}$$

Therefore every $\mathcal{F}_{r,s}$-transversal contains a member of $\mathcal{A}_{r,s}$.

Because $\mathcal{A}_{r,s}$ is an antichain, every A_i is an inclusion-minimal $\mathcal{F}_{r,s}$-transversal. Conversely, every minimal $\mathcal{F}_{r,s}$-transversal contains some A_i, which is itself a transversal, and hence equals A_i. This proves

$$b(\mathcal{F}_{r,s})=\mathcal{A}_{r,s}.$$

Applying Lemma 4.4 gives

$$b(\mathcal{A}_{r,s})=b(b(\mathcal{F}_{r,s}))=\mathcal{F}_{r,s}.$$

Thus the generator and component supports obstruct balancedness in both directions, and their blocker relationship is mutual with no additional minimal transversals on either side. □

5 The apex-cone boundary

The following height bound is [Sta26, Tag 0BBZ, Commutative Algebra, Lemma 10.60.12].

Theorem 5.1 (generalized Krull height theorem)**.** *Let A be a Noetherian ring, let $h\geq 1$, let $f_1,\dots,f_h\in A$, and let $\mathfrak{p}$ be a prime ideal minimal over $(f_1,\dots,f_h)$. Then*

$$\operatorname{ht}(\mathfrak{p})\leq h.$$

We retain the notation of Section 2 and the base family of Construction 4.1. The following elementary lemma records how monomial ideals generated in the original five variables interact with extension to the polynomial ring and multiplication by the new variable.

Lemma 5.2 (new-variable bookkeeping)**.** *Let $S=k[a,b,c,d,z]$ and let $R=S[t]$. Let $(\mathfrak{a}_\lambda)_{\lambda\in\Lambda}$ be a finite family of nonzero proper monomial ideals of S, and let $\mathfrak{a}\subseteq S$ be a nonzero proper monomial ideal. For every $m\in\mathbb{Z}_{>0}$:*

(a)

$$\Big(\bigcap_{\lambda\in\Lambda}\mathfrak{a}_\lambda\Big)R=\bigcap_{\lambda\in\Lambda}(\mathfrak{a}_\lambda R);$$

(b)

$$(t^m)\cap\mathfrak{a}R=t^m(\mathfrak{a}R);$$

(c) *the minimal monomial generators of* $\mathfrak{a}R$*, taken in* R*, are the members of* $\mathrm{G}(\mathfrak{a})$*, viewed as monomials of* R*, and*

$$\mathrm{G}(t(\mathfrak{a}R)) = \{tu \,|\, u \in \mathrm{G}(\mathfrak{a})\}\,.$$

Proof. For every monomial ideal $\mathfrak{b} \subseteq S$, its monomial basis gives the k-vector-space decomposition

$$\mathfrak{b}R = \bigoplus_{\eta\in\mathbb{N}} t^\eta \mathfrak{b}.$$

A polynomial belongs to $\bigcap_\lambda(\mathfrak{a}_\lambda R)$ exactly when each of its t^η-coefficients belongs to every $\mathfrak{a}_\lambda$, which proves (a). Moreover,

$$(t^m) = \bigoplus_{\eta\geq m} t^\eta S,$$

so intersecting this decomposition with that of $\mathfrak{a}R$ gives

$$(t^m) \cap \mathfrak{a}R = \bigoplus_{\eta\geq m} t^\eta \mathfrak{a} = t^m(\mathfrak{a}R).$$

The monomials of $\mathfrak{a}R$ are precisely the monomials $t^\eta u$ with u a monomial of $\mathfrak{a}$. If $\eta > 0$, then $u \in \mathfrak{a}R$ divides $t^\eta u$. Thus every minimal generator of $\mathfrak{a}R$ is t-free. Divisibility between t-free monomials is unchanged by extension from S to R, so the minimal generators of $\mathfrak{a}R$ are exactly those of $\mathfrak{a}$.

Every monomial of $t(\mathfrak{a}R)$ has the form $t^{\eta+1}u$ with u a monomial of $\mathfrak{a}$. If $\eta > 0$, it is divisible by $tu \in t(\mathfrak{a}R)$. It remains to compare the monomials tu. For monomials $u, v \in S$,

$$tu \mid tv \quad\Longleftrightarrow\quad u \mid v,$$

so tu is minimal among these monomials exactly when u is a minimal generator of $\mathfrak{a}$. This proves the displayed generating set. □

Theorem 5.3 (apex-cone boundary)**.** *Let* $r, s \in \mathbb{Z}_{>0}$*, retain the ideals* $I_{r,s}$*,* Q_j*,* P_j *for* $1 \leq j \leq 4$*, and the ideal* M *of Construction 4.1, and put*

$$R = k[a, b, c, d, z, t], \qquad K_{r,s} = tI_{r,s}R = (tac, tbd, tabz^r, tcdz^s).$$

Regard Q_j *and* P_j *as ideals of* R *for* $1 \leq j \leq 4$*, and put*

$$Q_0 = (t), \qquad P_0 = (t).$$

For $m \geq 1$*, put* $D_m = \bigcap_{j=1}^4 {Q_j}^m$*, and put*

$$\mathcal{G} = \mathcal{F}(K_{r,s}), \qquad \mathcal{B} = \{\{t\}, \{a, b, z\}, \{a, d\}, \{b, c\}, \{c, d, z\}\}.$$

Then the following hold:

(i) For every $m \geq 1$*,*

$${I_{r,s}}^m R = D_m = \bigcap_{j=1}^4 {Q_j}^m.$$

Consequently $D_1 = I_{r,s}R$*,* $\sqrt{I_{r,s}R} = MR$*, and*

$$\mathcal{F}(I_{r,s}R) = \mathcal{F}(MR).$$

(ii)

$$K_{r,s} = Q_0 \cap Q_1 \cap Q_2 \cap Q_3 \cap Q_4$$

is the unique irredundant irreducible monomial decomposition of $K_{r,s}$, up to reordering. Its component-radical vector is

$$(P_0, P_1, P_2, P_3, P_4).$$

These five primes are pairwise distinct and pairwise incomparable, have respective heights

$$(1, 3, 2, 2, 3),$$

and

$$\mathrm{Ass}_R(R/K_{r,s}) = \mathrm{Min}(K_{r,s}) = \{P_0, P_1, P_2, P_3, P_4\}.$$

In particular, $K_{r,s}$ has no embedded primes and its irreducible decomposition is MPV-minimal.

(iii) For every $m \geq 1$,

$$K_{r,s}{}^{(m)} = (t^m) \cap \bigcap_{j=1}^{4} Q_j{}^m = K_{r,s}{}^m.$$

Thus $K_{r,s}$ is Simis.

(iv)

$$\mathrm{G}(K_{r,s}) = \{tac, tbd, tabz^r, tcdz^s\}, \qquad \sqrt{K_{r,s}} = (tac, tbd, tabz, tcdz),$$

and

$$\mathcal{G} = \mathcal{F}(\sqrt{K_{r,s}}) = \{\{a, c, t\}, \{b, d, t\}, \{a, b, z, t\}, \{c, d, z, t\}\}.$$

(v) $\Delta(K_{r,s})$ is the cone with apex t over $\Delta(I_{r,s})$. It is contractible, and for every coefficient field

$$\beta_1(\Delta(K_{r,s})) = 0.$$

(vi) There are a squarefree monomial ideal $J \subseteq R$ and $d \in \mathbb{Z}_{>0}{}^6$ such that $K_{r,s} = J^{\delta_d}$ if and only if $r = s$. If $r = s = \rho$, the unique squarefree source is

$$J = (tac, tbd, tabz, tcdz) = tM,$$

and the weights on (a, b, c, d, z, t) are forced to be

$$d = (1, 1, 1, 1, \rho, 1).$$

(vii) If $r \neq s$, then $K_{r,s}$ satisfies the three original MPV hypotheses, has no squarefree standard-positive-diagonal root in R, and additionally satisfies

$$\mathcal{F}(K_{r,s}) = \mathcal{F}(\sqrt{K_{r,s}}), \qquad \beta_1(\Delta(K_{r,s})) = 0.$$

Consequently adding support preservation and $\beta_1(\Delta(L)) \leq 1$ to the three original MPV hypotheses does not yield the MPV conclusion.

(viii) Let $\mathcal{H}$ be a clutter on a finite vertex set V, let $t \notin V$, and put

$$\mathcal{H}^+ = \{E \cup \{t\} \mid E \in \mathcal{H}\}.$$

Deleting t and adding t are inverse length-preserving bijections between the strong cycles of $\mathcal{H}^+$ and those of $\mathcal{H}$. In particular, $\mathcal{H}$ is balanced if and only if $\mathcal{H}^+$ is balanced. Here $\mathcal{G} = (\mathcal{F}_{r,s})^+$.

(ix) The sequence

$$a, \{a, c, t\}, c, \{c, d, z, t\}, z, \{a, b, z, t\}, a$$

is a strong 3*-cycle in* $\mathcal{G}$*, while*

$$\mathcal{B} = \{\{t\}, \{a, b, z\}, \{a, d\}, \{b, c\}, \{c, d, z\}\}$$

contains the strong 3*-cycle*

$$a, \{a, d\}, d, \{c, d, z\}, z, \{a, b, z\}, a.$$

Thus the coned family refutes the proposed $\beta_1 \leq 1$ *strengthening, but it does not refute a balanced-support strengthening requiring the generator-support clutter, the component-support clutter, or both, to be balanced.*

The first three clauses identify the component and symbolic-power structure of the cone. Clauses (iv)–(vii) show that the apparent support-equality and low-dimensional homology repairs leave the diagonal-root obstruction unchanged, while the final two clauses show that this obstruction is still visible to strong-cycle hypotheses.

Proof of Theorem 5.3. We first extend the base-family power identity and identify the five components of the cone. We then compute its generators, support complex, and diagonal roots, and use one off-diagonal parameter pair for the strengthened MPV repair. Finally, we prove the general strong-cycle bijection induced by coning and inspect the two resulting cycles. The component and power computations control (i)–(iii), the generator, cone, root, and off-diagonal computations control (iv)–(vii), and the cycle computation controls (viii)–(ix).

Put $S = k[a, b, c, d, z]$, $R = S[t]$, $I = I_{r,s} \subseteq S$, and $K = K_{r,s} \subseteq R$. For $1 \leq j \leq 4$, write Q_j^S and P_j^S for the ideals of S constructed in Construction 4.1; in R set $Q_j = Q_j^S R$ and $P_j = P_j^S R$. Put $Q_0 = P_0 = (t)$.

Extended base powers and their supports. Theorem 4.3 gives

$$I^m = \bigcap_{j=1}^{4} (Q_j^S)^m \qquad (m \geq 1).$$

Extending this identity to R and using Lemma 5.2(a) give

$$I^m R = \Big(\bigcap_{j=1}^{4} (Q_j^S)^m\Big) R = \bigcap_{j=1}^{4} ((Q_j^S)^m R) = \bigcap_{j=1}^{4} Q_j^m = D_m.$$

Taking $m = 1$ gives $D_1 = IR = \bigcap_{j=1}^{4} Q_j$. Consequently

$$\sqrt{IR} = \bigcap_{j=1}^{4} P_j.$$

A monomial belongs to $\sqrt{I}$ exactly when its support contains the support of one of the four minimal generators of I. Replacing each generator by its squarefree support therefore gives $\sqrt{I} = M$. On the other hand, taking radicals in $I = \bigcap_j Q_j^S$ gives

$$M = \bigcap_{j=1}^{4} P_j^S.$$

Extending the last intersection and again applying the lemma give

$$MR = \bigcap_{j=1}^{4} P_j,$$

and therefore $\sqrt{IR} = MR$.

The ideals I and M are generated in S. Their four displayed generators have the respective supports

$$\{a, c\}, \qquad \{b, d\}, \qquad \{a, b, z\}, \qquad \{c, d, z\},$$

which form an antichain. Thus, for the squarefree ideal M, the four displayed monomials are its minimal monomial generators. The lemma preserves the minimal monomial generators of both ideals under extension to R, and the two generating lists have the same supports. Hence $\mathcal{F}(IR) = \mathcal{F}(MR)$. This proves (i).

The five irreducible components and their radicals. Lemma 5.2(b), with $m = 1$, gives

$$K = tIR = (t) \cap IR = Q_0 \cap Q_1 \cap Q_2 \cap Q_3 \cap Q_4.$$

Every Q_i is generated by positive powers of the variables in its support, so Proposition 3.2 shows that Q_i is irreducible, P_i-primary, and has radical P_i.

The displayed intersection is irredundant. Omitting Q_0 leaves IR, which contains ac, whereas every monomial of $K = tIR$ is divisible by t. For $1 \le j \le 4$, irredundancy of $I = \bigcap_{k=1}^{4} Q_k^S$ supplies a monomial

$$u_j \in \bigcap_{\substack{1 \le k \le 4 \\ k \ne j}} Q_k^S \setminus Q_j^S.$$

Then tu_j belongs to Q_0 and to every Q_k with $k \ne j$, but not to Q_j, because the latter four components do not involve t. Thus no component can be omitted. The displayed decomposition is therefore the unique irredundant irreducible monomial decomposition of K, up to reordering, by the uniqueness recalled in Section 2.

The supports of P_0, P_1, P_2, P_3, P_4 are

$$\{t\}, \qquad \{a, b, z\}, \qquad \{a, d\}, \qquad \{b, c\}, \qquad \{c, d, z\}.$$

They form an antichain, so the five primes are pairwise distinct and pairwise incomparable. The chains

$$\begin{gathered} (0) \subset P_0, \\ (0) \subset (a) \subset (a, b) \subset P_1, \\ (0) \subset (a) \subset P_2, \\ (0) \subset (b) \subset P_3, \\ (0) \subset (c) \subset (c, d) \subset P_4 \end{gathered}$$

are chains of prime ideals. Consequently

$$\operatorname{ht} P_0 \ge 1, \qquad \operatorname{ht} P_1 \ge 3, \qquad \operatorname{ht} P_2 \ge 2, \qquad \operatorname{ht} P_3 \ge 2, \qquad \operatorname{ht} P_4 \ge 3.$$

For the reverse inequalities, put

$$(h_0, h_1, h_2, h_3, h_4) = (1, 3, 2, 2, 3).$$

The coordinate primes are

$$P_0 = (t), \qquad P_1 = (a, b, z), \qquad P_2 = (a, d), \qquad P_3 = (b, c), \qquad P_4 = (c, d, z).$$

Thus each P_i is generated by h_i of the displayed variables.

We first verify that the ambient polynomial ring R satisfies the Noetherian hypothesis. We prove the needed closure property directly. Let U be a Noetherian ring, let ξ be an indeterminate, and let $\mathfrak{c} \subseteq U[\xi]$ be an ideal. For $\ell \geq 0$, let $\Lambda_\ell \subseteq U$ consist of zero and the leading coefficients, with respect to ξ, of polynomials in $\mathfrak{c}$ of degree at most ℓ. Each Λ_ℓ is an ideal. For addition, apart from the zero cases, take $f, g \in \mathfrak{c}$ with leading coefficients α, β and with $\deg g \leq \deg f \leq \ell$. If $\alpha + \beta \neq 0$, then

$$f + \xi^{\deg f - \deg g} g$$

has leading coefficient $\alpha + \beta$; if $\alpha + \beta = 0$, the desired sum is zero. For $\gamma \in U$, either $\gamma\alpha = 0$, or $\gamma\alpha$ is the leading coefficient of γf. Hence closure under multiplication follows as well.

The ideals Λ_ℓ are ascending, and their union Λ is therefore an ideal of U. Since U is Noetherian, finitely many elements of Λ generate it. All those generators lie in one Λ_{ℓ_0}, unless $\Lambda = 0$, in which case take $\ell_0 = 0$. Hence

$$\Lambda_\ell = \Lambda_{\ell_0} \qquad (\ell \geq \ell_0).$$

For every $\ell \leq \ell_0$, the ideal Λ_ℓ is finitely generated. Multiplying witnesses in $\mathfrak{c}$ by suitable powers of ξ, choose polynomials

$$f_{\ell,1}, \ldots, f_{\ell,N_\ell} \in \mathfrak{c}$$

of degree exactly ℓ whose leading coefficients $\lambda_{\ell,1}, \ldots, \lambda_{\ell,N_\ell}$ generate Λ_ℓ; take $N_\ell = 0$ when $\Lambda_\ell = 0$. Let $\mathfrak{c}' \subseteq \mathfrak{c}$ be the ideal generated by this finite collection, indexed by $0 \leq \ell \leq \ell_0$.

We prove by induction on $\ell = \deg \varphi$ that every nonzero $\varphi \in \mathfrak{c}$ belongs to $\mathfrak{c}'$. Its leading coefficient λ lies in Λ_ℓ. If $\ell \leq \ell_0$, write $\lambda = \sum_{p=1}^{N_\ell} \mu_p \lambda_{\ell,p}$ with $\mu_p \in U$. Then

$$\varphi - \sum_{p=1}^{N_\ell} \mu_p f_{\ell,p}$$

has degree less than ℓ, unless it is zero. If $\ell > \ell_0$, use $\Lambda_\ell = \Lambda_{\ell_0}$, write λ using the $\lambda_{\ell_0,p}$, and subtract the corresponding combination of the polynomials $\xi^{\ell-\ell_0} f_{\ell_0,p}$. The induction hypothesis applies to the difference, so $\varphi \in \mathfrak{c}'$. Thus $\mathfrak{c} = \mathfrak{c}'$ is finitely generated, and $U[\xi]$ is Noetherian. The field k is Noetherian, since every nonzero ideal contains 1. Applying the closure property just proved while adjoining a, b, c, d, z, t in succession therefore shows that R is Noetherian.

Since P_i is prime, it is minimal over itself, which is the ideal generated by its displayed h_i variables. Apply Theorem 5.1 for each $0 \leq i \leq 4$, with $A = R$, with the h_i displayed generators of P_i, and with $\mathfrak{p} = P_i$. This gives

$$\operatorname{ht} P_0 \leq 1, \qquad \operatorname{ht} P_1 \leq 3, \qquad \operatorname{ht} P_2 \leq 2, \qquad \operatorname{ht} P_3 \leq 2, \qquad \operatorname{ht} P_4 \leq 3.$$

Together with the chains above, this proves that the componentwise height vector is

$$(1, 3, 2, 2, 3),$$

as asserted in (ii).

Proposition 3.2, applied to the displayed decomposition, gives

$$\operatorname{Ass}_R(R/K) = \{P_0, P_1, P_2, P_3, P_4\}.$$

The product of the five Q_i is contained in K, so every prime containing K contains one of the P_i. Pairwise incomparability therefore makes every P_i minimal over K and forces every prime minimal over K to equal one of them. Hence

$$\mathrm{Min}(K) = \mathrm{Ass}_R(R/K) = \{P_0, P_1, P_2, P_3, P_4\}.$$

Thus K has no embedded primes, and the five distinct component radicals make its irreducible decomposition MPV-minimal. This proves (ii).

Symbolic and ordinary powers. By (ii), every component radical occurs once in the canonical minimal primary decomposition. Thus the canonical P_i-primary component of Theorem 3.3 is precisely Q_i, and $\mathrm{Min}(K) = \mathrm{Ass}_R(R/K)$. For every $m \geq 1$, that theorem and the first step give

$$K^{(m)} = \bigcap_{i=0}^{4} Q_i^m = (t^m) \cap D_m = (t^m) \cap I^m R = t^m I^m R.$$

The penultimate equality comes from $D_m = I^m R$, and the last from Lemma 5.2(b) applied to the ideal I^m. Finally,

$$t^m I^m R = (tIR)^m = K^m.$$

This proves (iii), and hence K is Simis.

Generators, radical, and support equality. The minimal generators of I are listed in Theorem 4.3. Applying the last part of the lemma to $\mathfrak{a} = I$ gives

$$\mathrm{G}(K) = \{tac, tbd, tabz^r, tcdz^s\}.$$

The identity $K = (t) \cap IR$, together with (i) and Lemma 5.2(b) with $m = 1$, gives

$$\sqrt{K} = (t) \cap \sqrt{IR} = (t) \cap MR = tMR = (tac, tbd, tabz, tcdz).$$

The four generator supports, in either displayed list, are

$$\{a, c, t\}, \qquad \{b, d, t\}, \qquad \{a, b, z, t\}, \qquad \{c, d, z, t\}.$$

They form an antichain because adding the same new element t to each member of an antichain preserves incomparability. Hence the displayed monomials are minimal in both ideals and

$$\mathcal{G} = \mathcal{F}(\sqrt{K}) = \{\{a, c, t\}, \{b, d, t\}, \{a, b, z, t\}, \{c, d, z, t\}\}.$$

This proves (iv).

The contractible support cone. Let $X = \Delta(I)$, regarded as a complex on $\{a, b, c, d, z\}$. The facets of X are the members of $\mathcal{F}_{r,s}$, and the facets of $\Delta(K)$ are obtained by adjoining t to every one of them. A subset of $F \cup \{t\}$ either omits t and is a subset of F, or contains t and is the union of $\{t\}$ with a subset of F. Therefore

$$\Delta(K) = \{\sigma \mid \sigma \in X\} \cup \{\sigma \cup \{t\} \mid \sigma \in X\}.$$

This is the cone with apex t over X. Its geometric realization is contractible, so its positive-dimensional reduced homology over every coefficient field vanishes. In particular,

$$\beta_1(\Delta(K)) = 0.$$

This proves (v).

The exact root diagonal and the forced data. For a variable y, let C_y^{comp} be the occurrence set of y in the five irreducible components, and let O_y^{gen} be its positive-exponent occurrence set among the minimal generators of K. Directly from the two displays above,

$$C_a^{\mathrm{comp}} = C_b^{\mathrm{comp}} = C_c^{\mathrm{comp}} = C_d^{\mathrm{comp}} = C_t^{\mathrm{comp}} = \{1\}, \qquad C_z^{\mathrm{comp}} = \{s, r\},$$

and

$$O_a^{\mathrm{gen}} = O_b^{\mathrm{gen}} = O_c^{\mathrm{gen}} = O_d^{\mathrm{gen}} = O_t^{\mathrm{gen}} = \{1\}, \qquad O_z^{\mathrm{gen}} = \{r, s\}.$$

If $r \neq s$, then $|C_z^{\mathrm{comp}}| = 2$. Theorem 3.7, applied to the irreducible decomposition of K, therefore rules out every representation $K = J^{\delta_d}$ with J squarefree and d positive. That criterion assumes no source-Simis hypothesis, so it also rules out representations even when source Simisness is omitted.

Suppose now that $r = s = \rho$. Then every O_y^{gen} is a singleton, and every variable occurs in a minimal generator of K. The supportification in Theorem 3.6 is

$$J_* = (tac, tbd, tabz, tcdz) = tM.$$

The criterion first gives the existence of a squarefree positive-diagonal representation. Its uniqueness and weight assertions then give, with no unused coordinate,

$$J = J_* = tM, \qquad d = (1, 1, 1, 1, \rho, 1).$$

Thus a squarefree source exists exactly when $r = s$, and both the source and every weight coordinate are forced on the diagonal. This proves (vi).

Failure of the support-plus-homology repair. Instantiate the conclusions already proved at $(r, s) = (1, 2)$:

$$K_{1,2} = (tac, tbd, tabz, tcdz^2).$$

This ideal has no embedded primes, its five component radicals are pairwise distinct in its irreducible decomposition, and it is Simis. It therefore satisfies the three original MPV hypotheses. It also satisfies

$$\mathcal{F}(K_{1,2}) = \mathcal{F}(\sqrt{K_{1,2}}), \qquad \beta_1(\Delta(K_{1,2})) = 0,$$

while (vi) gives no squarefree standard-positive-diagonal source in R at all. A fortiori, it has no such source that is Simis. Adding support preservation and $\beta_1(\Delta(L)) \leq 1$ to the three original MPV hypotheses therefore does not yield the MPV conclusion. This proves (vii).

Strong cycles under adding and deleting the apex. Let $\mathcal{H}$ be a clutter on a finite vertex set V, let $t \notin V$, and put

$$\mathcal{H}^+ = \{E \cup \{t\} \mid E \in \mathcal{H}\}.$$

Adding t preserves both distinctness and incomparability of edges, so $\mathcal{H}^+$ is again a clutter, on $V \cup \{t\}$. Fix $q \geq 3$ and let

$$(v_1, E_1, v_2, E_2, \ldots, v_q, E_q, v_1)$$

be a strong q-cycle in $\mathcal{H}$, with cycle-vertex set V_0. Since $V_0 \subseteq V$ and $t \notin V$,

$$(E_i \cup \{t\}) \cap V_0 = E_i \cap V_0 = \{v_i, v_{i+1}\}$$

for every i. Thus adding t to every edge produces a strong q-cycle in $\mathcal{H}^+$.

Conversely, let

$$(v_1, F_1, v_2, F_2, \ldots, v_q, F_q, v_1)$$

be a strong q-cycle in $\mathcal{H}^+$, and write $F_i = E_i \cup \{t\}$ with $E_i \in \mathcal{H}$. Let V_0 be its cycle-vertex set. We claim that $t \notin V_0$. If $t = v_j$, choose $i \notin \{j-1, j\}$ modulo q, which is possible because $q \geq 3$. Then $v_j \notin \{v_i, v_{i+1}\}$, while $v_j = t \in F_i$, contradicting

$$F_i \cap V_0 = \{v_i, v_{i+1}\}.$$

Hence $t \notin V_0$. The edges $E_i = F_i \setminus \{t\}$ are distinct, and

$$E_i \cap V_0 = F_i \cap V_0 = \{v_i, v_{i+1}\}$$

for every i. They therefore form a strong q-cycle in $\mathcal{H}$. Deleting t from edges and adding t to edges are inverse operations on these cycles, preserve their length, and fix their vertex sequence. Consequently $\mathcal{H}$ is balanced if and only if $\mathcal{H}^+$ is balanced.

The support computation above gives

$$\mathcal{G} = (\mathcal{F}_{r,s})^+.$$

This proves the general cone assertion in (viii). The component supports also give

$$\mathcal{B} = \{\{t\}\} \cup \mathcal{A}_{r,s}.$$

The two surviving strong triangles. For the first displayed sequence set $W = \{a, c, z\}$. Its three edges satisfy

$$\{a, c, t\} \cap W = \{a, c\}, \qquad \{c, d, z, t\} \cap W = \{c, z\}, \qquad \{a, b, z, t\} \cap W = \{a, z\}.$$

The three vertices and three edges are distinct, so

$$a, \{a, c, t\}, c, \{c, d, z, t\}, z, \{a, b, z, t\}, a$$

is a strong 3-cycle in $\mathcal{G}$.

For the component-support cycle set $W' = \{a, d, z\}$. The three relevant members of $\mathcal{B}$ satisfy

$$\{a, d\} \cap W' = \{a, d\}, \qquad \{c, d, z\} \cap W' = \{d, z\}, \qquad \{a, b, z\} \cap W' = \{a, z\}.$$

Thus

$$a, \{a, d\}, d, \{c, d, z\}, z, \{a, b, z\}, a$$

is a strong 3-cycle in $\mathcal{B}$. The other two members $\{t\}$ and $\{b, c\}$ do not enter the defining conditions for this cycle. This proves (ix).

For every $r, s \in \mathbb{Z}_{>0}$, both $\mathcal{G}$ and $\mathcal{B}$ contain a strong 3-cycle. Hence no member of the family satisfies a balancedness requirement on the generator-support clutter, on the component-support clutter, or on both. The off-diagonal member $K_{1,2}$ contradicts the support-plus-$\beta_1 \leq 1$ repair. The two strong triangles show that the same member does not satisfy any of the balanced-support hypotheses listed above. This proves the final assertion. □

6 The no-strong-triangle theorem

We retain the notation of Section 2. The theorem below turns the absence of a strong 3-cycle into a rigidity statement for the exponents on an incident variable. The proof has three layers: Simisness controls the number of minimal generators on each support; a minimal unequal incidence can be chosen isolated unless a strong 3-cycle is already present; and an isolated unequal incidence carries an explicit monomial obstruction to the second symbolic identity.

Theorem 6.1 (no strong triangle)**.** *Let $I \subsetneq S$ be a nonzero monomial ideal, and let*

$$I = \bigcap_{j=1}^{m} Q_j$$

be its irredundant irreducible monomial decomposition. Assume that:

(a) $\mathcal{F}(I) = \mathcal{F}(\sqrt{I})$, and $\Delta(I)$ has the members of this common clutter as its facets;

(b) the radicals $\sqrt{Q_1}, \dots, \sqrt{Q_m}$ are pairwise distinct;

(c) $\operatorname{Ass}(S/I) = \operatorname{Min}(I)$;

(d) I is Simis;

(e) the clutter $\mathcal{H} = \mathcal{F}(I)$ has no strong 3-cycle.

For each $F \in \mathcal{H}$, let u_F denote the minimal monomial generator of I whose support is F. Then, for every variable x_i occurring in at least one member of $\mathcal{H}$, the exponent of x_i in u_F is independent of the facet $F \in \mathcal{H}$ containing x_i; denote its common positive value by d_i. For a variable x_i occurring in no member of $\mathcal{H}$, put $d_i = 1$, and let $d = (d_1, \dots, d_n)$. Moreover,

$$J \coloneqq \sqrt{I} = \Big(\Big\{ \prod_{x_i \in F} x_i \;\Big|\; F \in \mathcal{H} \Big\} \Big)$$

is a squarefree Simis monomial ideal in S, and

$$I = J^{\delta_d}.$$

Consequently the MPV conclusion holds for I. In particular, since every balanced clutter has no strong odd cycle and hence no strong 3-cycle, the same conclusion holds when hypothesis (e) is replaced by the stronger hypothesis that $\mathcal{H}$ is balanced.

Thus the symbolic equality at level two forces the support map on minimal generators to be injective before the combinatorics of the clutter is used. The combinatorial hypothesis is then needed only to compare exponents on distinct facets through a common variable.

Lemma 6.2 (one generator on each support)**.** *Let $I \subsetneq S$ be a nonzero monomial ideal, and let*

$$I = \bigcap_{h=1}^{m} Q_h$$

be its irredundant irreducible monomial decomposition. Assume that $\sqrt{Q_1}, \dots, \sqrt{Q_m}$ are pairwise distinct, that

$$\mathcal{F}(I) = \mathcal{F}(\sqrt{I}),$$

and that I is Simis. Then

$$|\{u \in \mathrm{G}(I) \,|\, \operatorname{supp}(u) = F\}| = 1 \qquad \text{for every } F \in \mathcal{F}(I).$$

Proof. The proof has three parts. We first record the componentwise form of the second symbolic power and show that the common support clutter is an antichain. We then construct a witness from two generators on one support. Finally, we prove that the witness belongs to $I^{(2)}$ but not to I^2.

Components and supports. We identify every set of variables with its set of indices. Write

$$Q_h = (x_i^{c_{h,i}} : i \in E_h), \qquad P_h = \sqrt{Q_h} = (x_i : i \in E_h),$$

where $E_h \neq \emptyset$ and every $c_{h,i}$ is positive. Proposition 3.2 gives

$$\operatorname{Ass}(S/I) = \{P_1, \dots, P_m\}.$$

In Theorem 3.3, the canonical primary component indexed by P_h is Q_h, because the P_h are pairwise distinct. Hence

$$I^{(2)} = \bigcap_{P_h \in \operatorname{Min}(I)} Q_h^2. \tag{6.1}$$

For a set of indices H, put $x_H = \prod_{i \in H} x_i$. Then

$$\sqrt{I} = (x_H : H \in \mathcal{F}(I)).$$

Indeed, if $x^\alpha \in \mathrm{G}(I)$ has support H and $N \geq \max_{i \in H} \alpha_i$, then $x^\alpha \mid x_H^N$, so $x_H \in \sqrt{I}$. Conversely, if $x^z \in \sqrt{I}$, then $x^{Nz} \in I$ for some $N > 0$. Some $x^\alpha \in \mathrm{G}(I)$ divides x^{Nz}; writing $H = \operatorname{supp}(x^\alpha)$, we have $H \subseteq \operatorname{supp}(x^z)$, and hence $x_H \mid x^z$. Among squarefree monomials, divisibility is inclusion of supports, so $\mathcal{F}(\sqrt{I})$ consists of the inclusion-minimal members of $\mathcal{F}(I)$. The assumed equality $\mathcal{F}(I) = \mathcal{F}(\sqrt{I})$ therefore gives

$$H, H' \in \mathcal{F}(I),\ H \subseteq H' \quad \Longrightarrow \quad H = H'. \tag{6.2}$$

Suppose now that some $F \in \mathcal{F}(I)$ supports $r \geq 2$ distinct minimal monomial generators. Enumerate them as

$$x^{a^1}, \dots, x^{a^r}, \qquad |a^1| \leq \cdots \leq |a^r|,$$

where $|a| = \sum_{i=1}^n a_i$, and put

$$\beta = a^1, \qquad \gamma = a^2, \qquad u = x^\beta, \qquad v = x^\gamma.$$

Every a^s is positive on F and zero outside F.

The degree-deficient witness. Because u and v are distinct minimal generators, neither divides the other. Thus there are $p, q \in F$ such that

$$\beta_p > \gamma_p, \qquad \beta_q < \gamma_q.$$

Indeed, absence of such a p would give $\beta_i \leq \gamma_i$ for every i and hence $u \mid v$; the assertion for q follows symmetrically. In particular $p \neq q$ and, for every $i \in F$, the exponents β_i and γ_i are positive.

For $i \in F$, put

$$m_i = \min\{\beta_i, \gamma_i\}, \qquad M_i = \max\{\beta_i, \gamma_i\}, \qquad e_i = \max\{2m_i, M_i\},$$

and put $e_i = 0$ for $i \notin F$. Let $f = x^e$. Then $\operatorname{supp}(f) = F$. Since

$$2m_i \leq m_i + M_i = \beta_i + \gamma_i, \qquad M_i < m_i + M_i = \beta_i + \gamma_i,$$

we have $e_i \le \beta_i + \gamma_i$ for every $i \in F$. At p, where $m_p = \gamma_p$ and $M_p = \beta_p$, both inequalities are strict:

$$2\gamma_p < \beta_p + \gamma_p, \qquad \beta_p < \beta_p + \gamma_p.$$

Therefore

$$e_p < \beta_p + \gamma_p, \qquad |e| < |\beta| + |\gamma|. \tag{6.3}$$

The witness is not in the ordinary square. At p we have $e_p = \max\{2\gamma_p, \beta_p\}$. Because $\beta_p > \gamma_p > 0$,

$$2\beta_p > \max\{2\gamma_p, \beta_p\} = e_p, \qquad \beta_p + \gamma_p > \max\{2\gamma_p, \beta_p\} = e_p.$$

Hence $u^2 \nmid f$ and $uv \nmid f$. At q we have $e_q = \max\{2\beta_q, \gamma_q\}$, and $\gamma_q > \beta_q > 0$ gives

$$2\gamma_q > \max\{2\beta_q, \gamma_q\} = e_q.$$

Thus $v^2 \nmid f$.

Suppose that $f \in I^2$. Since I^2 is generated by products of pairs from $\mathrm{G}(I)$, there are $x^\mu, x^\rho \in \mathrm{G}(I)$ such that $x^{\mu+\rho} \mid f$. Both $\operatorname{supp}(x^\mu)$ and $\operatorname{supp}(x^\rho)$ are contained in $\operatorname{supp}(f) = F$, so (6.2) forces

$$\operatorname{supp}(x^\mu) = \operatorname{supp}(x^\rho) = F.$$

Consequently $\mu = a^s$ and $\rho = a^t$ for some $s, t \in \{1, \dots, r\}$.

If $s, t \in \{1, 2\}$, then $x^{\mu+\rho}$ is one of u^2, uv, v^2, each of which we have just excluded. Otherwise at least one of s, t is at least 3. If one index is 1, the other is at least 3, and the degree ordering gives

$$|a^s| + |a^t| \ge |a^1| + |a^2| = |\beta| + |\gamma|.$$

If neither index is 1, both are at least 2 in the remaining case, so

$$|a^s| + |a^t| \ge 2|a^2| \ge |a^1| + |a^2|.$$

On the other hand, $x^{a^s + a^t} \mid f$ implies

$$|a^s| + |a^t| = |a^s + a^t| \le |e|,$$

contrary to (6.3). Therefore

$$f \notin I^2.$$

The witness lies in the second symbolic power. Fix h with $P_h \in \operatorname{Min}(I)$, and suppress h from the notation:

$$Q = (x_i^{w_i} : i \in E).$$

A monomial x^z belongs to Q exactly when $z_i \ge w_i$ for some $i \in E$. Since $u, v \in I \subseteq Q$, there are $j, \ell \in E$ such that

$$\beta_j \ge w_j, \qquad \gamma_\ell \ge w_\ell.$$

Both j and ℓ belong to F, because the corresponding coordinates of β and γ vanish outside F while the w_i are positive.

If $j = \ell$, then $w_j \le m_j$, and the jth exponent of f satisfies

$$e_j \ge 2m_j \ge 2w_j.$$

Thus $x_j^{2w_j} \in Q^2$ divides f. If $j \neq \ell$, then

$$e_j \geq M_j \geq \beta_j \geq w_j, \qquad e_\ell \geq M_\ell \geq \gamma_\ell \geq w_\ell.$$

Thus $x_j^{w_j} x_\ell^{w_\ell} \in Q^2$ divides f. In either case $f \in Q_h^2$, and (6.1) gives $f \in I^{(2)}$.

Because I is Simis, $f \in I^{(2)} = I^2$, contradicting the preceding part of the proof. Hence no support carries two distinct minimal generators. Every member of $\mathcal{F}(I)$ is the support of at least one minimal generator by definition, so each supports exactly one. □

Lemma 6.3 (isolation of an unequal incidence)**.** *Let $\mathcal{H}$ be a clutter on a finite vertex set V, and assume that $\mathcal{H}$ has no strong 3-cycle. Fix $z \in V$, put*

$$\mathcal{H}_z = \{E \in \mathcal{H} \,|\, z \in E\},$$

and let $\lambda : \mathcal{H}_z \to \Lambda$ be a labeling of the edges through z by elements of a set Λ. If λ is nonconstant, then there are $F, G \in \mathcal{H}_z$ such that

$$\lambda(F) \neq \lambda(G), \qquad \{E \in \mathcal{H} \,|\, E \subseteq F \cup G\} = \{F, G\}.$$

Proof. We choose an unequal-label pair with minimal union and examine a hypothetical third edge in that union. Each possible position of such an edge either contradicts minimality by producing a smaller unequal-label pair or completes a strong 3-cycle with three vertices drawn from the three disjoint sets P, Q, D.

Among all pairs $A, B \in \mathcal{H}_z$ with $\lambda(A) \neq \lambda(B)$, choose one for which $|A \cup B|$ is as small as possible. Such a pair exists because λ is nonconstant. Put

$$P = A \setminus B, \qquad Q = B \setminus A, \qquad D = A \cap B.$$

The sets P and Q are nonempty: if, for instance, P were empty, then A would be properly contained in B, contrary to the clutter property. Moreover $z \in D$.

Suppose that

$$C \in \mathcal{H}, \qquad C \subseteq A \cup B, \qquad C \notin \{A, B\}.$$

The edge C is not contained in A: otherwise $C \subsetneq A$, again contrary to the clutter property. Hence there is a vertex

$$p \in C \setminus A \subseteq Q.$$

Symmetrically, C is not contained in B, so there is a vertex

$$q \in C \setminus B \subseteq P.$$

The vertices p, q, z lie in the disjoint sets Q, P, D, respectively, and the edges A, C, B are distinct.

A third edge in the union contains the fixed vertex. If $z \notin C$, then

$$A \cap \{z, q, p\} = \{z, q\}, \qquad C \cap \{z, q, p\} = \{q, p\}, \qquad B \cap \{z, q, p\} = \{p, z\}.$$

Thus

$$(z, A, q, C, p, B, z)$$

is a strong 3-cycle, contradicting the hypothesis. Therefore

$$z \in C. \tag{6.4}$$

We shall repeatedly use the following observation. If

$$d \in D \setminus C, \qquad q' \in P \cap C, \qquad p' \in Q \cap C,$$

then d, q', p' are distinct and

$$A \cap \{d, q', p'\} = \{d, q'\}, \qquad C \cap \{d, q', p'\} = \{q', p'\}, \qquad B \cap \{d, q', p'\} = \{p', d\}.$$

Consequently

$$(d, A, q', C, p', B, d) \tag{6.5}$$

is a strong 3-cycle. This is impossible, so no such triple d, q', p' can occur.

A third edge must omit at least one nonshared vertex. Suppose that $P \cup Q \subseteq C$. If also $D \subseteq C$, then

$$C = A \cup B.$$

Since $Q \neq \emptyset$, this gives $A \subsetneq C$, contrary to the clutter property. Hence there is some $d \in D \setminus C$. Our original choices satisfy $q \in P \cap C$ and $p \in Q \cap C$, so the triple (d, q, p) is a triple of the kind just excluded. The cycle (6.5), with $q' = q$ and $p' = p$, is a contradiction. Thus C omits at least one vertex of $P \cup Q$.

A third edge cannot omit vertices from both nonshared parts. Suppose that there are

$$q_0 \in P \setminus C, \qquad p_0 \in Q \setminus C.$$

Then

$$B \cup C \subsetneq A \cup B$$

because $q_0 \notin B \cup C$, and

$$A \cup C \subsetneq A \cup B$$

because $p_0 \notin A \cup C$. By (6.4), the pairs B, C and A, C both consist of edges through z. If $\lambda(B) \neq \lambda(C)$, then B, C is an unequal-label pair with union $B \cup C$, contradicting the minimality of $|A \cup B|$. Hence $\lambda(B) = \lambda(C)$. The same argument applied to A, C gives $\lambda(A) = \lambda(C)$, and therefore $\lambda(A) = \lambda(B)$, a contradiction.

A third edge omitting vertices from exactly one nonshared part gives a triangle. There remain two symmetric cases. Suppose first that C omits a vertex of P but no vertex of Q, so that $Q \subseteq C$. Since B and C are distinct members of a clutter, B is not contained in C. Choose

$$d \in B \setminus C.$$

Because $Q \subseteq C$ and $B = D \cup Q$, we have $d \in D \setminus C$. The original vertices $q \in P \cap C$ and $p \in Q \cap C$ therefore form with d a triple of the kind excluded by (6.5). This contradiction eliminates the case in which exactly P has an omitted vertex.

Finally, suppose that C omits a vertex of Q but no vertex of P, so that $P \subseteq C$. Since A and C are distinct members of a clutter, A is not contained in C. Choose $d \in A \setminus C$. Then $d \in D \setminus C$, while the original $q \in P \cap C$ and $p \in Q \cap C$ again give the forbidden triple and the strong 3-cycle (6.5). This contradiction eliminates the last case.

Every possible position of C is now excluded. Hence no edge other than A or B is contained in $A \cup B$. Taking $F = A$ and $G = B$ proves the lemma. □

Lemma 6.4 (equal exponents on an isolated pair)**.** *Let $I \subsetneq S$ be a nonzero monomial ideal, and let*

$$I = \bigcap_{h=1}^{m} Q_h$$

be its irredundant irreducible monomial decomposition. Assume that the radicals $\sqrt{Q_1}, \ldots, \sqrt{Q_m}$ are pairwise distinct, that

$$\mathrm{Ass}(S/I) = \mathrm{Min}(I),$$

and that I is Simis. Let $\mathcal{H} = \mathcal{F}(I)$, assume that $\mathcal{H}$ is a clutter, and let $F \neq G$ be members of $\mathcal{H}$ satisfying

$$\{E \in \mathcal{H} \,|\, E \subseteq F \cup G\} = \{F, G\}.$$

Assume that F and G each support exactly one minimal monomial generator of I, and denote these generators by u_F and u_G. Then, for every $x_i \in F \cap G$, the exponent of x_i in u_F equals the exponent of x_i in u_G.

Proof. We identify sets of variables with their sets of indices. Write

$$u_F = x^a, \qquad u_G = x^b, \qquad \mathrm{supp}(u_F) = F, \qquad \mathrm{supp}(u_G) = G.$$

Because $\mathcal{H}$ is a clutter and $F \neq G$, both $F \setminus G$ and $G \setminus F$ are nonempty. Suppose, toward a contradiction, that some $x_r \in F \cap G$ satisfies $a_r \neq b_r$.

For each $i \in F \cap G$, put

$$m_i = \min\{a_i, b_i\}, \qquad M_i = \max\{a_i, b_i\}.$$

Define

$$f = \Big(\prod_{x_i \in F \cap G} x_i^{\max\{2m_i, M_i\}}\Big)\Big(\prod_{x_i \in F \setminus G} x_i^{a_i}\Big)\Big(\prod_{x_i \in G \setminus F} x_i^{b_i}\Big). \tag{6.6}$$

Every displayed exponent is positive: $a_i, b_i > 0$ on $F \cap G$, $a_i > 0$ on $F \setminus G$, and $b_i > 0$ on $G \setminus F$. Moreover f has no other variable in its support, so

$$\mathrm{supp}(f) = F \cup G.$$

We first exclude every possible divisor of f among the products of two minimal generators of I. Since I^2 is generated by such products, suppose that $x^c, x^{c'} \in \mathrm{G}(I)$ satisfy $x^{c+c'} \mid f$. Then

$$\mathrm{supp}(x^c), \mathrm{supp}(x^{c'}) \subseteq F \cup G.$$

Both supports belong to $\mathcal{H}$, so the isolation hypothesis forces each to be F or G. Since each of these supports carries exactly one minimal generator, each of x^c and $x^{c'}$ is u_F or u_G.

Choose $x_p \in F \setminus G$. At p, the exponent of f is a_p, while the exponent of u_F^2 is $2a_p > a_p$. Thus $u_F^2 \nmid f$. Choose $x_q \in G \setminus F$. At q, the exponent of f is b_q, while the exponent of u_G^2 is $2b_q > b_q$. Thus $u_G^2 \nmid f$.

At the shared coordinate r, the positive integers a_r and b_r are unequal. Hence

$$m_r = \min\{a_r, b_r\} > 0, \qquad m_r < M_r,$$

and therefore

$$2m_r < m_r + M_r = a_r + b_r, \qquad M_r < m_r + M_r = a_r + b_r.$$

Consequently

$$\max\{2m_r, M_r\} < a_r + b_r.$$

The exponent of x_r in $u_F u_G$ is $a_r + b_r$, while its exponent in f is the left-hand side of this inequality. Thus $u_F u_G \nmid f$. The possible ordered products of two members of $\{u_F, u_G\}$ are $u_F^2, u_F u_G, u_G u_F, u_G^2$, and the middle two coincide, so none divides f. Therefore

$$f \notin I^2.$$

It remains to test f against every component of the second symbolic power. Write

$$Q_h = (x_i^{w_i} : i \in E_h), \qquad P_h = \sqrt{Q_h}\,,$$

with every w_i positive. By Proposition 3.2,

$$\operatorname{Ass}(S/I) = \{P_1, \dots, P_m\}.$$

Since the P_h are pairwise distinct, Theorem 3.3 shows that the canonical primary component indexed by P_h is Q_h. Since $\operatorname{Ass}(S/I) = \operatorname{Min}(I)$, every P_h is minimal, and hence

$$I^{(2)} = \bigcap_{h=1}^{m} Q_h^2. \tag{6.7}$$

Fix h and suppress it from the notation, writing $Q = (x_i^{w_i} : i \in E)$. A monomial x^z belongs to Q exactly when $z_i \geq w_i$ for some $i \in E$. Because $u_F, u_G \in I \subseteq Q$, there are $j, \ell \in E$ such that

$$a_j \geq w_j, \qquad b_\ell \geq w_\ell.$$

The coordinates of x^a and x^b vanish outside F and G, respectively, while the w_i are positive. Therefore

$$j \in E \cap F, \qquad \ell \in E \cap G.$$

Suppose first that $j = \ell$. Then $x_j \in F \cap G$, and the two displayed inequalities give

$$w_j \leq \min\{a_j, b_j\} = m_j.$$

The exponent of x_j in (6.6) is $\max\{2m_j, M_j\}$, which is at least $2m_j$ and hence at least $2w_j$. Thus $x_j^{2w_j} \in Q^2$ divides f.

Suppose instead that $j \neq \ell$. We check the exponent of x_j in each of its two possible positions. If $x_j \in F \setminus G$, then that exponent is a_j, which is at least w_j. If $x_j \in F \cap G$, then it is

$$\max\{2m_j, M_j\} \geq M_j \geq a_j \geq w_j.$$

Thus the exponent of x_j in f is at least w_j. Symmetrically, if $x_\ell \in G \setminus F$, its exponent is $b_\ell \geq w_\ell$, while if $x_\ell \in F \cap G$, it is

$$\max\{2m_\ell, M_\ell\} \geq M_\ell \geq b_\ell \geq w_\ell.$$

Therefore $x_j^{w_j} x_\ell^{w_\ell}$ divides f. Since $j \neq \ell$, this monomial is the product of two minimal monomial generators of Q and belongs to Q^2. Hence $f \in Q_h^2$ in this case as well.

Varying h and using (6.7), we obtain $f \in I^{(2)}$. Simisness gives $I^{(2)} = I^2$, contrary to $f \notin I^2$. The assumption that some shared coordinate has unequal exponents is therefore false, and the proof is complete. □

Proof of Theorem 6.1. The three lemmas control successive parts of the conclusion. Lemma 6.2 supplies one minimal generator u_F on each facet. Lemma 6.3 turns any variation of an incident exponent into an isolated unequal pair, and Lemma 6.4 rules that pair out. The resulting uniform exponents then meet the generator root criterion, and the Simis property transfers from the weighted ideal back to its root.

Put $\mathcal{H} = \mathcal{F}(I)$. By hypothesis (a), $\mathcal{H}$ is a clutter. Lemma 6.2, applied using hypotheses (a), (b), and (d), shows that

$$\{u \in \mathrm{G}(I) \,|\, \mathrm{supp}(u) = F\} = \{u_F\} \qquad (F \in \mathcal{H}).$$

For each $F \in \mathcal{H}$, write

$$u_F = x^{\alpha(F)}.$$

Fix an index i for which x_i occurs in some member of $\mathcal{H}$, and put

$$\mathcal{H}_i = \{F \in \mathcal{H} \,|\, x_i \in F\}.$$

Define a labeling $\lambda_i : \mathcal{H}_i \to \mathbb{Z}_{>0}$ by

$$\lambda_i(F) = \alpha(F)_i.$$

Suppose that λ_i is nonconstant. By hypothesis (e) and Lemma 6.3, there are $F, G \in \mathcal{H}_i$ such that

$$\lambda_i(F) \neq \lambda_i(G), \qquad \{E \in \mathcal{H} \,|\, E \subseteq F \cup G\} = \{F, G\}.$$

Every facet carries a unique minimal generator, and hypotheses (b), (c), and (d) give the remaining hypotheses of Lemma 6.4. Since $x_i \in F \cap G$, that lemma gives

$$\lambda_i(F) = \alpha(F)_i = \alpha(G)_i = \lambda_i(G),$$

a contradiction. Hence λ_i is constant. Let d_i denote its common positive value. For every index i such that x_i occurs in no member of $\mathcal{H}$, put $d_i = 1$, and set $d = (d_1, \ldots, d_n)$. Thus the exponent of every used variable is independent of the incident facet, as asserted.

We now identify the root. Let

$$U = \{i \,|\, x_i \text{ occurs in some member of } \mathcal{H}\},$$

and let $D_i(I)$ be the occurrence set of x_i among the minimal monomial generators of I, as in Theorem 3.6. The preceding paragraph and the uniqueness of u_F give

$$D_i(I) = \begin{cases} \{d_i\}, & i \in U, \\ \emptyset, & i \notin U. \end{cases}$$

In particular $|D_i(I)| \leq 1$ for every i.

For $F \in \mathcal{H}$, put $x_F = \prod_{x_i \in F} x_i$. The ideal J_* attached to I in Theorem 3.6 is

$$J_* = \left(\left\{x^{\mathrm{supp}(\nu)} \;\middle|\; x^\nu \in \mathrm{G}(I)\right\}\right) = \left(\{x_F \,|\, F \in \mathcal{H}\}\right).$$

The second equality uses the bijection $F \leftrightarrow u_F$ between facets and minimal generators. By hypothesis (a),

$$\mathcal{F}(\sqrt{I}) = \mathcal{H}.$$

A squarefree monomial is determined by its support, so the minimal monomial generating set of the squarefree monomial ideal $\sqrt{I}$ is precisely $\{x_F \mid F \in \mathcal{H}\}$. Hence

$$J_* = \sqrt{I} = J.$$

The final assertion of Theorem 3.6, applied with the positive vector d, now gives

$$I = {J_*}^{\delta_d} = J^{\delta_d}.$$

Its coordinates on unused variables are immaterial, because no minimal generator of J involves those variables; the prescribed value $d_i = 1$ is therefore admissible.

Finally,

$$J^{\delta_d} = I$$

is Simis by hypothesis (d). By [MPV25, Corollary 5.5(b)], stated as Theorem 3.5, the ideals J and J^{δ_d} are Simis together. Hence J is Simis. Thus J is a squarefree Simis monomial ideal in S and $I = J^{\delta_d}$, which is the MPV conclusion for I.

If the hypothesis that $\mathcal{H}$ has no strong 3-cycle is replaced by the hypothesis that $\mathcal{H}$ is balanced, then $\mathcal{H}$ has no strong cycle of odd length and hence no strong 3-cycle. The case already proved therefore applies. □

7 A strict support-four family

We retain the notation of Section 2. This section first records the two positive cases to which the construction is compared and the balanced-to-Mengerian interface. We then construct a cyclic family whose simplicial complexes are minimally nonforest and prove that every positive diagonal weighting gives a strict quantified example.

Theorem 7.1 (BDK forest case)**.** *Let $L \subset k[y_1, \dots, y_N]$ be a nonzero proper monomial ideal satisfying the three original MPV hypotheses. Assume further that*

$$\mathcal{F}(L) = \mathcal{F}(\sqrt{L})$$

and that $\Delta(L)$ is a simplicial forest. Then the MPV conclusion holds for L.

The connected tree case is [BDK26a, Theorem 2.3]; the displayed forest consequence of the main results of that paper gives the disconnected case recorded above.

Theorem 7.2 (BDK support-three case)**.** *Let $L \subset k[y_1, \dots, y_N]$ be a nonzero proper monomial ideal. The ideal L is of* support 3 *if every $u \in \mathrm{G}(L)$ has the form*

$$u = y_i^a y_j^b y_k^c$$

with i, j, k pairwise distinct and $a, b, c \geq 1$; thus every minimal generator has support cardinality exactly 3. Every support-3 ideal satisfies

$$\mathcal{F}(L) = \mathcal{F}(\sqrt{L}).$$

If a support-3 ideal satisfies the three original MPV hypotheses, then the MPV conclusion holds for it. An ideal having a minimal generator of support cardinality 4 belongs neither to this class nor to the broader class in which every minimal generator has support cardinality at most 3.

This is [BDK26a, Section 3 and Theorem 3.10]. In the terminology used throughout this section, support 3 means support cardinality exactly 3, rather than the cumulative support-at-most-3 condition mentioned in the last sentence of the theorem.

Theorem 7.3 (balanced clutters are Mengerian)**.** *Let Δ be a finite simplicial complex. If its facet clutter $\mathcal{F}(\Delta)$ is balanced, then Δ is Mengerian.*

This implication is the main theorem of [FHO74], quoted in [HHTZ08, Section 2]; no theorem number for the original result is assigned there.

Theorem 7.4 (Mengerian complexes and Simis facet ideals)**.** *Let Δ be a finite simplicial complex on a vertex set V, let $S = k[V]$, and let $I = I(\Delta)$. The following are equivalent:*

(i) *Δ is Mengerian;*

(ii) *$I^{(q)} = I^q$ for every $q \geq 1$;*

(iii) *the associated graded ring $\operatorname{gr}_I(S)$ is reduced;*

(iv) *I is normally torsion-free, equivalently*

$$\operatorname{Ass}(S/I^q) = \operatorname{Ass}(S/I) \qquad \text{for every } q \geq 1.$$

These equivalences are [HHTZ08, Corollary 1.6]; the additional normal Cohen–Macaulay Rees consequence proved there is not used. Thus Theorems 7.3 and 7.4 form the Fulkerson–Hoffman–Oppenheim to Herzog–Hibi–Trung–Zheng lineage used below.

We also use two diagonal-weighting results already recorded in Section 3: weight-primary compatibility is [MPV25, Corollary 4.6], recorded in Theorem 3.4, while Simis transfer is [MPV25, Corollary 5.5(b)], recorded in Theorem 3.5. The latter result, rather than an independent argument, carries the Simis property from a squarefree root to each of its positive diagonal weightings.

Corollary 7.5 (balanced squarefree ideals are Simis)**.** *Let Δ be a finite simplicial complex on a vertex set V, let $S = k[V]$, and put $I = I(\Delta)$. If $\mathcal{F}(\Delta)$ is balanced, then I is Simis and normally torsion-free, and*

$$\operatorname{Ass}(S/I^q) = \operatorname{Ass}(S/I) \qquad \text{for every } q \geq 1.$$

Proof. Balancedness of $\mathcal{F}(\Delta)$ makes Δ Mengerian by Theorem 7.3. In Theorem 7.4, condition (i) therefore implies each of conditions (ii) and (iv). Condition (ii) says that I is Simis, while condition (iv) gives normal torsion-freeness and the displayed equality of associated primes. □

Definition 7.6 (strict quantified example)**.** Let L be a nonzero proper monomial ideal. The *balanced-support strengthening* for L consists of the three original MPV hypotheses,

$$\mathcal{F}(L) = \mathcal{F}(\sqrt{L}),$$

and balancedness of $\mathcal{F}(L)$. The *one-facet-deletion strengthening* consists of the three original MPV hypotheses, the same support equality, and the requirement that

$$\langle \mathcal{F}(L) \setminus \{F\} \rangle$$

is a simplicial forest for every $F \in \mathcal{F}(L)$ with $\mathcal{F}(L)\setminus\{F\} \neq \varnothing$. Given either proposed strengthening, L is a *strict quantified example* if it satisfies the complete antecedent of that strengthening and satisfies the MPV conclusion, while for each of Theorems 7.1 and 7.2 it fails at least one antecedent which that theorem adds to the three original MPV hypotheses.

Construction 7.7 (even-cycle family)**.** Fix an integer $m \geq 2$, put $n = 2m$, and read all subscripts below modulo n. Let

$$V_m = \{x_1, \ldots, x_n, y_1, \ldots, y_n, z_1, \ldots, z_n\}$$

be a set of $3n = 6m$ algebraically independent variables over k, and put $S_m = k[V_m]$. For $1 \leq i \leq n$, put

$$E_i = \{x_i, x_{i+1}, y_i, z_i\}, \qquad \mathcal{H}_m = \{E_1, \ldots, E_n\}, \qquad \Delta_m = \langle E_1, \ldots, E_n \rangle.$$

For $F \subseteq V_m$, write $u_F = \prod_{v \in F} v$. Fix an arbitrary family $d = (d_v)_{v \in V_m} \in {\mathbb{Z}_{>0}}^{V_m}$, put

$$u_F{}^{\langle d \rangle} = \prod_{v \in F} v^{d_v}, \qquad L^{\langle d \rangle} = L^{\delta_d}$$

for every monomial ideal $L \subseteq S_m$, and define

$$J_m = (u_{E_1}, \ldots, u_{E_n}), \qquad I_{m,d} = \left(u_{E_1}{}^{\langle d \rangle}, \ldots, u_{E_n}{}^{\langle d \rangle}\right).$$

Finally, let $\mathcal{C}_m = b(\mathcal{H}_m)$ and, for $C \in \mathcal{C}_m$, put

$$P_C = (v : v \in C), \qquad Q_C(d) = (v^{d_v} : v \in C).$$

Each facet has two private vertices, while consecutive facets meet in one x-variable and nonconsecutive facets are disjoint. This incidence pattern is the source of both balancedness and minimal nonforestness.

Theorem 7.8 (even-cycle family)**.** *Retain Construction 7.7. For every $m \geq 2$ and every $d \in {\mathbb{Z}_{>0}}^{V_m}$, the following hold:*

(1) The displayed monomials $u_{E_i}{}^{\langle d \rangle}$ for $1 \leq i \leq n$ are precisely the minimal monomial generators of $I_{m,d}$. Thus $I_{m,d}$ has exactly $2m$ minimal generators and

$$\mathcal{F}(I_{m,d}) = \mathcal{H}_m.$$

(2)

$$\sqrt{I_{m,d}} = J_m = I_{m,\mathbf{1}}, \qquad \mathcal{F}(I_{m,d}) = \mathcal{F}(\sqrt{I_{m,d}}) = \mathcal{H}_m.$$

(3) The decompositions

$$J_m = \bigcap_{C \in \mathcal{C}_m} P_C, \qquad I_{m,d} = \bigcap_{C \in \mathcal{C}_m} Q_C(d)$$

are irredundant, their component radicals P_C are pairwise distinct, and

$$\operatorname{Ass}_{S_m}(S_m/I_{m,d}) = \operatorname{Min}(I_{m,d}) = \{P_C \,|\, C \in \mathcal{C}_m\}.$$

Thus $I_{m,d}$ has no embedded primes and its irreducible decomposition is MPV-minimal.

(4)

$$I_{m,d} = {J_m}^{\delta_d}.$$

Both J_m and $I_{m,d}$ are Simis and normally torsion-free, and for every $q \geq 1$

$${I_{m,d}}^{(q)} = {I_{m,d}}^q, \qquad \operatorname{Ass}_{S_m}(S_m/{I_{m,d}}^q) = \{P_C \,|\, C \in \mathcal{C}_m\}.$$

(5) *$\mathcal{H}_m$ is a balanced 4-uniform clutter, while Δ_m is not a simplicial forest.*

(6) *Every nonempty proper facet subcomplex of Δ_m is a simplicial forest. Equivalently,*

$$\Gamma_i = \langle \mathcal{H}_m \setminus \{E_i\} \rangle$$

is a simplicial forest for every $1 \le i \le n$, and Δ_m is minimally nonforest.

Consequently, $\{I_{m,d} \,|\, m \ge 2,\ d \ne \mathbf{1}\}$ is an infinite family of nonsquarefree, no-embedded-prime, MPV-minimal, Simis, normally torsion-free ideals all of whose minimal generators have support cardinality 4. Every member satisfies the antecedent and conclusion of each of the balanced-support and one-facet-deletion strengthenings. Every member fails the forest antecedent of Theorem 7.1 and fails both the exact-support-3 and cumulative-support-at-most-3 antecedents relevant to Theorem 7.2. Hence every such member is a strict quantified example for both strengthenings in the sense of Definition 7.6. Finally, for each $m \ge 2$ let $d_{y_1}^{(m)} = 2$ and $d_v^{(m)} = 1$ for all $v \ne y_1$. As m varies, the ambient pairs $(S_m, I_{m,d^{(m)}})$ are pairwise nonisomorphic, because S_m has $6m$ variables.

Proof. Fix $m \ge 2$ and $d \in {\mathbb{Z}_{>0}}^{V_m}$, and put $n = 2m$, so that $n \ge 4$. The proof first determines the cycles, leaves, and facet deletions of Δ_m. It then derives both irreducible decompositions from minimal vertex covers, transports the squarefree decomposition by the diagonal weighting, and finally obtains the Simis and associated-prime assertions through the cited balanced and weighted interfaces. The facet-incidence step proves (5) and (6), the next three steps prove (1)–(4), and the last step assembles the quantified strictness conclusion.

Facet incidence controls both cycles and leaves. For every index i, the variables y_i and z_i occur only in E_i, while x_i occurs exactly in E_{i-1} and E_i. Hence every E_i has cardinality 4, and $y_i \in E_i \setminus E_j$ whenever $j \ne i$. Thus $\mathcal{H}_m$ is a 4-uniform clutter with $n = 2m$ distinct members. For distinct i and j,

$$E_i \cap E_j = \begin{cases} \{x_i\}, & j = i-1, \\ \{x_{i+1}\}, & j = i+1, \\ \varnothing, & j \notin \{i-1, i, i+1\}. \end{cases} \tag{1}$$

Here $i-1$, i, and $i+1$ are distinct modulo n because $n \ge 4$.

We show that every strong cycle has length n. Let

$$(v_1, F_1, v_2, F_2, \ldots, v_q, F_q, v_1)$$

be a strong q-cycle in $\mathcal{H}_m$, with the indices of its vertices and edges read modulo q. Each v_t lies in the two distinct edges F_{t-1} and F_t. No y_i or z_i can therefore occur among the v_t, since each such variable belongs to only one edge of $\mathcal{H}_m$. Write $v_t = x_{p_t}$ for distinct residues $p_t \in \mathbb{Z}/n\mathbb{Z}$.

If $F_t = E_{j_t}$, then F_t contains x_{p_t} and $x_{p_{t+1}}$, while its only x-variables are x_{j_t} and x_{j_t+1}. Consequently

$$\{p_t, p_{t+1}\} = \{j_t, j_t + 1\}, \qquad \varepsilon_t := p_{t+1} - p_t \in \{1, -1\} \quad \text{in } \mathbb{Z}/n\mathbb{Z}.$$

Two consecutive signs cannot differ. For example, if $\varepsilon_t = 1$ and $\varepsilon_{t+1} = -1$, then

$$p_{t+2} = p_{t+1} - 1 = p_t,$$

contrary to the distinctness of the cycle vertices; the other change of sign is identical, and the same calculation applies across the cyclic boundary. Hence all ε_t are equal. After reversing the cycle if necessary, assume that they all equal 1. Then

$$p_t \equiv p_1 + (t-1) \pmod{n} \qquad (1 \le t \le q+1),$$

where $p_{q+1} = p_1$. Closure gives $n \mid q$, while the distinctness of $p_1, \dots, p_q$ gives $q \le n$. Thus $q = n = 2m$, which is even. Therefore $\mathcal{H}_m$ contains no strong odd cycle and is balanced.

The full complex Δ_m has no leaf. Indeed, suppose that E_i is a leaf with joint E_j. Applying the leaf condition to E_{i-1} and using (1) gives

$$\{x_i\} = E_i \cap E_{i-1} \subseteq E_i \cap E_j.$$

Thus $x_i \in E_j$. Since x_i lies only in E_{i-1} and E_i, and $j \neq i$, this forces $j = i-1$. Applying the leaf condition to E_{i+1} similarly forces $j = i+1$. These conclusions contradict $i-1 \neq i+1$ modulo n. Hence no facet of Δ_m is a leaf, and Δ_m, being a nonempty facet subcomplex of itself, is not a forest.

It remains to prove that every nonempty proper facet subcomplex is a forest. Let $\emptyset \neq \mathcal{A} \subsetneq \mathcal{H}_m$, and after a cyclic relabeling suppose that $E_n \notin \mathcal{A}$. Let $\mathcal{B}$ be a nonempty subset of $\mathcal{A}$; because $\mathcal{H}_m$ is a uniform clutter, the members of $\mathcal{B}$ are the facets of the subcomplex it generates. List the remaining cyclic indices linearly as $1, \dots, n-1$, and choose a maximal consecutive block

$$E_a, E_{a+1}, \dots, E_b$$

of members of $\mathcal{B}$. At the endpoints we use that $E_0 = E_n$ is absent.

Suppose first that $a = b$. Maximality, including the endpoint convention, shows that neither E_{a-1} nor E_{a+1} belongs to $\mathcal{B}$. By (1), E_a is disjoint from every other member of $\mathcal{B}$. It is therefore a leaf: if it is the only facet this is part of the definition, and otherwise any other facet can serve as a joint. Suppose instead that $a < b$. Then $E_{a-1} \notin \mathcal{B}$ and $E_{a+1} \in \mathcal{B}$, while every other member of $\mathcal{B}$ is different from both neighbors of E_a. Equation (1) gives

$$E_a \cap E_{a+1} = \{x_{a+1}\}, \qquad E_a \cap H = \varnothing \quad \text{for } H \in \mathcal{B} \setminus \{E_a, E_{a+1}\}.$$

Hence E_a is a leaf with joint E_{a+1}. Thus every nonempty facet subcomplex of $\langle \mathcal{A} \rangle$ has a leaf, so $\langle \mathcal{A} \rangle$ is a forest. Since $\mathcal{A}$ was an arbitrary nonempty proper subset of $\mathcal{H}_m$, every nonempty proper facet subcomplex of Δ_m is a forest. In particular every Γ_i is a forest, and Δ_m is minimally nonforest. The displayed one-facet formulation is equivalent because every nonempty proper facet subcomplex omits some E_i and is therefore a facet subcomplex of Γ_i. This proves (5) and (6).
The generators acquire arbitrary positive weights without changing their supports. Put

$$g_i = {u_{E_i}}^{\langle d \rangle} \qquad (1 \le i \le n).$$

Since the E_i are distinct 4-sets, no one is contained in another. If $i \neq j$ and g_j divided g_i, then $E_j \subseteq E_i$, because all weights are positive. This is impossible. Hence $g_1, \dots, g_n$ are pairwise nondivisible and, as they generate $I_{m,d}$, are precisely its minimal monomial generators. Their supports are the E_i, so $\mathcal{F}(I_{m,d}) = \mathcal{H}_m$ and $\Delta(I_{m,d}) = \Delta_m$. The same antichain argument, applied directly to the defining generators $u_{E_1}, \dots, u_{E_n}$, shows that these are precisely the minimal generators of J_m, that $\mathcal{F}(J_m) = \mathcal{H}_m$, and that $\Delta(J_m) = \Delta_m$. Hence $J_m = I_{m,\mathbf{1}} = I(\Delta_m)$.

Every monomial of J_m is divisible by one of the u_{E_i}, and the weight map $\alpha \mapsto \delta_d(\alpha)$ is coordinatewise positive and additive. Therefore the defining generating set of ${J_m}^{\delta_d}$ consists of monomials each divisible by some g_i and contains every g_i. Hence

$$I_{m,d} = {J_m}^{\delta_d}. \tag{2}$$

The same argument applied to q-fold products of monomials of J_m gives, for every $q \geq 1$,

$${I_{m,d}}^q = ({J_m}^q)^{\delta_d}. \tag{3}$$

Indeed, a monomial of ${J_m}^q$ is divisible by a product $u_{E_{i_1}} \cdots u_{E_{i_q}}$, and the weighted image of such a product is $g_{i_1} \cdots g_{i_q}$. Conversely, each such product of the $u_{E_{i_r}}$ lies in ${J_m}^q$. These are exactly the two divisibility inclusions needed for (3).

Minimal vertex covers determine both irreducible decompositions. Let M be a monomial and let $A = \operatorname{supp}(M)$. Because J_m is generated by the squarefree monomials u_{E_i},

$$M \in J_m \quad \Longleftrightarrow \quad E_i \subseteq A \text{ for some } i.$$

The last condition is equivalent to $A \cap C \neq \varnothing$ for every $C \in \mathcal{C}_m$. One direction follows because every vertex cover meets every E_i. Conversely, if no E_i is contained in A, then $V_m \setminus A$ meets every E_i and hence is a vertex cover; it contains a minimal vertex cover disjoint from A. Since

$$M \in P_C \quad \Longleftrightarrow \quad A \cap C \neq \varnothing,$$

we obtain

$$J_m = \bigcap_{C \in \mathcal{C}_m} P_C. \tag{4}$$

The members of $\mathcal{C}_m$ form an antichain. Fix $C \in \mathcal{C}_m$, and for every $D \neq C$ choose

$$v_{C,D} \in D \setminus C.$$

The monomial

$$w_C = \prod_{D \neq C} v_{C,D}$$

lies in every P_D with $D \neq C$, while its support is disjoint from C, so $w_C \notin P_C$. Thus (4) is irredundant. Each $P_C = Q(C, \mathbf{1})$ is irreducible and P_C-primary by Proposition 3.2, which also gives

$$\operatorname{Ass}_{S_m}(S_m/J_m) = \{P_C \,|\, C \in \mathcal{C}_m\}.$$

The covers, and hence the primes P_C, are pairwise incomparable. To see that all these primes are minimal, let P be a prime containing $J_m = \bigcap_D P_D$. Since $\prod_D P_D \subseteq J_m \subseteq P$, primality gives $P_D \subseteq P$ for some D. If P is minimal over J_m, then P_D is a prime containing J_m and $P_D \subseteq P$, so $P = P_D$. If instead $P \subseteq P_C$ for some C, then $P_D \subseteq P_C$, so $D = C$ and $P = P_C$. Thus every minimal prime is one of the P_C, while no P_C properly contains another prime over J_m. Thus

$$\operatorname{Ass}_{S_m}(S_m/J_m) = \operatorname{Min}(J_m) = \{P_C \,|\, C \in \mathcal{C}_m\}. \tag{5}$$

In particular J_m has no embedded primes.

Because the radicals in (4) are pairwise distinct, Theorem 3.3 identifies (4) with the canonical minimal primary decomposition of J_m. For every C,

$${P_C}^{\delta_d} = (v^{d_v} : v \in C) = Q_C(d).$$

Applying Theorem 3.4 to (2) and (4) therefore gives

$$I_{m,d} = \bigcap_{C \in \mathcal{C}_m} Q_C(d) \tag{6}$$

and

$$\operatorname{Ass}_{S_m}(S_m/I_{m,d}) = \operatorname{Ass}_{S_m}(S_m/J_m) = \{P_C \,|\, C \in \mathcal{C}_m\}\,. \tag{7}$$

The second decomposition is irredundant as well. Retain the elements $v_{C,D} \in D \setminus C$ and put

$$\widetilde{w}_C = \prod_{D \neq C} v_{C,D}^{\,d_{v_{C,D}}}.$$

For every $D \neq C$, the factor indexed by D puts the variable $v_{C,D}$ in $\widetilde{w}_C$ with exponent at least $d_{v_{C,D}}$, so $\widetilde{w}_C \in Q_D(d)$. Every variable occurring in $\widetilde{w}_C$ lies outside C, so $\widetilde{w}_C \notin Q_C(d)$. Thus no term of (6) can be omitted. By Proposition 3.2, each $Q_C(d)$ is an irreducible P_C-primary monomial ideal. Consequently (6) is the irredundant irreducible monomial decomposition of $I_{m,d}$, and its component radicals are pairwise distinct; the decomposition is therefore MPV-minimal. The primes in (7) remain pairwise incomparable. Since every minimal prime is associated, (7) gives

$$\operatorname{Ass}_{S_m}(S_m/I_{m,d}) = \operatorname{Min}(I_{m,d}) = \{P_C \,|\, C \in \mathcal{C}_m\}\,,$$

so $I_{m,d}$ has no embedded primes. This proves (3).

Taking radicals in (6) and using (4) gives

$$\sqrt{I_{m,d}} = \bigcap_{C \in \mathcal{C}_m} \sqrt{Q_C(d)} = \bigcap_{C \in \mathcal{C}_m} P_C = J_m.$$

We already saw that $J_m = I_{m,\mathbf{1}}$ and that both ideals have generator-support clutter $\mathcal{H}_m$. Together with the minimal-generator paragraph, this proves (1), (2), and the displayed weighting equality in (4).

The cited balanced and diagonal interfaces supply the Simis and power assertions. Since $\mathcal{H}_m$ is the facet clutter of Δ_m and is balanced, Corollary 7.5, applied to $J_m = I(\Delta_m)$, gives

$$J_m{}^{(q)} = J_m{}^q, \qquad \operatorname{Ass}_{S_m}(S_m/J_m{}^q) = \{P_C \,|\, C \in \mathcal{C}_m\} \tag{8}$$

for every $q \geq 1$, and shows that J_m is normally torsion-free.

By (2) and Theorem 3.5, $I_{m,d}$ is Simis because J_m is Simis. Equivalently,

$$I_{m,d}{}^{(q)} = I_{m,d}{}^q \qquad (q \geq 1).$$

For the associated primes of its powers, apply Theorem 3.4 to the monomial ideal $L = J_m{}^q$. Using (3) and (8), we obtain

$$\operatorname{Ass}_{S_m}(S_m/I_{m,d}{}^q) = \operatorname{Ass}_{S_m}(S_m/(J_m{}^q)^{\delta_d}) = \operatorname{Ass}_{S_m}(S_m/J_m{}^q) = \{P_C \,|\, C \in \mathcal{C}_m\} \tag{9}$$

for every $q \geq 1$. At $q = 1$ this is (7), so $I_{m,d}$ is normally torsion-free as well. The squarefree Simis, normal-torsion-freeness, and power-associated-prime assertions for J_m come from the FHO74/HHTZ08 chain through Theorems 7.3 and 7.4; the descendant assertions for $I_{m,d}$ come from the MPV25 Simis transfer [MPV25, Corollary 5.5(b)] and weight-primary compatibility [MPV25, Corollary 4.6]. In particular, J_m is not presented here as a new or independent squarefree Simis construction. This completes the proof of (4).

Every nonsquarefree member is a strict quantified example. Assume now that $d \neq \mathbf{1}$, and choose $v \in V_m$ with $d_v > 1$. The incidence description at the beginning of the proof shows that v lies in some facet $F = E_i$. We claim that $u_F \notin I_{m,d}$. If one of the minimal generators ${u_{E_j}}^{\langle d \rangle}$ divided u_F, then $E_j \subseteq F$. Both sets have cardinality 4, so $E_j = F$ and $j = i$; but the exponent of v in ${u_F}^{\langle d \rangle}$ is $d_v > 1$, while its exponent in u_F is 1. Hence $u_F \in J_m \setminus I_{m,d}$. Since J_m is the radical of $I_{m,d}$, the latter is not radical and therefore is not a squarefree monomial ideal.

Every minimal generator of $I_{m,d}$ has support cardinality 4. The properties already proved give the three original MPV hypotheses for $I_{m,d}$: (3) gives the absence of embedded primes and MPV-minimality, while (4) gives Simisness. Parts (2), (4), (5), and (6) give, respectively, the support equality, the MPV conclusion with the squarefree Simis root J_m in the same ring, balancedness of the common support clutter, and the one-facet-deletion forest condition. Thus every member with $d \neq \mathbf{1}$ satisfies both proposed strengthenings and their conclusion.

For Theorem 7.1, the member fails the forest antecedent because $\Delta(I_{m,d}) = \Delta_m$ is not a simplicial forest by (5), although the support equality required there does hold. For Theorem 7.2, every minimal generator has support cardinality 4; the member consequently fails the exact-support-3 antecedent and also belongs neither to that class nor to the broader support-at-most-3 class. As $m \geq 2$ and $d \neq \mathbf{1}$ were arbitrary, every member of the displayed family is a strict quantified example for both strengthenings.

Finally, for each $r \geq 2$ let $e^{(r)} \in {\mathbb{Z}_{>0}}^{V_r}$ be defined by

$$e^{(r)}_{y_1} = 2, \qquad e^{(r)}_v = 1 \quad (v \neq y_1),$$

and put

$$L_r = I_{r,e^{(r)}} \subseteq S_r.$$

Each L_r is one of the nonsquarefree examples above. The ambient rings S_r have $6r$ variables and hence distinct Krull dimensions, so the ambient examples (S_r, L_r) are pairwise nonisomorphic as r varies. This proves that the displayed family is infinite without asserting any classification of distinct weights for a fixed value of r. □

8 Boundaries and nonnecessity

We retain the notation of the preceding sections. This section determines what remains of the root criteria when their incidence bounds are weakened, and then separates several sufficient antecedents by an explicit component-incidence family. We begin with two elementary calculations. The first identifies the contribution of one irreducible component to a symbolic power; the second is the finite monomial obstruction underlying the sharp intervening-support bound.

Lemma 8.1 (pure-power contraction)**.** *Let Y and Z be disjoint finite sets of variables, with $Y \neq \varnothing$, put $S = k[Y \sqcup Z]$, and fix a family $e = (e_y)_{y \in Y}$ of positive integers. For a nonempty subset $E \subseteq Y$, put*

$$Q_E(e) = (y^{e_y} : y \in E)S, \qquad P_E = (y : y \in E)S.$$

Then, for every $t \geq 1$, the ideal $Q_E(e)^t$ is P_E-primary and

$$(Q_E(e)^t S_{P_E}) \cap S = Q_E(e)^t.$$

Proof. Write $Y_0 = E$ and $Z_0 = (Y \setminus E) \sqcup Z$. It suffices to treat the case $E = Y$, after replacing Y, Z by Y_0, Z_0 and adjoining the variables in $Y \setminus E$ to Z_0. Thus put

$$Q = (y^{e_y} : y \in Y)S, \qquad P = (Y)S, \qquad M = Q^t.$$

Let $M_0 = Q_0^t \subseteq k[Y]$, where $Q_0 = (y^{e_y} : y \in Y)$. The ring $B = k[Y]/M_0$ is a finite-dimensional local k-algebra. Its maximal ideal $\mathfrak{n} = (Y)B$ is nilpotent, because the image of every $y \in Y$ is nilpotent. Identify

$$S/M \cong B \otimes_k k[Z]$$

and filter this module by

$$F^q = \mathfrak{n}^q B \otimes_k k[Z] \qquad (q \geq 0).$$

The filtration is finite, and every F^q/F^{q+1} is a finite free $k[Z]$-module.

Let $s \in S \setminus P$. Write $s = s_0 + s_+$, where $s_0 \in k[Z]$ and $s_+ \in (Y)S$. Then $s_0 \neq 0$. Multiplication by s on F^q/F^{q+1} is multiplication by s_0, because the variables in Y act by zero on that subquotient; it is therefore injective. If $sg = 0$ for some nonzero $g \in S/M$, choose q with $g \in F^q \setminus F^{q+1}$. Injectivity on F^q/F^{q+1} gives $sg \in F^q \setminus F^{q+1}$, a contradiction. Thus every $s \notin P$ is a nonzerodivisor on S/M.

If $g \in MS_P \cap S$, some $s \notin P$ satisfies $sg \in M$, and the preceding injectivity gives $g \in M$. This proves the displayed contraction. Moreover, if $fg \in M$ and $g \notin P$, then $f \in M$. Since $\sqrt{M} = P$, this is exactly the condition that M be P-primary. □

Lemma 8.2 (core square obstruction)**.** *Let $R = k[y_0, y_1, y_2]$, let $\lambda, \mu, \alpha, \beta$ be positive integers with $\lambda < \mu$, and put*

$$\mathfrak{A} = (y_0^\lambda, y_1^\alpha), \qquad \mathfrak{B} = (y_0^\mu, y_2^\beta).$$

Then

$$(\mathfrak{A} \cap \mathfrak{B})^2 \subsetneq \mathfrak{A}^2 \cap \mathfrak{B}^2.$$

If γ, δ are positive integers and either

$$\mathfrak{C} = (y_1^\gamma, y_2^\delta),$$

or θ is a positive integer and

$$\mathfrak{C} = (y_0^\theta, y_1^\gamma, y_2^\delta),$$

then

$$(\mathfrak{A} \cap \mathfrak{B} \cap \mathfrak{C})^2 \subsetneq \mathfrak{A}^2 \cap \mathfrak{B}^2 \cap \mathfrak{C}^2.$$

Proof. The proof exhibits, in each incidence case, a monomial in the intersection of the squares which is not in the square of the intersection.

The two-component witness. Put $M = \max\{2\lambda, \mu\}$ and $f = y_0^M y_2^\beta$. Since $y_0^{2\lambda} \mid f$ and $y_0^\mu y_2^\beta \mid f$, we have $f \in \mathfrak{A}^2 \cap \mathfrak{B}^2$. The only generators of $\mathfrak{A} \cap \mathfrak{B}$ that divide f are

$$y_0^\mu \qquad \text{and} \qquad y_0^\lambda y_2^\beta.$$

Their pairwise products have exponent triples

$$(2\mu, 0, 0), \qquad (\mu + \lambda, 0, \beta), \qquad (2\lambda, 0, 2\beta)$$

in the variables y_0, y_1, y_2. None divides f: the first coordinates 2μ and $\mu + \lambda$ exceed M, because $\lambda < \mu$, while $2\beta > \beta$ in the third product. Hence $f \notin (\mathfrak{A} \cap \mathfrak{B})^2$.

The three-component witness with no small exponent at y_0. Now let $\mathfrak{C}$ be present, and first suppose either that it has the two-generator form, or that it has the three-generator form with $\theta > \mu$. Put

$$\widehat{\alpha} = \max\{\alpha, \gamma\}, \qquad \widehat{\beta} = \max\{\beta, \delta\},$$

and

$$F = y_0^{\mu} y_1^{\widehat{\alpha}} y_2^{\widehat{\beta}}.$$

This monomial lies in $\mathfrak{A}^2 \cap \mathfrak{B}^2 \cap \mathfrak{C}^2$: it is respectively divisible by

$$y_0^{\lambda} y_1^{\alpha}, \qquad y_0^{\mu} y_2^{\beta}, \qquad y_1^{\gamma} y_2^{\delta}.$$

Put $K = \mathfrak{A} \cap \mathfrak{B} \cap \mathfrak{C}$. If $F \notin K^2$, then F is already the required witness. For the rest of this case assume

$$F \in K^2.$$

A minimal generator of K is the least common multiple obtained by choosing one displayed pure-power generator from each of the three ideals. When $\theta > \mu$, the generator y_0^{θ} of $\mathfrak{C}$ cannot occur, because every resulting y_0-exponent would exceed that of F. Recording exponent vectors in the order (y_0, y_1, y_2), the intersection generators that can divide F are therefore among

support	exponent vectors	
$\{y_0, y_1\}$	$(\mu, \gamma, 0)$	$(\mu, \widehat{\alpha}, 0)$
$\{y_0, y_2\}$	$(\mu, 0, \delta)$	$(\lambda, 0, \widehat{\beta})$
$\{y_1, y_2\}$	$(0, \widehat{\alpha}, \beta)$	$(0, \alpha, \widehat{\beta})$
$\{y_0, y_1, y_2\}$	$T = (\lambda, \gamma, \beta)$	$U = (\mu, \alpha, \delta)$.

A product of two candidates divides F only if their coordinatewise sum is at most $(\mu, \widehat{\alpha}, \widehat{\beta})$. If one candidate has first coordinate μ and the other has positive first coordinate, the first coordinate of their sum is greater than μ. Otherwise the second candidate has first coordinate 0 and lies in the third row. Pairing either first-row candidate with a third-row candidate makes the second coordinate larger than $\widehat{\alpha}$: every such sum is positive and either contains $\widehat{\alpha}$ as a strict summand or is $\gamma + \alpha > \max\{\gamma, \alpha\}$. For the remaining two candidates with first coordinate μ, namely $(\mu, 0, \delta)$ and $U = (\mu, \alpha, \delta)$, the third-coordinate sum with either third-row vector is $\delta + \beta$ or $\delta + \widehat{\beta}$, and both are larger than $\widehat{\beta} = \max\{\beta, \delta\}$. Thus no candidate with first coordinate μ can occur.

The remaining candidates are

$$D = (\lambda, 0, \widehat{\beta}), \qquad T = (\lambda, \gamma, \beta), \qquad E = (0, \widehat{\alpha}, \beta), \qquad W = (0, \alpha, \widehat{\beta}).$$

Any product involving D has third coordinate larger than $\widehat{\beta}$. The product $T + E$ has second coordinate $\gamma + \widehat{\alpha} > \widehat{\alpha}$, while $T + W$ has third coordinate $\beta + \widehat{\beta} > \widehat{\beta}$. Finally, $E + E$ and $E + W$ exceed the second coordinate, and $W + W$ exceeds the third. Consequently the only formally possible product is $T + T$. Since this product must divide F, we obtain

$$2\lambda \le \mu, \qquad 2\gamma \le \widehat{\alpha}, \qquad 2\beta \le \widehat{\beta}.$$

The last two inequalities force

$$\widehat{\alpha} = \alpha \ge 2\gamma, \qquad \widehat{\beta} = \delta \ge 2\beta.$$

Now put

$$G = y_0^{\lambda} y_1^{\widehat{\alpha}} y_2^{\widehat{\beta}}.$$

It belongs to $\mathfrak{A}^2 \cap \mathfrak{B}^2 \cap \mathfrak{C}^2$, because it is divisible respectively by

$$y_0^{\lambda} y_1^{\alpha} = y_0^{\lambda} y_1^{\widehat{\alpha}}, \qquad y_2^{2\beta}, \qquad y_1^{\gamma} y_2^{\delta} = y_1^{\gamma} y_2^{\widehat{\beta}}.$$

Any lcm candidate that divides G and contains y_0 must assign y_0 to $\mathfrak{A}$, since $\lambda < \mu$ and, in the three-generator case, $\lambda < \theta$. The lcm candidates dividing G are therefore

$$D = (\lambda, 0, \widehat{\beta}), \qquad T = (\lambda, \gamma, \beta), \qquad E = (0, \widehat{\alpha}, \beta), \qquad W = (0, \widehat{\alpha}, \widehat{\beta}).$$

Since E divides W, replacing every occurrence of W in a candidate product by E gives a divisor of that product. Hence if a product of two candidates divides G, there is one drawn only from D, E, T that also divides G. It suffices to rule out those products. Each has an excess coordinate:

$$D + D,\ D + E,\ D + T \quad \text{have } y_2\text{-exponent} > \widehat{\beta},$$

$$E + E,\ E + T \quad \text{have } y_1\text{-exponent} > \widehat{\alpha},$$

and $T + T$ has y_0-exponent $2\lambda > \lambda$. Hence $G \notin K^2$. It is the required witness in the remaining case.

The three-component witness with $\theta \leq \mu$. It remains to take

$$\mathfrak{C} = (y_0^{\theta}, y_1^{\gamma}, y_2^{\delta}), \qquad \theta \leq \mu.$$

Put

$$M = \max\{2\lambda, \mu\}, \qquad N = \begin{cases} \beta, & 2\theta \leq M, \\ \widehat{\beta}, & 2\theta > M, \end{cases} \qquad H = y_0^M y_2^N.$$

The monomial H lies in $\mathfrak{A}^2$, because $2\lambda \leq M$, and in $\mathfrak{B}^2$, because $y_0^{\mu} y_2^{\beta} \mid H$. If $2\theta \leq M$, then $y_0^{2\theta} \mid H$. If $2\theta > M$, then $\theta \leq \mu \leq M$ and $N = \widehat{\beta} \geq \delta$, so $y_0^{\theta} y_2^{\delta} \mid H$. Thus $H \in \mathfrak{A}^2 \cap \mathfrak{B}^2 \cap \mathfrak{C}^2$.

The intersection generators dividing H are supported on $\{y_0, y_2\}$. With $h = \max\{\lambda, \theta\}$, their exponent vectors are among

$$D_0 = (\mu, 0, 0), \qquad D_1 = (\lambda, 0, \widehat{\beta}), \qquad D_2 = (\mu, 0, \delta), \qquad D_3 = (h, 0, \beta).$$

Every listed first coordinate is at least λ, while

$$\mu + \lambda > M \qquad \text{and} \qquad 2\mu > M.$$

Hence any product involving D_0 or D_2 has first coordinate greater than M. The products $D_1 + D_1$ and $D_1 + D_3$ have third coordinates $2\widehat{\beta}$ and $\widehat{\beta} + \beta$, both larger than N, because $N \leq \widehat{\beta}$ and $\beta > 0$. Only $D_3 + D_3$ remains. If $2\theta \leq M$, then $N = \beta$ by the definition of N, and $D_3 + D_3$ has third coordinate $2\beta > N$. If $2\theta > M$, then $M \geq 2\lambda$ gives $\theta > \lambda$, so $h = \theta$, and $D_3 + D_3$ has first coordinate $2\theta > M$. Thus no product of two intersection generators divides H; hence H is the required witness in this final case. All possibilities for $\mathfrak{C}$ have now been exhausted. □

Theorem 8.3 (one-intervenor root criterion)**.** *Let $I \subsetneq S$ be a nonzero monomial ideal with no embedded primes, and let*

$$I = \bigcap_{j=1}^{m} q_j, \qquad q_j = (x_r^{a_{j,r}} : r \in A_j),$$

be an irredundant irreducible decomposition such that $m \geq 1$, every A_j is a nonempty subset of $\{1, \ldots, n\}$, every $a_{j,r}$ is a positive integer, and the A_j are pairwise distinct. For $k < \ell$, put

$$N(k, \ell) = \{j \notin \{k, \ell\} \mid A_j \subseteq A_k \cup A_\ell\}.$$

Assume

$$|N(k,\ell)| \le 1 \quad \textit{whenever } A_k \cap A_\ell \neq \varnothing.$$

If

$$I^{(2)} = I^2,$$

then there are a squarefree monomial ideal $J \subseteq S$ and $d \in {\mathbb{Z}_{>0}}^n$ such that $I = J^{\delta_d}$. More precisely, for every i put

$$C_i = \{a_{j,i} \,|\, 1 \le j \le m,\ i \in A_j\}, \qquad V = \bigcup_{j=1}^{m} A_j.$$

If $i \in V$, then $|C_i| = 1$; let c_i be its unique element. If $i \notin V$, let c_i be an arbitrary positive integer, and set $c = (c_1, \dots, c_n)$. With

$$P_j = (x_r : r \in A_j), \qquad J = \bigcap_{j=1}^{m} P_j,$$

the displayed decomposition of J is irredundant and

$$I = J^{\delta_c}.$$

For another $d \in {\mathbb{Z}_{>0}}^n$,

$$I = J^{\delta_d} \quad \Longleftrightarrow \quad d_i = c_i \textit{ for every } i \in V;$$

the coordinates d_i with $i \notin V$ are arbitrary. Moreover, J is the unique squarefree monomial ideal admitting a positive diagonal representation of I. The intervening-support inequality holds automatically when $|A_j| = 2$ for every j.

The bound 1 is sharp. If the inequality is relaxed to $|N(k,\ell)| \le 2$ for all intersecting pairs, then for every pair of distinct $r, s \in \mathbb{Z}_{>0}$ the ideal

$$I_{r,s} = (ac, bd, abz^r, cdz^s) \subset k[a,b,c,d,z]$$

satisfies all the resulting hypotheses and

$${I_{r,s}}^{(2)} = {I_{r,s}}^2,$$

but its mismatching pair (Q_1, Q_4) has exactly the two intervening supports indexed by 2 and 3, and $I_{r,s}$ admits no representation J^{δ_d} with J squarefree and d positive.

Proof. The argument has four parts. We first identify the second symbolic square as the intersection of the component squares. We then localize an assumed exponent mismatch to a core of at most three components and apply Lemma 8.2. This proves uniform occurrence exponents, after which Theorem 3.7 supplies the forced root. The final part checks the height-two consequence and sharpness.

The second symbolic square is the component square intersection. By Proposition 3.2,

$$\operatorname{Ass}(S/I) = \{P_1, \dots, P_m\}.$$

Since I has no embedded primes, these are precisely the minimal primes of I. They are pairwise incomparable: the A_j are pairwise distinct, and distinct minimal primes cannot contain one another.

Fix an index r. For $j \neq r$, incomparability gives $h \in A_j \setminus A_r$. Then $x_h \notin P_r$, while $x_h^{a_{j,h}} \in q_j$, so $(q_j)_{P_r} = S_{P_r}$. Consequently

$$I_{P_r} = (q_r)_{P_r}, \qquad (I^2)_{P_r} = (q_r^2)_{P_r}.$$

Lemma 8.1, applied with $t = 2$, gives $(q_r^2)_{P_r} \cap S = q_r^2$. Therefore

$$I^{(2)} = \bigcap_{r=1}^{m} ((I^2)_{P_r} \cap S) = \bigcap_{r=1}^{m} q_r^2. \tag{8.1}$$

An exponent mismatch reduces to a core with at most one auxiliary component. Suppose that some two components have unequal exponents at a common variable. Interchanging them if needed, denote their supports by A_k and A_ℓ, and choose a variable $x_i \in A_k \cap A_\ell$ such that

$$\lambda = a_{k,i} < \mu = a_{\ell,i}.$$

Put $A = A_k$, $B = A_\ell$, and $U = A \cup B$. Since $A \cap B \neq \varnothing$, the intervening-support hypothesis shows that at most one index j outside $\{k, \ell\}$ satisfies $A_j \subseteq U$.

Let W be the multiplicative set of monomials in the variables outside U, put

$$R = k[x_u : u \in U], \qquad T = S_W,$$

and let Q_A, Q_B, and the possible Q_C, be the irreducible ideals in R obtained from the surviving components. Every other component contains a variable outside U; a pure power of that variable belongs to the component and becomes invertible in T, so that component becomes the unit ideal. If $L \subseteq R$ is a monomial ideal, then

$$LT \cap R = L.$$

Indeed, if $x^\gamma \in W$ and $x^\gamma g \in LS$ with $g \in R$, then dividing the monomial terms of $x^\gamma g$ by x^γ shows that $g \in L$.

Localizing the equality $I^{(2)} = I^2$, using (8.1), and contracting to R therefore gives the core equality

$$\mathfrak{L}^2 = \mathfrak{R}, \tag{8.2}$$

where

$$\mathfrak{L} = Q_A \cap Q_B, \qquad \mathfrak{R} = Q_A^2 \cap Q_B^2$$

if there is no auxiliary component, and

$$\mathfrak{L} = Q_A \cap Q_B \cap Q_C, \qquad \mathfrak{R} = Q_A^2 \cap Q_B^2 \cap Q_C^2$$

if C is present.

Choose distinct indices i_0, i_1, i_2 and put $y_\epsilon = x_{i_\epsilon}$ for $\epsilon = 0, 1, 2$. Set $i_0 = i$. If C is absent, incomparability of A and B lets us choose

$$i_1 \in A \setminus B, \qquad i_2 \in B \setminus A.$$

If C is present, then $C \subseteq A \cup B$, while C is incomparable with each of A and B. Hence we may choose

$$i_1 \in (A \cap C) \setminus B, \qquad i_2 \in (B \cap C) \setminus A.$$

These indices, together with i_0, are distinct because $i_0 \in A \cap B$, $i_1 \in A \setminus B$, and $i_2 \in B \setminus A$. In either case set

$$\alpha = a_{k,i_1}, \qquad \beta = a_{\ell,i_2},$$

and, when $C = A_g$ is present, set

$$\gamma = a_{g,i_1}, \qquad \delta = a_{g,i_2}, \qquad \theta = a_{g,i_0} \quad \text{if } i_0 \in C.$$

Thus, for monomials supported only on y_0, y_1, y_2, the components behave as

$$\mathfrak{A}^\circ = (y_0^\lambda, y_1^\alpha), \qquad \mathfrak{B}^\circ = (y_0^\mu, y_2^\beta),$$

together with either no third ideal,

$$\mathfrak{C}^\circ = (y_1^\gamma, y_2^\delta) \quad \text{when } i_0 \notin C,$$

or

$$\mathfrak{C}^\circ = (y_0^\theta, y_1^\gamma, y_2^\delta) \quad \text{when } i_0 \in C.$$

The same membership statement holds after squaring, since a product involving any other component generator has positive exponent at a variable outside $\{y_0, y_1, y_2\}$. Equality (8.2), restricted to monomials supported on these three variables, is therefore incompatible with Lemma 8.2. No exponent mismatch can occur.

Uniform exponents give the unique diagonal source. For every i, all exponents $a_{j,i}$ with $i \in A_j$ are consequently equal. Thus condition (ii) of Theorem 3.7 holds for the given decomposition. That theorem now proves all the more precise conclusions concerning C_i, the weights c_i, the ideals P_j, the representation $I = J^{\delta_c}$, the rigidity of the weights on V, and the uniqueness of the squarefree source J.

The height-two case and the sharp next bound. Suppose that $|A_j| = 2$ for every j. For distinct intersecting supports we may write

$$A_k = \{i, j\}, \qquad A_\ell = \{i, h\}, \qquad j \neq h.$$

The only two-element subset of $A_k \cup A_\ell$ different from these two sets is $\{j, h\}$. Pairwise distinctness of the supports therefore gives $|N(k, \ell)| \leq 1$.

Now fix distinct $r, s \in \mathbb{Z}_{>0}$ and retain Construction 4.1. In the ring $R_0 = k[a, b, c, d, z]$, put $L = I_{r,s}$. By Theorem 4.3, the displayed decomposition

$$L = Q_1 \cap Q_2 \cap Q_3 \cap Q_4$$

is irredundant. In the variable order (a, b, c, d, z), its component support sets are

$$E_1 = \{1, 2, 5\}, \quad E_2 = \{1, 4\}, \quad E_3 = \{2, 3\}, \quad E_4 = \{3, 4, 5\}.$$

These sets are pairwise distinct, L has no embedded primes, and

$$L^{(q)} = L^q \qquad \text{for every } q \geq 1.$$

In particular, L is all-power Simis and satisfies the degree-two equality

$$L^{(2)} = L^2.$$

Since there are only four components, the intervening-support set $N_L(k,\ell)$ for L has cardinality at most 2 for every pair $k<\ell$. Thus L satisfies every hypothesis obtained from the theorem by replacing the intervening-support bound 1 by 2.

The supports E_1 and E_4 meet exactly in the index 5 of the variable z. The corresponding z-exponents in Q_1 and Q_4 are s and r, respectively, so this is an exponent-mismatching pair because $r\neq s$. Their union is the full index set, while E_2 and E_3 are both contained in that union. Hence

$$N_L(1,4)=\{2,3\}.$$

Finally, Theorem 4.3 shows that L has a standard positive diagonal squarefree source if and only if $r=s$, even when no Simis condition is imposed on the source. Since $r\neq s$, no such source exists. This proves both the sharpness of the bound 1 and the rootlessness assertion. □

The separation at the next bound is therefore not caused by a failure of the power identities: the witnessing ideals $I_{r,s}$ with $r\neq s$ are Simis in every degree. They fail representability while having two, rather than at most one, intervening components for a single exponent-mismatching pair.

Corollary 8.4 (one-intervenor MPV conclusion)**.** *Retain all hypotheses and notation of Theorem 8.3, including $I^{(2)}=I^2$, and assume in addition that I is Simis. Then the forced squarefree source J of that theorem is Simis and*

$$I=J^{\delta_c}.$$

Consequently the MPV conclusion holds for I.

Proof. Theorem 8.3 gives J and c with $I=J^{\delta_c}$. Since $c\in\mathbb{Z}_{>0}{}^n$, Theorem 3.5 says that J is Simis if and only if J^{δ_c} is Simis. The latter ideal is I, which is Simis by the additional hypothesis. Hence J is Simis, and the displayed representation is the MPV conclusion for I. □

Construction 8.5 (disjoint-block family)**.** Let $m\geq 1$, let $B_1,\dots,B_m$ be pairwise disjoint finite nonempty sets of variables, put

$$V=\bigsqcup_{h=1}^{m}B_h,\qquad S=k[V],$$

and fix a family $a=(a_z)_{z\in V}$ of positive integers. For $E\subseteq V$, put $P(E)=(z:z\in E)S$, and define

$$P_h=P(B_h),\qquad Q_h(a)=(z^{a_z}:z\in B_h)S,$$
$$I_a=\bigcap_{h=1}^{m}Q_h(a),\qquad J=I_{\mathbf{1}}=\bigcap_{h=1}^{m}P_h.$$

Theorem 8.6 (disjoint-block family)**.** *Retain Construction 8.5, and let $\Delta=\Delta(I_a)=\Delta(J)$. Then:*

(i) Each $Q_h(a)$ is irreducible, and

$$I_a=\bigcap_{h=1}^{m}Q_h(a)$$

is an irredundant irreducible decomposition. Moreover

$$\operatorname{Ass}(S/I_a)=\operatorname{Min}(I_a)=\{P_1,\dots,P_m\}.$$

Thus I_a has no embedded primes and is MPV-minimal; its component heights are $|B_1|,\dots,|B_m|$. The same assertions hold for J.

(ii) *For every* $t \geq 1$,

$$I_a^t = I_a{}^{(t)} = \bigcap_{h=1}^{m} Q_h(a)^t, \qquad J^t = J^{(t)} = \bigcap_{h=1}^{m} P_h^t,$$

and

$$\operatorname{Ass}(S/I_a^t) = \operatorname{Ass}(S/J^t) = \{P_1, \ldots, P_m\}.$$

In particular I_a *and* J *are Simis and normally torsion-free.*

(iii)

$$\mathrm{G}(J) = \{z_1 \cdots z_m \,|\, z_h \in B_h\}, \qquad \mathrm{G}(I_a) = \left\{ z_1^{a_{z_1}} \cdots z_m^{a_{z_m}} \;\middle|\; z_h \in B_h \right\}.$$

Each displayed generating set has $\prod_{h=1}^{m} |B_h|$ *members, every displayed monomial has support cardinality exactly* m*, and*

$$\sqrt{I_a} = J, \qquad \mathcal{F}(I_a) = \mathcal{F}(J).$$

(iv)

$$I_a = J^{\delta_a}.$$

The ideal J *is the unique squarefree monomial ideal representing* I_a *by a positive diagonal weighting, and*

$$I_a = J^{\delta_d} \quad \Longleftrightarrow \quad d = a$$

for $d \in {\mathbb{Z}_{>0}}^V$*. Consequently* I_a *is nonsquarefree exactly when* $a \neq \mathbf{1}$*. Whenever* $a \neq \mathbf{1}$*,* I_a *satisfies the full MPV conclusion with the squarefree Simis source* J*.*

(v) *The facets of* Δ *are the sets* $\{z_1, \ldots, z_m\}$ *with* $z_h \in B_h$*. If at least two blocks have cardinality at least* 2*, then* Δ *has no leaf and hence is not a simplicial forest. If at least three blocks have cardinality at least* 2*, then* $\mathcal{F}(I_a)$ *contains a strong triangle; consequently* $\mathcal{F}(I_a)$ *is neither balanced nor free of strong triangles.*

(vi) *For every* $k < \ell$,

$$N_a(k, \ell) = \{j \notin \{k, \ell\} \,|\, B_j \subseteq B_k \cup B_\ell\} = \varnothing$$

and

$$X_{I_a} = \varnothing.$$

Thus I_a *satisfies the at-most-one-intervening-support condition of Theorem 8.3 with the stronger value* 0*, independently forcing the same squarefree source* J*.*

Proof. We first turn the block intersections into products and compute the associated primes of every power. We then identify the minimal generators and the unique diagonal source. The leaf and strong-cycle assertions are verified directly from the common transversal description of the facets, and the final clause applies Theorem 8.3.

Component structure and irredundance. We use the following product identity. If $M_h \subseteq k[B_h]$ is a monomial ideal for each h, and each M_h is also regarded as extended to S, then

$$\bigcap_{h=1}^{m} M_h S = \prod_{h=1}^{m} M_h S.$$

The product is contained in every factor, hence in the intersection. Conversely, a monomial in the intersection has, for each h, a monomial generator $g_h \in M_h$ dividing it. The supports of the g_h

are pairwise disjoint, so $g_1 \cdots g_m$ divides the given monomial, which therefore lies in the product. Both sides are monomial ideals, proving the identity.

Each $Q_h(a)$ is irreducible by Proposition 3.2. To prove irredundance, choose one variable $y_\ell \in B_\ell$ for every ℓ, fix h, and put

$$f_h = \prod_{\ell \neq h} y_\ell^{a_{y_\ell}},$$

with the empty product interpreted as 1 when $m = 1$. For $\ell \neq h$, the monomial f_h is divisible by the generator $y_\ell^{a_{y_\ell}}$ of $Q_\ell(a)$, so $f_h \in Q_\ell(a)$. On the other hand, f_h involves no variable from B_h, whereas every monomial in $Q_h(a)$ is divisible by z^{a_z} for some $z \in B_h$. Thus

$$f_h \in \bigcap_{\ell \neq h} Q_\ell(a) \setminus Q_h(a),$$

so the displayed irreducible decomposition of I_a is irredundant.

Proposition 3.2 now gives

$$\operatorname{Ass}(S/I_a) = \{P_1, \ldots, P_m\}.$$

Because the nonempty blocks are pairwise disjoint, no P_h is contained in another. Hence every P_h is minimal, the minimal and associated primes coincide, and the decomposition is MPV-minimal. Its component heights are $|B_h|$. The product identity and the definition of I_a also give

$$I_a = \prod_{h=1}^{m} Q_h(a). \tag{8.3}$$

Taking $a = \mathbf{1}$ gives all the same conclusions for J and $J = \prod_h P_h$.

Powers, symbolic powers, and associated primes. From (8.3), commutativity of ideal multiplication, and the product identity applied to the ideals $Q_h(a)^t$, we obtain

$$I_a^t = \prod_{h=1}^{m} Q_h(a)^t = \bigcap_{h=1}^{m} Q_h(a)^t.$$

By Lemma 8.1, each $Q_h(a)^t$ is P_h-primary and satisfies

$$(Q_h(a)^t S_{P_h}) \cap S = Q_h(a)^t.$$

Fix j. If $h \neq j$, choose $y \in B_h$. Then $y \notin P_j$, while $y^{ta_y} \in Q_h(a)^t$, so $(Q_h(a)^t)_{P_j} = S_{P_j}$. Localizing the displayed power decomposition at P_j and contracting therefore yields

$$((I_a^t)_{P_j}) \cap S = Q_j(a)^t.$$

The primes P_j are exactly the minimal primes of I_a, and hence of I_a^t because $\sqrt{I_a^t} = \sqrt{I_a}$. The definition of symbolic powers gives

$$I_a{}^{(t)} = \bigcap_{j=1}^{m} ((I_a^t)_{P_j} \cap S) = \bigcap_{j=1}^{m} Q_j(a)^t = I_a^t.$$

It remains to identify the associated primes of these powers. For $g_{h,t} = f_h^t$ and $\ell \neq h$, the monomial $g_{h,t}$ is divisible by the product of t copies of $y_\ell^{a_{y_\ell}}$, so $g_{h,t} \in Q_\ell(a)^t$. It has zero total exponent in the variables of B_h. Every monomial in $Q_h(a)^t$ is a product of t monomials from

$Q_h(a)$; each factor has positive total exponent in B_h, so every such product has positive total exponent in B_h. Consequently

$$g_{h,t} \in \bigcap_{\ell \neq h} Q_\ell(a)^t \setminus Q_h(a)^t.$$

Thus the primary decomposition $I_a^t = \bigcap_h Q_h(a)^t$ is minimal. Its radicals P_h are therefore exactly the associated primes of S/I_a^t:

$$\operatorname{Ass}(S/I_a^t) = \{P_1, \ldots, P_m\} = \operatorname{Ass}(S/I_a) \qquad (t \geq 1).$$

This proves Simisness and normal torsion-freeness for I_a. Repeating the argument with $a = \mathbf{1}$, for which $Q_h(\mathbf{1}) = P_h$, gives

$$J^t = J^{(t)} = \bigcap_{h=1}^{m} P_h^t, \qquad \operatorname{Ass}(S/J^t) = \{P_1, \ldots, P_m\} = \operatorname{Ass}(S/J).$$

Minimal generators and the diagonal source. For each h,

$$\mathrm{G}(Q_h(a)) = \{z^{a_z} \mid z \in B_h\}.$$

Expanding the product (8.3), the monomials

$$z_1^{a_{z_1}} \cdots z_m^{a_{z_m}}, \qquad z_h \in B_h,$$

generate I_a. Two such monomials coming from different transversals are incomparable under divisibility: if their entries differ in B_h, each has positive exponent at its selected variable and zero exponent at the variable selected there by the other. Hence the displayed monomials are precisely the minimal monomial generators of I_a. Setting every $a_z = 1$ gives the displayed minimal generating set of J. Both sets have $\prod_{h=1}^m |B_h|$ members, and every member uses exactly one variable from each block, so every support has cardinality m.

Every generator of I_a is divisible by the corresponding squarefree generator of J, so $I_a \subseteq J$. The ideal J is an intersection of primes and is therefore radical, giving $\sqrt{I_a} \subseteq J$. Conversely, for a transversal $v = z_1 \cdots z_m$ take

$$E = \max\{a_{z_1}, \ldots, a_{z_m}\}.$$

Then v^E is divisible by $z_1^{a_{z_1}} \cdots z_m^{a_{z_m}}$, so $v \in \sqrt{I_a}$. These monomials generate J, hence $J \subseteq \sqrt{I_a}$. Thus $\sqrt{I_a} = J$, and the displayed minimal generating sets also give $\mathcal{F}(I_a) = \mathcal{F}(J)$.

Each variable $z \in V$ belongs to exactly one block, say B_h, so its only component occurrence exponent in the displayed decomposition is a_z. All occurrence-exponent sets are therefore singletons. Since the union of the blocks is the full ambient variable set V, Theorem 3.7 gives

$$I_a = J^{\delta_a},$$

shows that J is the unique squarefree source, and shows that a positive diagonal representation by J forces every weight to equal the corresponding entry of a. Thus

$$I_a = J^{\delta_d} \iff d = a \qquad (d \in {\mathbb{Z}_{>0}}^V).$$

If $a = \mathbf{1}$, then $I_a = J$ is squarefree. If $a \neq \mathbf{1}$, choose z with $a_z > 1$ and choose a transversal containing z. The corresponding minimal generator of I_a has z-exponent $a_z > 1$. If a squarefree generating set existed, one of its members would divide this minimal generator and hence would

be a proper divisor lying in I_a, which is impossible. Thus I_a is not squarefree. For every a, the ideal J is Simis by the preceding part and represents I_a; for $a \neq \mathbf{1}$ this is therefore a nonsquarefree instance of the full MPV conclusion with squarefree Simis source J.

Leaves and a strong triangle in the transversal complex. The common facets are the transversals

$$F = \{z_1, \dots, z_m\}, \qquad z_h \in B_h.$$

Assume that at least two blocks have cardinality at least 2. Fix a facet F and a facet $G \neq F$. Choose h for which the members selected from B_h by F and G are distinct. Then $|B_h| \geq 2$, and by hypothesis there is another block B_ℓ with $\ell \neq h$ and $|B_\ell| \geq 2$. Construct a facet H by retaining the entry of F in B_h, replacing its entry in B_ℓ by a different member of B_ℓ, and retaining every other entry of F. Then $H \neq F$. The member selected by F in B_h belongs to $F \cap H$, but not to G, because F and G select different members of B_h and every facet selects exactly one member of that block. Thus

$$F \cap H \not\subseteq F \cap G.$$

Since $G \neq F$ was arbitrary, F is not a leaf; since F was arbitrary, Δ has no leaf. The complex itself is a nonempty facet subcomplex of itself, so Δ is not a simplicial forest.

Now suppose at least three blocks have cardinality at least 2. After reordering the blocks, choose distinct $p_i, q_i \in B_i$ for $i = 1, 2, 3$, and choose $y_h \in B_h$ for $h \geq 4$. The three transversals

$$F_1 = \{p_1, p_2, q_3, y_4, \dots, y_m\},$$
$$F_2 = \{q_1, p_2, p_3, y_4, \dots, y_m\},$$
$$F_3 = \{p_1, q_2, p_3, y_4, \dots, y_m\}$$

are distinct facets. With $C = \{p_1, p_2, p_3\}$, their intersections with C are

$$F_1 \cap C = \{p_1, p_2\}, \qquad F_2 \cap C = \{p_2, p_3\}, \qquad F_3 \cap C = \{p_3, p_1\}.$$

Hence

$$(p_1, F_1, p_2, F_2, p_3, F_3, p_1)$$

is a strong triangle. Therefore the common generator-support clutter is not balanced and does not satisfy the no-strong-triangle condition.

The component-incidence invariants. Let $k < \ell$ and $j \notin \{k, \ell\}$. Since B_j is nonempty and disjoint from $B_k \cup B_\ell$,

$$B_j \not\subseteq B_k \cup B_\ell.$$

Thus $N_a(k, \ell) = \varnothing$. Distinct component supports are themselves disjoint, so no two components share a variable. For the exponent-mismatching pair set X_{I_a} of Theorem 3.7, this gives

$$X_{I_a} = \varnothing.$$

The ideal I_a has no embedded primes and pairwise distinct component supports by the first part, and its every $N_a(k, \ell)$ is empty. It is Simis by the second part, so in particular ${I_a}^{(2)} = I_a^2$. Theorem 8.3 therefore applies with the stronger value 0. Its source is

$$\bigcap_{h=1}^{m} P(B_h) = \bigcap_{h=1}^{m} P_h = J,$$

and, because every ambient variable belongs to some block, it forces every weight to equal the corresponding entry of a. This independently recovers the source and weight rigidity established above. □

The theorem deliberately keeps two conclusions separate. Every member of the family is represented by its radical, but the geometry of the common transversal complex becomes less forest-like as more blocks acquire size at least two. This allows the specialization below to separate those geometric and support-size antecedents from the component-incidence antecedent. Since the mismatch set X_{I_a} is empty by part (vi), it cannot enlarge a represented-Simis success class by relaxing a condition quantified only over exponent-mismatching pairs.

Corollary 8.7 (transversal $(3,2,2,2)$ specialization)**.** *In Construction 8.5, take*

$$B_1 = \{u_1, v_1, w_1\}, \quad B_2 = \{u_2, v_2\}, \quad B_3 = \{u_3, v_3\}, \quad B_4 = \{u_4, v_4\},$$

and $a_z = 2$ for all nine variables. Then I_a has 24 nonsquarefree minimal monomial generators, all of support cardinality 4, and its component heights are $(3,2,2,2)$. Its common facet complex $\mathcal{F}(I_a) = \mathcal{F}(J)$ is leaf-free and contains the strong triangle

$$F_1 = \{u_1, u_2, v_3, u_4\}, \quad F_2 = \{v_1, u_2, u_3, u_4\}, \quad F_3 = \{u_1, v_2, u_3, u_4\},$$

whose cycle vertices are u_1, u_2, u_3. The ideal satisfies the full MPV conclusion and the one-intervenor condition, while it lies outside each of the following criteria: every minimal generator has support cardinality at most 3, including the exact-support-3 criterion; the common facet complex is a forest; the generator clutter is balanced or has no strong triangle; and every component has height at most 2. In particular, the no-strong-triangle sufficient condition of Theorem 6.1 is not necessary for the MPV conclusion.

Moreover,

$$X_{I_a} = \varnothing.$$

Thus this specialization separates the component-incidence route from the support-size, forest, balanced/no-strong-triangle, and pure component-height-two sufficient routes at the level of hypotheses. Because every added hypothesis quantified only over exponent-mismatching pairs $(k,\ell) \in X_{I_a}$ is vacuous here, this specialization is not a success-class separator for any criterion obtained from its standing hypotheses solely by such an added quantified condition.

Proof. Specialize Theorem 8.6 to $m = 4$, to the displayed four blocks, and to $a_z = 2$ for every variable. The squarefree and weighted ideals have the product forms

$$J = (u_1, v_1, w_1)(u_2, v_2)(u_3, v_3)(u_4, v_4)$$

and

$$I_a = (u_1^2, v_1^2, w_1^2)(u_2^2, v_2^2)(u_3^2, v_3^2)(u_4^2, v_4^2).$$

The component supports have cardinalities

$$3, 2, 2, 2,$$

so these are exactly the component heights by part (i) of Theorem 8.6. The displayed minimal generating sets consist of one variable from each block. There are consequently

$$3 \cdot 2 \cdot 2 \cdot 2 = 24$$

minimal generators. Each generator of J has support cardinality 4, each generator of I_a has the same support and is a product of four squares, and

$$I_a = J^{\delta_a}.$$

The common generator supports form the transversal clutter on these four blocks.

All four blocks have cardinality at least 2, so the common facet complex is leaf-free by the leaf argument in Theorem 8.6. To make the strong triangle explicit, put $C = \{u_1, u_2, u_3\}$. The three displayed sets F_1, F_2, F_3 are transversal facets, and

$$F_1 \cap C = \{u_1, u_2\}, \qquad F_2 \cap C = \{u_2, u_3\}, \qquad F_3 \cap C = \{u_1, u_3\}.$$

Thus

$$(u_1, F_1, u_2, F_2, u_3, F_3, u_1)$$

is a strong 3-cycle. The common vertex u_4 of the three facets is not in the cycle-vertex set and therefore does not affect the strong-cycle condition.

Parts (i), (ii), (iii), and (iv) of Theorem 8.6 also show that I_a and J have no embedded primes, that their irreducible decompositions are MPV-minimal, that both are Simis, that

$$\mathcal{F}(I_a) = \mathcal{F}(J) = \mathcal{F}(\sqrt{I_a}),$$

and that J is the squarefree Simis source of I_a. The members of this common clutter are precisely the facets of $\Delta(I_a) = \Delta(J)$, because the common minimal generators form an antichain. Since every $a_z = 2 \neq 1$, this is a nonsquarefree instance of the full MPV conclusion. Part (vi) gives $N_a(k, \ell) = \varnothing$ for every pair and $X_{I_a} = \varnothing$, so the one-intervenor condition holds with the value 0.

Every minimal generator has support cardinality 4, so the specialization lies outside both the exact-support-3 class and the cumulative class in which all generator supports have cardinality at most 3. Its facet complex has no leaf, so it is not a forest; it contains the displayed strong triangle, so its generator clutter is neither balanced nor free of strong triangles; and one component has height 3, so the components are not all of height at most 2.

The only hypothesis of Theorem 6.1 that fails is the absence of a strong 3-cycle. The three original MPV hypotheses, the support equality, the pairwise distinctness of the component radicals, and Simisness all hold by the preceding paragraph, while the MPV conclusion holds through J. Hence the no-strong-triangle sufficient condition is not necessary for the MPV conclusion.

Finally, $X_{I_a} = \varnothing$ because distinct blocks are disjoint. Every condition whose universal quantification is only over pairs $(k, \ell) \in X_I$ is therefore satisfied vacuously by I_a. The displayed separations are thus separations between support-size, forest, cycle, and component-height antecedents on the one hand and the component-incidence route on the other. They do not make this represented Simis example a success-class separator for any criterion obtained from the same standing hypotheses solely by such an exponent-mismatching-pair condition. □

The remaining comparison concerns an external criterion whose additional incidence hypothesis is quantified only over exponent-mismatching component pairs. We state it in the exact convention of its source before drawing the posterior consequence.

The following criterion is [BK25, Notation 3.5, Set-up 4.1 and Theorem 4.7], stated in the notation of this paper.

Theorem 8.8 (BK Simis-equivalence criterion)**.** *Let $I \subsetneq S = k[x_1, \ldots, x_n]$ be a nonzero monomial ideal without embedded primes, and let*

$$I = \bigcap_{j=1}^{r} q_j, \qquad q_j = (x_i^{w_{j,i}} : i \in A_j), \qquad p_j = \sqrt{q_j} = (x_i : i \in A_j),$$

be an irredundant irreducible monomial decomposition, with $\{1,\dots,n\} = \bigcup_{j=1}^{r} A_j$. *Assume, as part of BK's minimality convention, that*

$$p_j \neq p_k \qquad (j \neq k),$$

equivalently that $A_j \neq A_k$ *for* $j \neq k$. *Together with irredundance, this is exactly the MPV-minimality convention used in this paper. Put*

$$X_I = \{(k,\ell) \mid 1 \leq k < \ell \leq r \text{ and some } i \in A_k \cap A_\ell \text{ satisfies } w_{k,i} \neq w_{\ell,i}\}.$$

Assume that

$$A_j \not\subseteq A_k \cup A_\ell \qquad \text{for every } (k,\ell) \in X_I \text{ and every } j \notin \{k,\ell\}.$$

Then I *is Simis if and only if there are a squarefree Simis monomial ideal* $J \subseteq S$ *and a standard positive diagonal weighting* $w \in \mathbb{Z}_{>0}{}^n$ *such that*

$$I = J^{\delta_w}.$$

In particular, if I *is Simis, then the MPV conclusion holds for* I.

The distinct-radical convention is essential, not a difference of terminology. Merely requiring the components q_j to be pairwise distinct does not suffice. Indeed, in $k[x,y]$ set

$$L = (x,y)^2 = (x^2,y) \cap (x,y^2), \qquad P = (x,y).$$

The displayed decomposition is irredundant and both component radicals equal P, while $\operatorname{Ass}(k[x,y]/L) = \operatorname{Min}(L) = \{P\}$. The two supports are both $\{x,y\}$ and exhaust the variables, and $X_L = \{(1,2)\}$; hence there is no third component and the intervening-support condition is vacuous. For every $t \geq 1$, $L^t = P^{2t}$ is P-primary, so L is Simis. On the other hand, Theorem 3.7 rules out a positive diagonal squarefree root because the x-exponents of the two components are $2,1$, and the y-exponents are $1,2$. Thus deleting $p_j \neq p_k$ from the standing hypotheses makes the displayed Simis equivalence false. BK's convention and MPV-minimality must not be separated.

Corollary 8.9 (posterior vacuity of the BK mismatch test)**.** *Let* I *satisfy any, and hence every, of the equivalent diagonal-representability conditions in Theorem 3.7. Then*

$$X_I = \varnothing,$$

and the supports A_j *are pairwise distinct and pairwise incomparable. If in addition* I *satisfies the three original MPV hypotheses, normalize as follows. Put* $V = \bigcup_j A_j$, *let* $Y = \{x_i \mid i \in V\}$ *and* $Z = \{x_i \mid i \notin V\}$, *and write* $I' \subseteq S' = k[Y]$ *for the ideal with the same component data. Then* I' *satisfies every standing hypothesis of Theorem 8.8, including exhaustion, and* $X_{I'} = \varnothing$; *moreover* I' *is Simis. Applying the corrected theorem in* S' *and then restoring the variables* Z *with arbitrary positive weights yields a squarefree Simis positive-diagonal root of* I *in* S. *The ambient ring* S *after restoration need not satisfy the exhaustion hypothesis, so the theorem is applied only before restoration. Consequently represented Simis MPV successes give no success-class separation from [BK25], Theorem 4.7, after the same unused-variable normalization. The supported novelty is instead the a priori support-only criterion and proof, the sharp degree-two obstruction, strict quantified coverage beyond the two named BDK26 positive classes, and hypothesis-level separation among sufficient antecedents.*

Proof. Retain the setup and decomposition of Theorem 3.7, and let

$$J = \bigcap_{j=1}^{m} P_j \subseteq S, \qquad P_j = (x_i : i \in A_j),$$

be its forced squarefree source. By that theorem $X_I = \varnothing$ and the displayed prime decomposition of J is irredundant. Hence the A_j are pairwise distinct. They are pairwise incomparable as well: if $A_j \subsetneq A_\ell$, then $P_j \subsetneq P_\ell$ and $\bigcap_{h\neq\ell} P_h \subseteq P_j \subsetneq P_\ell$, contradicting irredundance of the prime decomposition. Proposition 3.2 therefore gives

$$\operatorname{Ass}(S/I) = \{P_j\} = \operatorname{Min}(I), \qquad \operatorname{Ass}(S/J) = \{P_j\} = \operatorname{Min}(J).$$

Thus, in the represented class under discussion, absence of embedded primes and MPV-minimality already follow from representability and irredundance.

Now assume the three original MPV hypotheses. By the preceding paragraph their no-embedded and MPV-minimality clauses are automatic; the additional bundle supplies Simisness of I. Put

$$V = \bigcup_{j=1}^{m} A_j, \qquad Y = \{x_i \,|\, i \in V\}, \qquad Z = \{x_i \,|\, i \notin V\},$$

and, in $S' = k[Y]$, put

$$q'_j = (x_i^{a_{j,i}} : i \in A_j)S', \qquad p'_j = (x_i : i \in A_j)S', \qquad J' = \bigcap_{j=1}^{m} p'_j, \qquad I' = \bigcap_{j=1}^{m} q'_j.$$

For a monomial $m = m_Y z^\gamma$ with $m_Y \in S'$ and z^γ a monomial in Z,

$$m \in q_j \iff m_Y \in q'_j, \qquad m \in P_j \iff m_Y \in p'_j. \tag{8.4}$$

It follows that $I = I'S$ and $J = J'S$. The same monomial test makes irredundance descend and ascend: irredundance of an unprimed decomposition is immediate from a primed monomial witness by extension, while irredundance of the primed decomposition follows because a monomial witness in S can be contracted to its Y-part by (8.4). Hence $I' = \bigcap_j q'_j$ and $J' = \bigcap_j p'_j$ are irredundant. The exponents and supports are unchanged, so $X_{I'} = \varnothing$. Theorem 3.7, applied to I', gives

$$I' = J'^{\delta_c}$$

for the forced weights c on Y, and $J' = \bigcap_j p'_j$ is its unique squarefree source. Because the A_j are pairwise incomparable, Proposition 3.2 gives

$$\operatorname{Ass}(S'/I') = \left\{p'_j\right\} = \operatorname{Min}(I'), \qquad \operatorname{Ass}(S'/J') = \left\{p'_j\right\} = \operatorname{Min}(J').$$

The corresponding equalities after extension are

$$\operatorname{Ass}(S/I) = \{P_j\} = \operatorname{Min}(I), \qquad \operatorname{Ass}(S/J) = \{P_j\} = \operatorname{Min}(J).$$

Thus I' and J' have no embedded primes, independently of the additional no-embedded hypothesis on I; and I' satisfies exhaustion by the definition of Y.

We record one monomial-localization observation. Let L be a monomial ideal in a polynomial ring R, and let $\mathfrak{p}$ be a monomial prime. For a monomial w,

$$w \in LR_{\mathfrak{p}} \cap R \iff \text{some monomial } v \notin \mathfrak{p} \text{ satisfies } vw \in L. \tag{8.5}$$

One direction is immediate. Conversely, if $sw \in L$ with $s \notin \mathfrak{p}$, some term v of s lies outside $\mathfrak{p}$. Multiplication by the single monomial w injects the terms of s, so vw occurs in sw; since L is monomial, $vw \in L$.

The contraction $C = LR_{\mathfrak{p}} \cap R$ is monomial. Choose a lexicographic order $\prec$ with the variables of $\mathfrak{p}$ first and 1 least. Thus every monomial outside $\mathfrak{p}$ is smaller than every monomial in $\mathfrak{p}$, and the order is multiplicative. Let $g \in C$, choose $s \notin \mathfrak{p}$ with $sg \in L$, and let x^{a_0} and x^{b_0} be the least terms of s and g. For arbitrary terms x^a and x^b of s and g,

$$x^{a_0+b_0} \preceq x^{a_0+b} \preceq x^{a+b},$$

with a strict inequality unless $a = a_0$ and $b = b_0$. Hence $x^{a_0+b_0}$ occurs in sg with nonzero coefficient; monomiality of L and $x^{a_0} \notin \mathfrak{p}$ give $x^{b_0} \in C$ by (8.5). Subtracting this term and inducting on the number of terms of g proves that C is monomial.

Now let U be either I' or J'. The preceding prime calculations give $\operatorname{Min}_{S'}(U) = \left\{p'_j\right\}$ and $\operatorname{Min}_S(US) = \{P_j\}$. Fix $t \geq 1$. Since S is free over S' on the monomials in Z, a monomial $m = m_Y z^\gamma$ and a monomial multiplier $u = u_Y z^\delta$ satisfy

$$u \notin P_j \iff u_Y \notin p'_j, \qquad um \in U^t S \iff u_Y m_Y \in U^t.$$

By (8.5),

$$m \in (U^t S)_{P_j} \cap S \iff m_Y \in (U^t)_{p'_j} \cap S'.$$

Both contractions are monomial, so monomialwise equality gives

$$(U^t S)_{P_j} \cap S = ((U^t)_{p'_j} \cap S')S.$$

Intersecting over j and again using monomial membership in the free extension,

$$(US)^{(t)} = \bigcap_j ((U^t S)_{P_j} \cap S) = U^{(t)} S. \tag{8.6}$$

Taking $U = I'$ in (8.6), and using $I = I'S$ and Simisness of I, yields

$$I'^{(t)} S = I^{(t)} = I^t = (I')^t S.$$

Contracting by the free-module decomposition of S gives $I'^{(t)} = (I')^t$ for every t. Thus I' is Simis.

Consequently I' satisfies every standing hypothesis of the corrected Theorem 8.8: irredundance, pairwise distinct radicals, absence of embedded primes, exhaustion in S', $X_{I'} = \varnothing$, and Simisness. Applying that theorem in S' gives a squarefree Simis monomial ideal $K' \subseteq S'$ and weights $e \in {\mathbb{Z}_{>0}}^V$ such that

$$I' = K'^{\delta_e}.$$

The uniqueness clause in Theorem 3.7 identifies $K' = J'$; hence J' is Simis.

Choose arbitrary positive weights f on the variables in Z and write $\widetilde{e} = (e, f) \in {\mathbb{Z}_{>0}}^n$. A monomial $u_Y z^\gamma$ belongs to $J = J'S$ exactly when $u_Y \in J'$. Therefore $J^{\delta_{\widetilde{e}}}$ is generated by the monomials

$$(u_Y)^{\delta_e} z^{\delta_f(\gamma)} \qquad (u_Y \in J' \text{ a monomial},\ \gamma \in \mathbb{N}^Z).$$

Each is a multiple of one with $\gamma = 0$, and every $\gamma = 0$ term occurs, so

$$J^{\delta_{\widetilde{e}}} = J'^{\delta_e} S = I'S = I.$$

Taking $U = J'$ in (8.6) now gives

$$J^{(t)} = J'^{(t)}S = (J')^t S = J^t$$

for every t, because J' is Simis. Thus J is squarefree and Simis, and it is the unique squarefree source identified by the component criterion. The weights on Z occur in no component generator and are therefore arbitrary.

The exhaustion hypothesis was imposed only in S'. After restoring Z, the ambient ring S need not satisfy exhaustion; restoration is the final extension step, not a second application of the cited theorem in S. Since the original I was an arbitrary represented member satisfying the three original MPV hypotheses, every represented Simis MPV success is in the represented-success branch of Theorem 8.8.

The scope of the positive inclusion is exact. The a priori arguments in this paper are the generator-support criterion of Theorem 6.1, which derives the absence of exponent mismatches before representability is known, and the exponent-independent at-most-one-intervening support criterion of Theorem 8.3. Its sharp next-bound witnesses $I_{r,s}$ with $r \neq s$ are all-power Simis but have no diagonal squarefree source, so they do not enter the represented-success branch at all. Strict quantified coverage beyond the two named BDK26 positive classes is supplied by the nonsquarefree members of the family in Theorem 7.8: each satisfies the balanced-support and one-facet-deletion antecedents and the MPV conclusion while failing an added antecedent of each of Theorems 7.1 and 7.2. The transversal $(3, 2, 2, 2)$ specialization, by contrast, is used only for the hypothesis-level separation recorded in Corollary 8.7. These are inclusions among the criteria stated above; no priority or universal prior-art conclusion is being asserted. □

A Details of the use of generative AI

The research reported in this paper was carried out with the assistance of Pharos (website: github.com/frenzymath/Pharos), an automated mathematical reasoning system built on Rethlas [Ju+26]. Two projects were initiated on 18 October 2026 and 20 October 2026, respectively. The first project was supplied with the Méndez–Vaz Pinto–Villarreal conjecture and relevant references and produced a counterexample to the conjecture. The second project was supplied with this counterexample and tasked with finding additional hypotheses under which the conjectured conclusion holds, including cases beyond the positive classes treated in [BDK26a]. The present manuscript was prepared from the drafts produced by these two projects.

Each project initially comprised a coordinator and five theorem-producing workers. Pharos allows the coordinator to recruit additional helper and verifier agents autonomously, without human intervention in these operational decisions. All coordinators, workers, helpers and verifiers used internal model of Hy.

No additional mathematical guidance was supplied for the project after its initial input. The only subsequent user intervention was a message identifying some mistakes after the first draft had been produced.

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